

ERP PCA Toolkit 2.9974 Tutorial

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Thanks to Arild Hestvik for being a patient, long-suffering beta-tester, to Scott Miller for his extensive, helpful editing of the Tutorial document, to Dennis Molfese for his unstinting support for the development of the 2.00 version of the Toolkit, to Manny Donchin for encouraging me to venture out of my comfort zone by extending the PCA procedure to two steps, and Don Tucker for first suggesting to me that it might make some sort of mad sense to visualize the ERP PCAs by multiplying the factor loadings by the factor scores.

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About the ERP PCA (EP) Toolkit

This tutorial file doubles as a manual for the EP Toolkit. For this reason, it will go into quite some detail about the functions and how they work. For those more interested in a quick run-through of how to use the EP Toolkit, the most relevant paragraphs are colored blue. The paragraphs in black can be read as needed.

Uses

The EP Toolkit is a collection of software tools for enhancing analysis of ERP data. It runs in MATLAB and is meant to be an adjunct to the various ERP software suites currently used by researchers. The Toolkit contains functions for improved artifact correction, trial averaging, file editing, data viewing, factor analysis, and inferential analysis. The EP Toolkit has three major goals:

- 1) Facilitating multivariate decomposition of ERP data in order to better characterize their components. This is done by:
 - a) providing automated routines for conducting principal components analysis (PCA), including the two-step sequential PCA procedure.
 - b) providing a range of rotation options, including facilitated access to EEGLAB's independent components analysis (ICA) Infomax rotation.
 - c) translating PCA results back into voltage space so that they can be transparently evaluated as ERP waveforms and exported into various file formats for additional analysis.
- 2) Facilitating analysis of noisy data, including data from developmental and clinical research populations. This is made possible by:
 - a) multivariate techniques for automated removal of blink, saccade, and motion artifacts from session data using the Multiple Algorithm Artifact Correction (MAAC) procedure (Dien, 2024).
 - b) implementation of robust ANOVA statistics that are more resistant to noisy data than conventional ANOVA statistics.
 - c) use of trimmed means for trial averages and robust ANOVAs for inferential analyses.
 - d) provision of quality control (QC) measures that allow for easier detection of problematic data files.
- 3) Supporting analysis of multi-modal data and combined analysis techniques. This is made possible by:
 - a) application of fMRI artifact correction procedures.
 - b) provision of ERP, spectral, and time-frequency measures.
 - c) support for eye-tracking data.

The EP Toolkit is freely available under the GNU General Public License version 3. It is intended to be a research tool only and must not be used in clinical practice.

Citation

When used for a publication, please provide credit in the methods section and let me know that you have used it. I am hoping to eventually get some grant funding to support continued development of this toolbox. This in turn would benefit you the user. The appropriate citation is:

Dien, J. (2010). The ERP PCA Toolkit: An open source program for advanced statistical analysis of event-related potential data. *Journal of Neuroscience Methods* 187(1), 138-145.

Website

You can always download the latest version of the EP Toolkit from the following address: <http://sourceforge.net/projects/erppcatoolkit/>

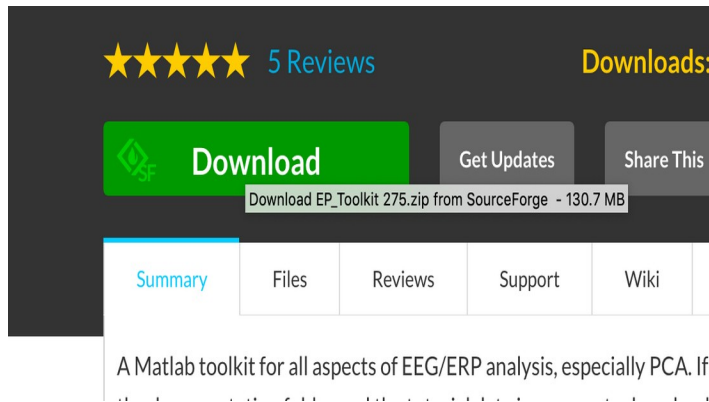
Mailing List

If you wish to be informed of new releases, please be sure to sign up for the mailing list: <https://lists.sourceforge.net/lists/listinfo/erppcatoolkit-support>

Suggestions and Bug Reports

I have not been able to obtain grant funding to support development of the EP Toolkit. I have therefore instituted a support program. While I will happily answer simple questions, fix bugs, and consider feature requests, this kind of support will only be available as time permits. Expedited bug fixes, extended explanations, and simple feature additions, are available for a \$1000 annual consulting honorarium. I would also be happy to provide an onsite 1-2 day workshop on the EP Toolkit and ERP methods in general in return for a \$1000 honorarium plus travel and lodging.

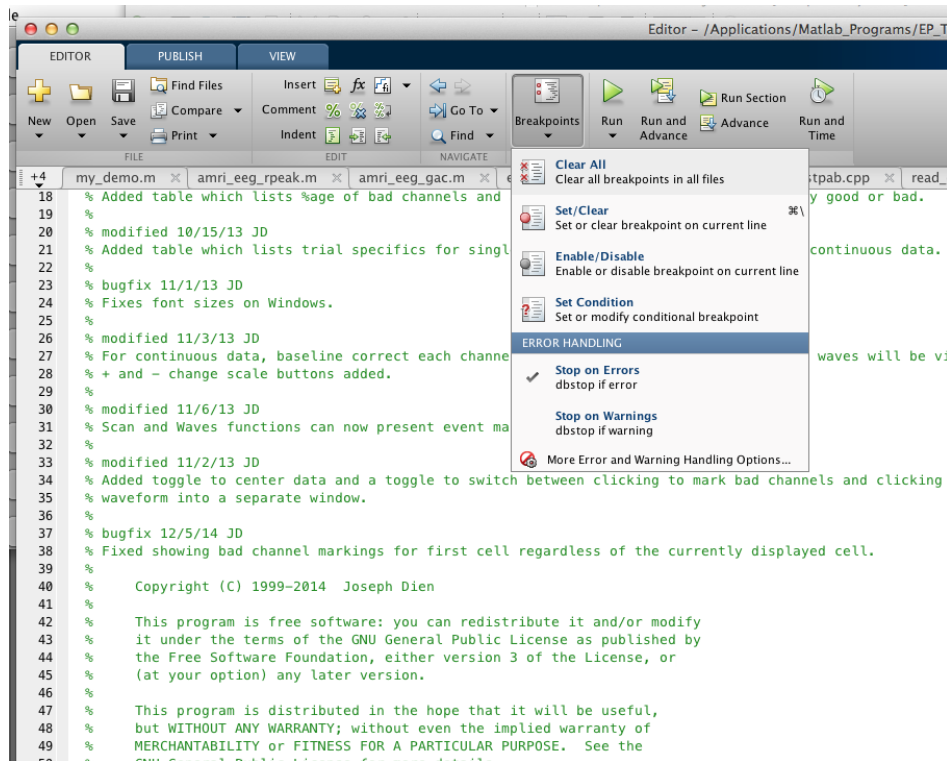
Your first step should ALWAYS be to check the version number of the EP Toolkit. It is pointless to report a bug with an old version of the EP Toolkit as it has likely already been fixed a long time ago. The version number appears whenever you start the Toolkit in both the splash page as well as the command window. The current version of the EP Toolkit can be found on the download page. Just mouse over the download button. The EP Toolkit should also be checking for updates every time you start it and should let you know when one is available.



Please feel free to send me suggestions or bug reports at: jdien07@mac.com. To troubleshoot, I will be needing to replicate the steps you made exactly so I can observe what the program is doing. I will therefore require all the files and all the information that will enable me to do so. When sending a bug report, please include the output you get from typing the command `ver` in the Matlab command window, the messages the EP toolkit generates when first starting it up, a copy of the file that you are trying to use, a screenshot with the settings you used, and an exact description of the problem and any error messages. When describing the problem, please do not say something vague like "so-and-so does not work." Please report the incident by first describing what you saw, then exactly what you did, then describing the results, all in sufficient detail that I can exactly replicate your steps and recreate the error.

To get a screenshot, on the Mac simultaneously press command, control, shift, and 4 (yeah, it's a bit tricky). The cursor will turn into a crosshairs. Use it to select the part of the screen you wish to include in the snapshot (it will generate a gray selection region as you click and drag). On a Windows computer, click on the window you wish to make a picture of (only it will be in the picture), then press alt and prtsrn. On a Linux computer, you can use a utility program like Screenshot. Regardless, once you have the screenshot, you should be able to paste it into the e-mail message.

It would be especially helpful if you can turn on the Stop on Errors setting and trigger the error again. At this point Matlab will be in a troubleshooting mode, denoted by a `K>>` prompt in the command window. Then type **save test;** This will have the effect of saving the Matlab session to a file named test.mat. If you are on a Windows computer, it may label it a microsoft access table but don't worry about it. What's going on is that Matlab produces data files that have .mat as the suffix. Windows computers think that any file that has .mat as a suffix is a microsoft access table shortcut and labels them that way but they're really not. Anyway, if you send me the file it'll be particularly helpful to me. The files will often be too large for e-mail. Internet services like Dropbox and Box and Google Drive can be used.



Please do not assume that I will be familiar with the problem that you have run into. Most bugs are triggered by a specific combination of circumstances and it is likely that I have never seen it before myself, otherwise I would already have fixed it. Also, please keep in mind that although I try hard to fix problems that arise, I literally have to do this work on my spare time as I do not have grant support and my time is limited.

Getting Started with Principal Components Analysis

You may find Dien and Frishkoff (2005) to be a useful introduction to the topic of applying PCA to ERP datasets. Dien (2012) provides a more advanced treatment, focusing on the different choices one must make as one progresses through a PCA of ERP data. Dien, Khoe, & Mangun (2007) provide a detailed consideration of ICA and how it contrasts with PCA rotations.

System Requirements

This package has been tested for use on a Macintosh under OS X 10.15.7 and Linux Pop!_OS 22.04 using version 24.2 (2024b) of MATLAB. The toolbox has also been used somewhat in Windows 10 but has not been fully tested so caveat emptor. I would be happy to help users troubleshoot platform incompatibilities if they are willing to work through them with me. I don't support anything earlier than Matlab 2020a as I just don't

have the time to juggle all the myriad differences between the versions; earlier versions of Matlab will crash the EP Toolkit due to the use of Matlab commands that didn't exist yet.

I mostly work with EGI and BrainVision files and so the Toolkit has mostly been tested with them, although it has received some use with other file formats. I would be happy to work with users to better support the existing list of file formats or to add new ones. The Toolkit functions also need significant amounts of RAM. I would suggest a minimum of 8GB and preferably at least 16GB, depending on the size of the datafiles. Also, the faster the computer the better as the algorithms it implements are quite computationally intensive.

Supported File Types

The EP Toolkit currently reads EGI Simple Binary (EGI .raw) including the epoch-marked variant for session files only, EGI EGIS (BESA .raw), EGI Matlab, EGI MFF, EEGLAB set/study including the Widmann variant, ERPlab erp, Neuroscan average, text, Biosemi BDF, Neuromag FIFF, BrainVision EEG/DAT/SEG, Psychophysiology Toolbox, FieldTrip, Matlab .mat, and EP Toolkit format files.

While the idea of being able to read Matlab .mat files may seem straightforward since the EP Toolkit is a Matlab program, it is important to understand that such data are missing all the meta-data that give the numbers meaning. When reading a .mat file with a data matrix (there must be only one matrix in the file), the EP Toolkit will present a window asking the user to specify what each of the Matrix dimensions corresponds to (e.g., channels). Any unused dimensions can be left as is. It will use the default sampling rate in the text file preferences and will assume there is no baseline period. Any additional information will need to be added manually via the Edit function and/or via .ced files if appropriate.

Text files can be read with either .txt or as .csv suffixes and a wide range of delimiters and unicode variants, all of which will be determined automatically. The regular .txt option will be for two-dimensional arrays where one axis is channels and one is time, typically used with the Single File Mode option, to be described later. The second option is the "stacked" text file, where each row is a waveform and the columns are the time points. There are seven ID columns to denote what the waveform corresponds to. For more information, first save an example data file to "stacked" format and then inspect it to see how it is organized.

The EP Toolkit can read some subject and trial data (like accuracy and reaction time and condition) from information encoded in the trigger channel of BrainVision files. An example E-Prime script (AP2-LD-Assoc-CB1.es2) and an example PsychToolbox script (DY3.m) are included in the Documentation directory. It can also read in the information encoded in E-Prime .txt files. It currently can write out EGI Simple Binary (EGI .raw), EGI EGIS, EEGLAB set/study, ERPlab erp, text, Neuromag FIFF, EGI MFF to a limited

degree, FieldTrip to a limited degree, Matlab .mat, and EP Toolkit format files. The support for both reading and writing FIFF and FieldTrip files are currently limited as they are both very complex file formats. There is also an option to save files in BESA .raw format, which is really the EGI EGIS file format, along with the .ela and .sfp files required by BESA.

Credits

Credit should be given to the algorithms used for the rotations:

Varimax

Kaiser, H. F. (1959). Computer program for varimax rotation in factor analysis. *Educational and Psychological Measurement*, 19(3), 413-420.

Promax

Hendrickson, A. E. & White, P. O. (1964). Promax: A quick method for rotation to oblique simple structure. *The British Journal of Statistical Psychology*, 17, 65-70.

Infomax

Delorme, A. & Makeig, S. (2004). EEGLAB: An open source toolbox for analysis of single-trial EEG dynamics including independent component analysis. *Journal of Neuroscience Methods*, 134, 9-21.

Gradient Projection method used for the following rotations:

QMAX - Quartimax

QMIN - Quartimin

Oblimin (gamma=0 tends to be recommended)

CRFE - Crawford-Ferguson family

MINE - minimum entropy

IPSC - Bentler's invariant pattern simplicity criterion

TIIC - Comrey's tandem II criterion

Geomin

MMER - McCammon minimum entropy ratio

Jennrich, R. I. (2001). A simple general procedure for orthogonal rotation. *Psychometrika*, 66, 289-306.

Jennrich, R. I. (2002). A simple general method for oblique rotation. *Psychometrika*, 67(1), 7-20.

Also, credit for the rotational criteria themselves:

Varimax

Kaiser, H. F. (1958). The varimax criterion for analytic rotation in factor analysis. *Psychometrika*, 23, 187-200.

Promax

Hendrickson, A. E. & White, P. O. (1964). Promax: A quick method for rotation to oblique simple structure. *The British Journal of Statistical Psychology*, 17, 65-70.

Infomax

Bell, A. J. & Sejnowski, T. J. (1995). An information-maximisation approach to blind separation and blind deconvolution. *Neural Computation*, 7(6), 1129-1159.

Quartimax

Carroll, J. B. (1953). Analytical solution for approximating simple structure in factor analysis. *Psychometrika*, 18, 23-38.

Quartimin

Carroll, J. B. (1953). Analytical solution for approximating simple structure in factor analysis. *Psychometrika*, 18, 23-38.

Oblimin

Carroll, J. B. (1957). Biquartimin Criterion for Rotation to Oblique Simple Structure in Factor Analysis. *Science*, 126(3283), 1114-1115.

Crawford-Ferguson family

Crawford, C. B. & Ferguson, G. A. (1970). A general rotation criterion and its use in orthogonal rotation. *Psychometrika*, 35, 321-332.

minimum entropy

Jennrich, R. I. (2004). Rotation to simple loadings using component loss functions: The orthogonal case. *Psychometrika*, 69, 257-273.

Bentler's invariant pattern simplicity criterion

Bentler, P. M. (1977). Factor simplicity index and transformations. *Psychometrika*, 42, 277-295.

Comrey's tandem II criterion

Comrey, A. L. (1967). Tandem criteria for analytic rotation in factor analysis. *Psychometrika*, 32, 277-295.

Geomin

Yates, A. (1987). *Multivariate exploratory data analysis: A perspective on exploratory factor analysis*. Albany, New York: State University of New York Press.

McCammon minimum entropy ratio

McCammon, R. B. (1966). Principal components analysis and its application in large-scale correlation studies. *Journal of Geology*, 74, 721-733.

Variable Oblimin

Gorsuch, R. L. (1983). *Factor analysis*, 2nd edition. Hillsdale, NJ:Lawrence Erlbaum Associates.

Extended Infomax

Lee, T.-W., Girolami, M., & Sejnowski, T. (1999). Independent component analysis using an extended infomax algorithm for mixed sub-Gaussian and super-Gaussian sources. *Neural Computation*, 11(2), 609–633.

JADE

Cardoso, J. F., & Souloumiac, A. (1993). Blind beamforming for non-gaussian signals. *IEE Proceedings F (Radar and Signal Processing)*, 140(6), 362–370.

SOBI

Belouchrani, A., Abed-Meraim, K., Cardoso, J. F., & Moulines, E. (1993). Second-order blind separation of temporally correlated sources. *Proc. Int. Conf. Digital Signal Processing*, 346–351.

fastICA

Hyvarinen, A. (1999). Fast and robust fixed-point algorithms for independent component analysis. *IEEE Transactions on Neural Networks*, 10(3), 626–634.

AMICA

Palmer, J. A., Kreutz-Delgado, K., & Makeig, S. (2012). AMICA: An adaptive mixture of independent component analyzers with shared components. *Swartz Center for Computational Neuroscience, University of California San Diego, Tech. Rep.*

Example Dataset

An example dataset is available for testing the EP Toolkit and is used in the tutorial in a separate file entitled `tutorial_dataset.zip` (<https://umd.box.com/s/m3rn0w0fpddy72kwmkekorqhf20kooxg>). This dataset is a very simple lexical decision task, collected by myself at the University of Maryland at College Park, using a BrainVision EEG system with 69 channels. This is a subset of an unpublished study, with only one of the conditions (long stimulus onset asynchrony or SOA associative priming) and only twenty of the participants. There were two blocks (LA1 and LA2) separated by a rest period. In each trial, a prime was presented in uppercase, followed by a target in lowercase at an SOA of 500 ms. The participants were instructed to indicate whether the targets were words or non-words (50/50) as fast as possible, while still being accurate. To further encourage controlled processing, there were also filler trials to increase the relationship proportion to 60%. Thus, the cells were LAWR (related word), LAWU (unrelated word), LANR (non-word with sublexical characteristics matched to the LAWR targets), LANU (non-word with sublexical characteristics matched to the LAWU targets), LANF (non-word filler trials), and LAWF (related word filler trials).

The data were collected using a PsychToolbox Matlab script. The script includes code to transmit session-specific and trial-specific information (e.g., stimuli words, accuracy, reaction time) over the trigger channel, which the EP Toolkit is then able to decode and include in the .ept files as additional meta-data. As will be seen, the availability of this

information can be quite helpful. A similar functionality is available for E-Prime. Examples of both are provided in the Documentation folder.

Set Up

Installation

In order to run the EP Toolkit, you will need to install the following programs:

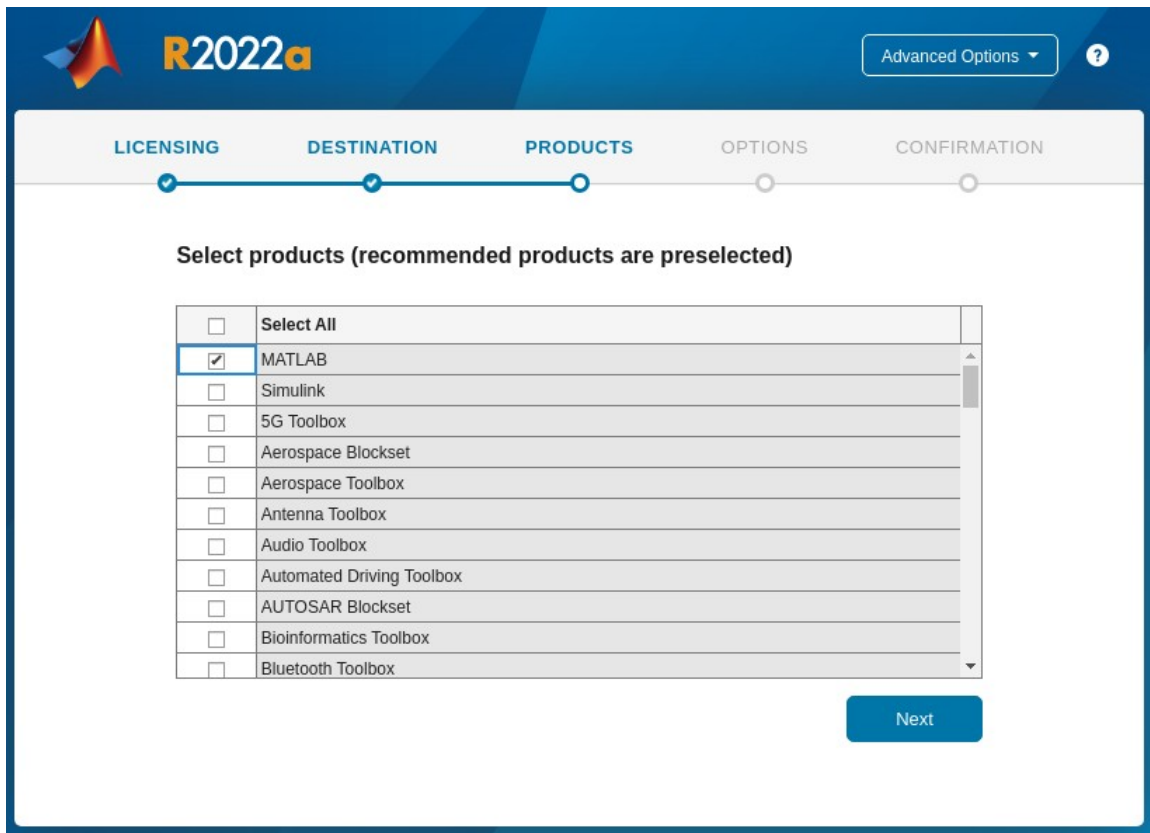
- (1) MATLAB
- (2) EP Toolkit
- (3) EEGLAB
- (4) FieldTrip
- (5) A C++ compiler (for dipole analysis)

All of these are freely available online except for MATLAB, which has affordable options for student and academic licensing. While installing EEGLab and FieldTrip takes some extra effort, I want to make sure that these software projects are getting download credit, rather than just including copies of their open-source files in my own project. While the EP Toolkit uses some of their functions (I see no reason to reinvent the wheel), most of it has been written by myself. The EP Toolkit will indicate in the Command window when functions from outside projects are being used and credits them.

1. Install MATLAB

MATLAB may be purchased directly from The Mathworks, Inc. (www.mathworks.com), but check with your academic institution first as it may already have a volume license or reduced pricing. Only the core MATLAB program is needed to run most of the EP Toolkit, so it is not necessary to purchase any MATLAB add-ons from Mathworks. A number of optional Matlab toolboxes can be helpful for specific functions however. The Statistics Toolbox is required for some of the sampleTest functions. For those who intend to use EEGLAB extensively, it will also run on the core MATLAB program, but the EEGLAB authors recommend that you install the Signal Processing Toolbox and Statistics Toolbox to improve the efficiency of some EEGLAB functions. The Signal Processing Toolkit is also needed for the fMRIB EEGLab plug-in (for correcting fMRI artifacts) but not for the AMRI fMRI artifact correction routines. The Signal Processing Toolkit also enhances a number of the FieldTrip functions used by the EP Toolkit, such as spectral decomposition. The Optimization Toolbox and the Parallel Computing Toolbox enhance the source analysis function. The Deep Learning Toolbox is required for the NIPALS option in the PCA function.

So, unless you have some reason not to, you should make sure the following toolboxes are checked during installation: MATLAB, Deep Learning Toolbox, Parallel Computing Toolbox, Optimization Toolbox, Signal Processing Toolbox, Statistics Toolbox. Conversely, you do not need any Simulink toolboxes.



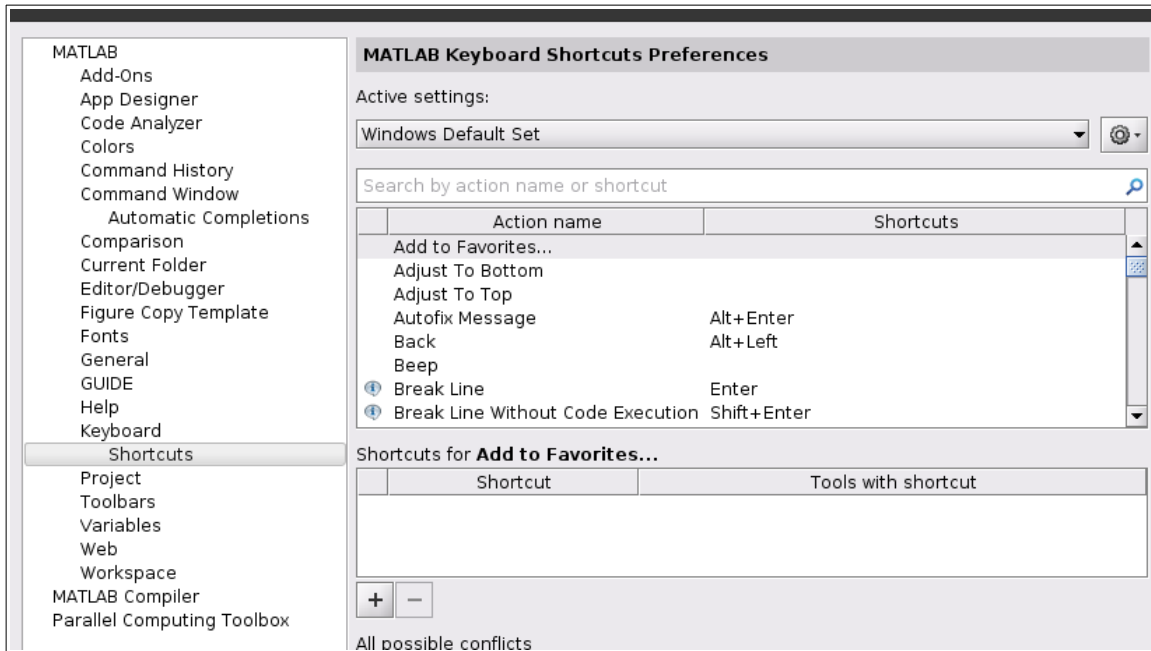
The image shows the MATLAB R2022a installation wizard interface. At the top, there is a blue header with the MATLAB logo and 'R2022a' text. To the right of the header is a button labeled 'Advanced Options' with a dropdown arrow and a help icon. Below the header is a progress bar with five steps: LICENSING, DESTINATION, PRODUCTS, OPTIONS, and CONFIRMATION. The 'PRODUCTS' step is currently selected, indicated by a blue circle and a line. Below the progress bar, the text 'Select products (recommended products are preselected)' is displayed. A table lists various products with checkboxes. The 'MATLAB' checkbox is checked, while all others are unchecked. A 'Next' button is located at the bottom right of the table.

Product	Selected
Select All	☐
MATLAB	☒
Simulink	☐
5G Toolbox	☐
Aerospace Blockset	☐
Aerospace Toolbox	☐
Antenna Toolbox	☐
Audio Toolbox	☐
Automated Driving Toolbox	☐
AUTOSAR Blockset	☐
Bioinformatics Toolbox	☐
Bluetooth Toolbox	☐

Next

Follow the instructions that come with MATLAB to install it.

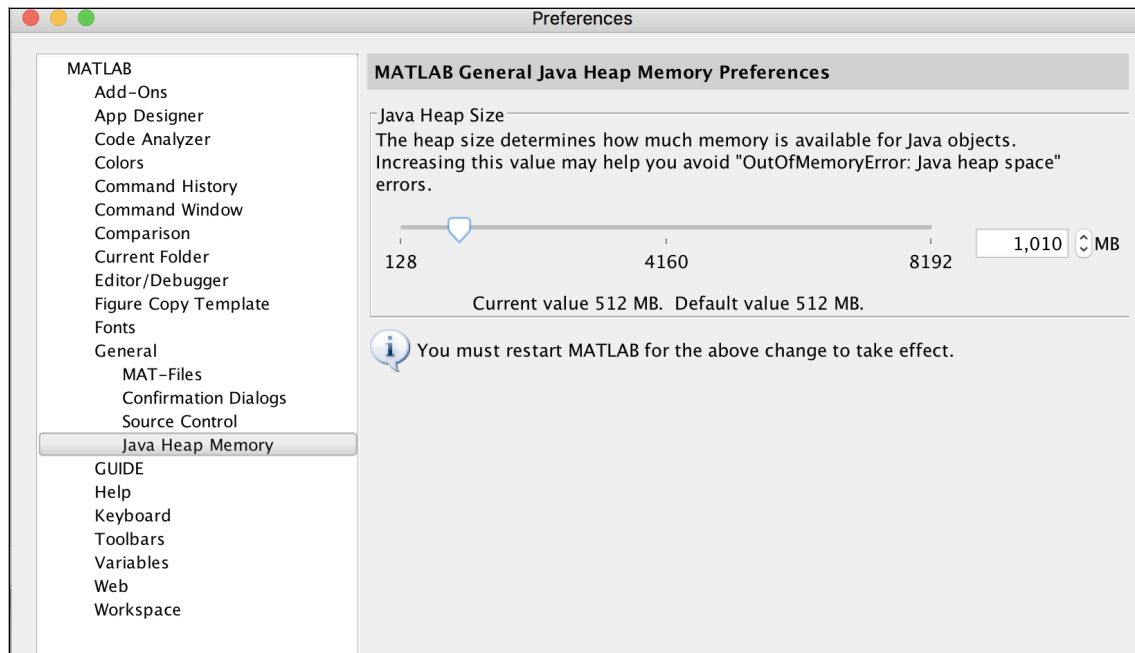
You will likely wish to change the keyboard shortcuts so that they correspond to standard PC convention (e.g., CTR-C means copy). To do so, click on Preferences and then bring up the Shortcuts pane under Keyboard. There, change Emacs Default Set to Windows Default Set and press OK.



At this point, I recommend creating a single folder on your computer to contain the EP Toolkit, EEGLAB, and FieldTrip, as well as any other MATLAB programs you may wish to run on your computer. Simply create the new folder, name it (I recommend “Matlab_Programs”), and make sure it is placed where you want it. This keeps the programs organized and provides a common path within MATLAB for accessing applications. Note that I used an underscore rather than a space in the name of the folder as the presence of spaces can confuse Matlab if the programmer of Matlab software was not careful.

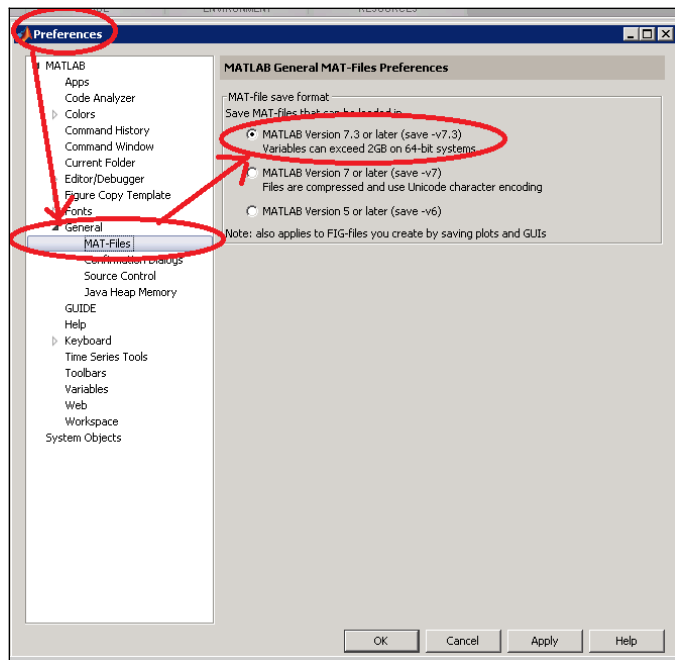
On the Mac, I would strongly recommend not putting this folder either on the Desktop or in the Documents folder. The reason is that the Mac, by default, backs up everything in these two locations to a cloud server named iCloud, and it only has a limited amount of space. A much better location would be the Applications folder. Alternatively, if you use DropBox or Box, you could locate the Matlab_Programs folder there, which is what I do.

In older versions of Matlab, the Java environment that Matlab runs in does not set aside enough memory to run large datasets. Go to the Matlab preferences and move the slider up, if it is less than 1GB, as shown:



which tells it to set aside 1GB of memory, up from the default of 256MB. Note that this setting is for what is called the heap space, which is a sort of a workspace that Matlab uses for whatever it is working on at the moment. This is aside from the more general memory storage represented by the Matlab workspace. Matlab can and will use more memory than the heap space (and virtual memory stored in available disk space as well) for general memory storage so it is not necessary for the heap space to be more than this to take full advantage of your installed RAM. You do, however, want to make sure that both your operating system and Matlab is 64-bit to make full use of your available memory space.

You should also make sure that the Matlab preferences are set to use the current version of .mat files. If it is set to use a legacy version then it may not be able to handle large datasets.



If the preferences are set to an older version, you may end up getting an error message like the following:

```

The name of the experiment is:
The pre-stimulus period of the data is: 0 msec
Warning: Variable 'EPdata' cannot be saved to a MAT-file whose version is older than 7.3.
To save this variable, use the -v7.3 switch.
Skipping...
> In ep_saveEPdataset at 95
   In ep at 2634
>> ver

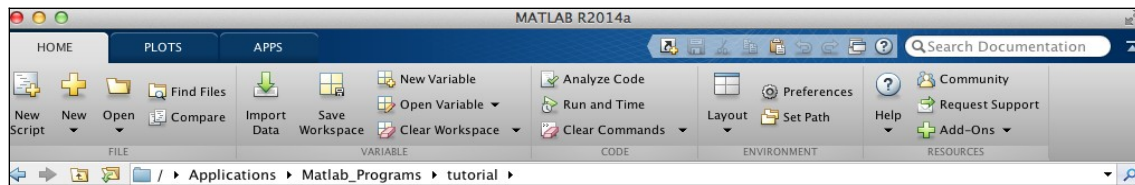
```

2. Install EP Toolkit

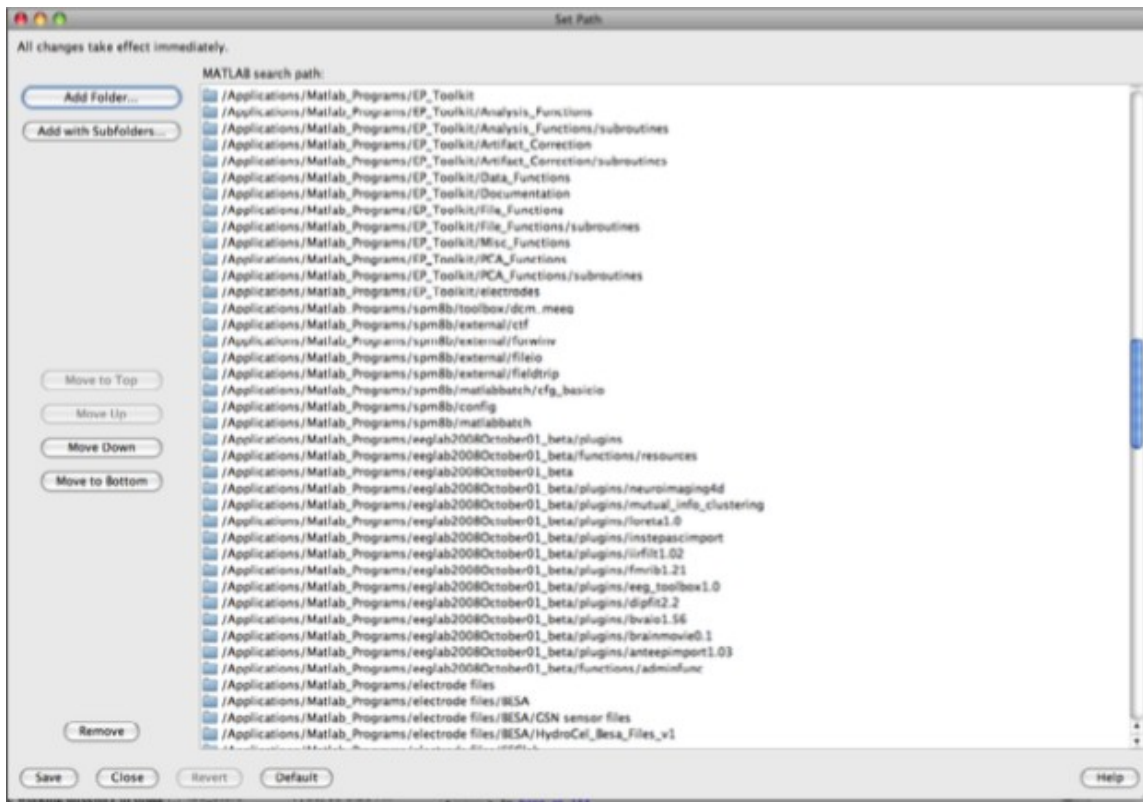
If you haven't already done so, download the latest version of the EP Toolkit from the project website listed above, and unzip the file. This will produce a folder named *EP_Toolkit* with several files and folders inside of it. On non-Mac platforms you may find a number of files starting with "_" which contain Mac-specific information and can be ignored.

Put this EP_Toolkit folder into the desired directory. On my computer, I place it in the Matlab_Programs folder that I created and placed in the Documents folder. **Note: Do not use OneDrive's Files On-Demand feature with this or any other of the Matlab programs that you will be installing.**

Now open MATLAB and add this folder to the MATLAB path. To do this, click on the Set Path button on the Home tab of the main Matlab window.

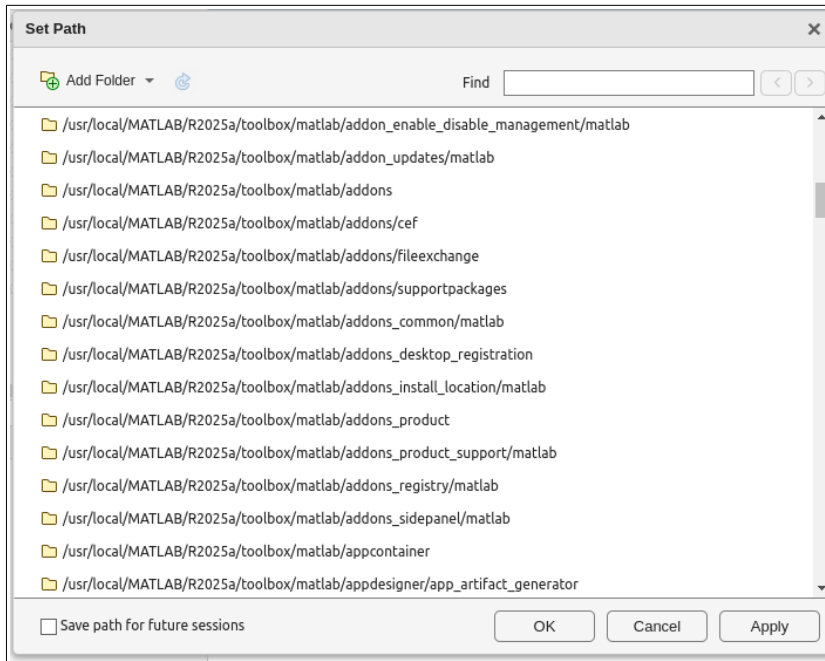


In the window that then appears, click on the "Add with Subfolders" button. A browser window will open. Navigate to the EP_Toolkit folder (e.g. on my Mac, I click on *Documents*, then *Matlab_Programs*, then *EP_Toolkit*). With the EP_Toolkit folder selected, click the "Open" button. The path list for the Toolkit should end up looking like this, with several lines, one for each subdirectory in the Toolkit folder:



Now save these changes by clicking "Save" at the bottom of the Set Path window. Many new users of MATLAB forget this critical step. If you close the window without saving, MATLAB will discard the path changes the next time you quit the program and you will have to repeat the process.

As of MATLAB 2025a, this window looks a bit different. If you click on Add Folder, it will again give you the choice of adding just the folder or also the subfolders. The other change is there is a checkbox instead of a save button. When I used this on a Linux computer, I got an error message that it could not save the path. Instead, you would have to use the Command Window to do: `savepath('~/pathdef.m')`



On OS X, as of version 10.15 (Catalina), you will need to perform an extra step to allow what are called "mex" files to work. The instructions can be found here: https://www.fieldtriptoolbox.org/faq/mexmaci64_cannot_be_opened_because_the_devel_oper_cannot_be_verified/

Basically, you need to find the path of the EP Toolkit directory. So navigate Matlab's Current Folder (click on the little folder icon with the green arrow, so it'll bring up a file navigation window) to the folder where you put the EP Toolkit and then enter the EP Toolkit directory. Then type pwd in the Command Window. A typical path might look like: /Applications/Matlab_Programs/EP_Toolkit
If it doesn't have "EP_Toolkit" at the end of it, you didn't do it right.

Then start up the Terminal app, located in the Utilities folder in the Applications folder. Copy and paste the following into the Terminal's window:

```
sudo xattr -r -d com.apple.quarantine LOCATION_OF_EP-TOOLKIT
```

replacing LOCATION_OF_EP-TOOLKIT with the location of the directory. If you kept it in the example location, it would be:

```
sudo xattr -r -d com.apple.quarantine /Applications/Matlab_Programs/EP_Toolkit
```

You will then be asked to provide the admin password, so be sure to have that ready.

Do all of this by copy and pasting, as you will otherwise likely make mistakes. Note that there are three ways to copy and paste: 1) right-click to get a menu. 2) Command-C and

Command-V. 3) Use the application's menu at the top of the screen. If one doesn't work, try another. For example, by default, Matlab maps Command-C and Command-V to actions other than copy and paste, for whatever reason, but the other two will work. Sometimes you have to use the right combination of copying from one application to another (like using right-click both times or whatever). You may need to experiment.

It will ask you for your Mac's admin password.

Then type:

```
sudo find LOCATION_OF_EP-TOOLKIT -name \*.mexmaci64 -exec spctl --add {} \;
```

So if you keep it in the example location, it would be:

```
sudo find /Applications/Matlab_Programs/EP_Toolkit -name \*.mexmaci64 -exec spctl --add {} \;
```

3. Install EEGLAB

You will also need Delorme and Makeig's excellent EEGLAB toolbox (<http://sccn.ucsd.edu/eeglab/index.html>) if you wish to run the InfoMax rotation (ICA) and to make use of their 2D and 3D plotting functions for visualizing the data. They recommend using the latest "stable" version. As of "eeglab2024.2.1" this worked fine.

1. Download the compressed EEGLAB distribution below into an eeglab directory/folder

What version should I download? If you do not have Matlab, download the compiled version. If you are just starting with EEGLAB, download the stable version (EEGLAB 9 or 10). If you are planning to do extensive multi-subject analysis, download the development version (EEGLAB 11).

EEGLAB v11 most current (32MB) Download link	EEGLAB most current. For developers and advanced users only (tutorial demo STUDY, separately available in downloads below). This version is also recommended if you are actively working with EEGLAB studies. Use the latest generated development version in the FTP folder . The current version contains the most recent developments and bug fixes. Instead of downloading the ZIP, it is also possible to use SVN (SubVersion). See the download wiki page for more information. Revision details are available on the EEGLAB wiki .
EEGLAB v10 stable (32MB) Download link	EEGLABv10. This version is still supported. Use the latest generated EEGLAB version 10 in the FTP folder (latest is at the bottom). The latest changes made to this version are detailed here . Older EEGLAB versions are available here .
EEGLAB compiled (182MB) Download link	EEGLAB compiled version for windows OS 32-bit. This version does not require Matlab. If you have access to Matlab though, we recommend the version above. More information about the compiled version of EEGLAB is available here .

After downloading and unzipping EEGLAB, move the folder to the desired directory (e.g., Matlab_Programs folder). Then open MATLAB and add the EEGLAB folder to the path. Do so by first adding just the root directory (without the subdirectories) using the Add Folder... button rather than the "Add with Subfolders" button in the "Add Path..." window. Then type "eeglab" in the command window. EEGLab will add appropriate subdirectories to the path. Make sure to have only one copy of EEGLab in your path list. **Save the new additions to the path!**

4. Install FieldTrip

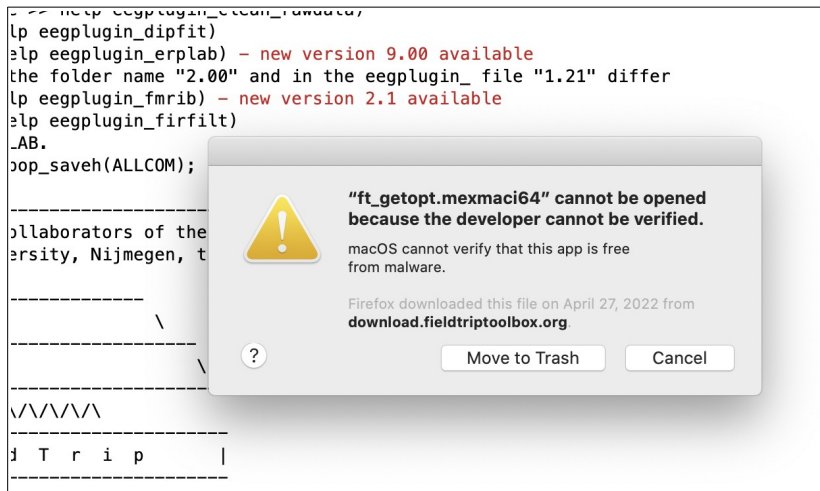
You will also need Robert Oostenveld's excellent FieldTrip toolbox (<http://www.ru.nl/fcdonders/fieldtrip/>) in order to be able to read in files (I contributed to some of the code), CSD, filtering, spectral analyses, and a simple dipole source analysis routine. Go to the website and register. They will email you a website URL to download the file from. The site will contain a list that will look like the figure below. The number is the date of the file (YYYY-MM-DD). There are two versions for each, the full FieldTrip and the Lite version. I use the full version for all my testing and cannot

guarantee that the EP Toolkit will work properly with the Lite version. Note that the latest version is at the BOTTOM of the list, at least as of this time.

Index of /			
../			
example/	30-Nov-2021 12:16	-	
historical/	15-Apr-2022 15:31	-	
modules/	04-May-2022 21:46	-	
test/	11-Mar-2022 15:13	-	
tutorial/	11-Mar-2022 15:32	-	
workshop/	20-Sep-2021 16:08	-	
00README.txt	15-Apr-2022 10:55		1751
fieldtrip-20220104.zip	05-Jan-2022 22:46	418092971	
fieldtrip-20220113.zip	20-Jan-2022 22:46	418094901	
fieldtrip-20220125.zip	26-Jan-2022 22:46	418094913	
fieldtrip-20220126.zip	27-Jan-2022 22:46	418094913	
fieldtrip-20220128.zip	29-Jan-2022 22:46	418095899	
fieldtrip-20220131.zip	31-Jan-2022 22:46	418095899	
fieldtrip-20220202.zip	02-Feb-2022 22:46	418096622	
fieldtrip-20220203.zip	03-Feb-2022 22:46	418096681	
fieldtrip-20220204.zip	04-Feb-2022 22:46	418096794	
fieldtrip-20220206.zip	06-Feb-2022 22:46	418098858	
fieldtrip-20220208.zip	08-Feb-2022 22:46	418099225	
fieldtrip-20220212.zip	12-Feb-2022 22:46	418098618	
fieldtrip-20220214.zip	14-Feb-2022 22:46	418098643	
fieldtrip-20220215.zip	15-Feb-2022 22:46	418101399	
fieldtrip-20220216.zip	16-Feb-2022 22:46	418101401	

Download the file, unzip it, place it in the desired directory (e.g. Matlab_Programs). Fieldtrip will not work if it is put in some Windows shared folders like OneDrive or lab servers, although it seems to work on DropBox and Box. Remember to add it to the MATLAB path too. Do so by first adding just the root directory (without the subdirectories) using the Add Folder... button rather than the "Add with Subfolders" button in the "Set Path" window. Then type "ft_defaults" in the command window (if using OS X, see next paragraph before doing so). FieldTrip will add appropriate subdirectories to the path. It's no longer necessary to worry about what is in the "external" directory. Make sure to have only one copy of FieldTrip in your path list. **Save the new additions to the path!**

On OS X, you will need to again perform the process to allow FieldTrip to mex files, as you did for the EP Toolkit. If you forget to do so, you will get the following error message when you try to use ft_defaults.



When this comes up, click on Cancel and then perform the procedure described for the EP Toolkit to eliminate this problem.

```
sudo find LOCATION_OF_FIELDTRIP -name \*.mexmaci64 -exec spctl --add {} \;
```

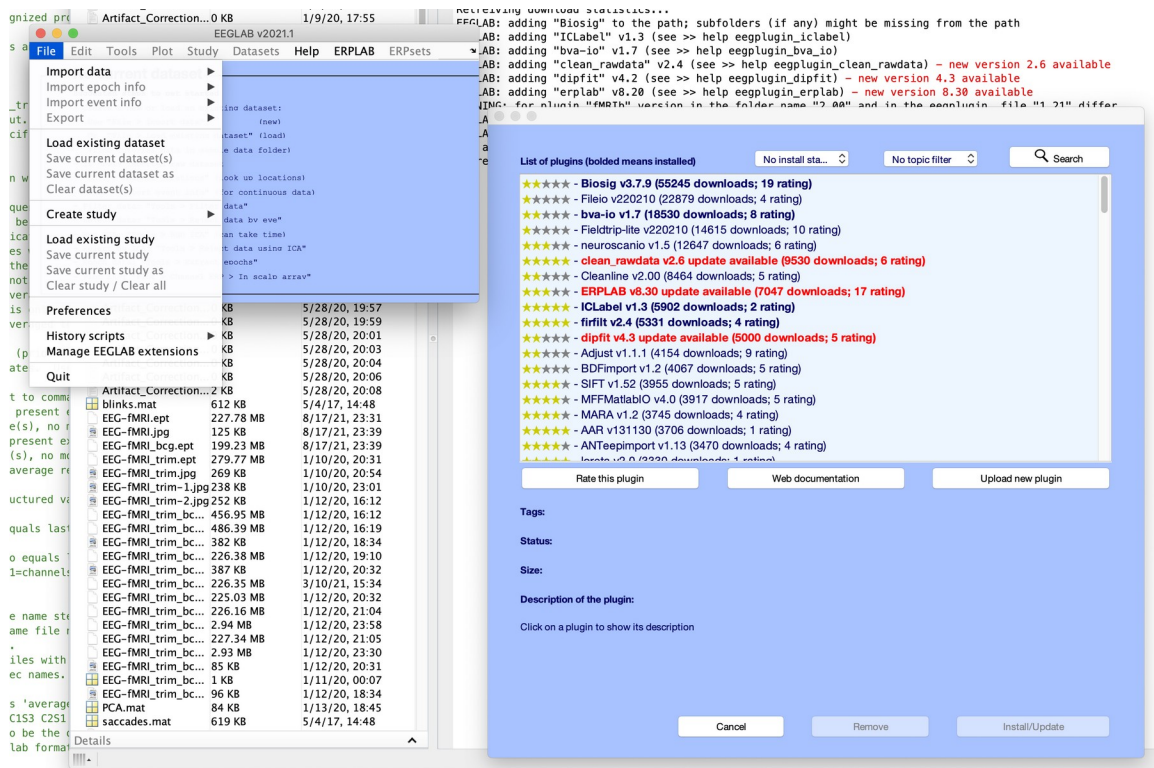
If your Mac is one of the newer ones with the M-series CPU (Apple Silicon), then you will also need to do the following command in Terminal:

```
sudo find LOCATION_OF_FIELDTRIP -name \*.mexmac64 -exec spctl --add {} \;
```

In order for the dipole analysis feature to work, it will also be necessary to have a copy of the gcc or clang compiler installed on your computer. On the Mac just go to the app store and download the free Xcode application. Linux distros tend to have it preinstalled. I'm not really certain how to install gcc on Windows but it should be available from the internet for free as it is the open source compiler (there are lots of hits on google for this topic).

5. Add *fMRI* artifact correction routines

If you wish to perform fMRI artifact correction, install the fMRIB EEGLab plug-in by first starting up EEGLab by typing "eeglab" in the command window and then choosing from its File menu the "Manage EEGLab Extensions " option. I made some fixes to it and they were accepted by EEGLab so make sure you have the 2.0 (or later) version installed. Afterwards, save the path again using Matlab's Set Path button.



An alternative fMRI artifact correction routine, which relies primarily on the ICA approach rather than the OBS template approach and does not require the Signal Processing Toolbox, can also be installed. To do so, go to the following website: <http://amri.ninds.nih.gov/cgi-bin/software>

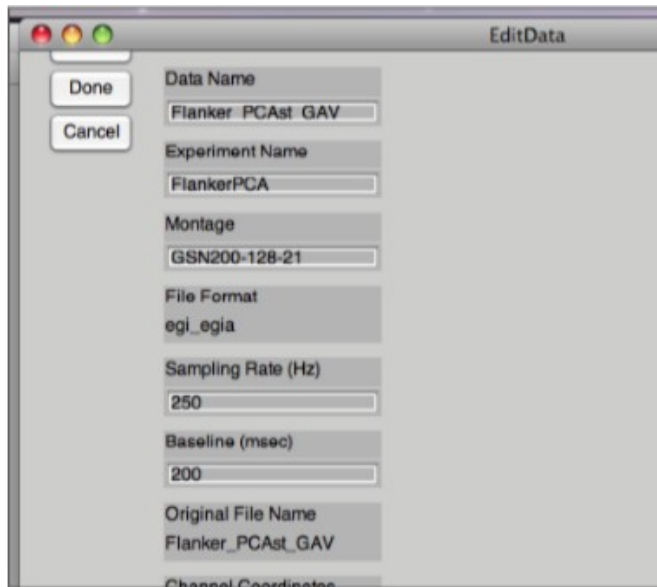
Create an account and then use the resulting password to download the available software. The current release (amri_eegfmri_toolbox.20130731.v0.1.4.tar) crashes if the final gradient artifact is too near the end. A fix for this is included in the externals directory of the EP Toolkit. Either substitute it for the original or make sure that the EP Toolkit is higher on the path list than the AMRI package. Also, the “mi” subdirectory is currently missing compiled mex files for Intel Macs. These mex files are also provided in the EP Toolkit’s external directory.

Avoiding Common Problems

Monitor Resolution

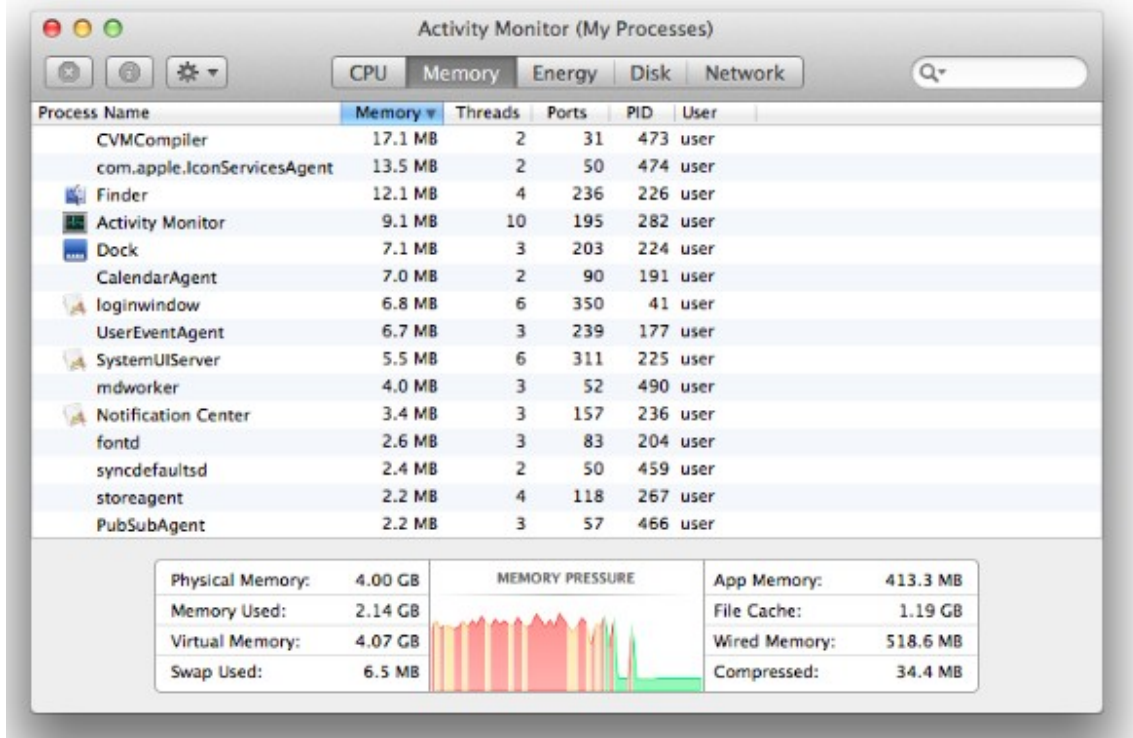
On a Mac with a smaller monitor, make sure to set your Dock to appear on the right or left side of the screen. If it is set to be at the bottom and is abutting onto the EP Toolkit pane then it may throw off its contents, as in this picture. Note how the contents of the pane have been shifted upwards, resulting in some things disappearing off the screen. The Task Bar can cause the same problem on Windows machines. On Windows 10, you

can move the Task Bar to the side of the screen by right-clicking the Task Bar and choosing Task Bar Settings. You can also use the monitor preferences (to be described) to specify a smaller window region that excludes the Dock/Task Bar area.



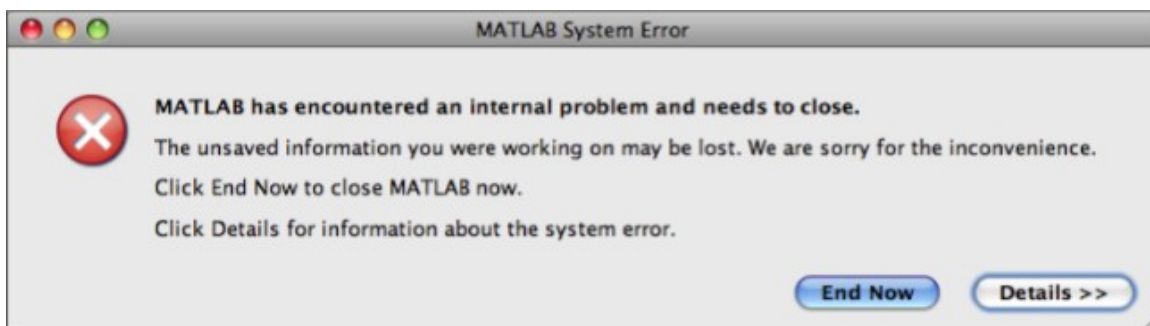
Memory Issues

On OS X, it is highly recommended that you use the Activity Monitor (or the Task Manager on Windows). On OS X you'll want to keep the main window open on the Memory pane, where it'll show you an overall "memory pressure" graph, which you'll want to have in the green zone.



One reason you want to keep an eye on this is that MATLAB at least used to have very poor memory management. You can think of memory like a large room where objects are placed for later use. Objects are then removed when no longer needed. The problem is that when a new object is added, MATLAB cannot rearrange the existing objects to make efficient use of the floor space. If it tries to store a large object and there is no existing space large enough to accommodate it, then the program will crash. You will start getting alarming red messages about being OUT OF MEMORY in the command line and the EP Toolkit will stop working. When there is enough overall memory but it has simply been cut into too many small pieces to handle the new object (like a new data file), then this is called "memory fragmentation."

Examples of memory errors are:



Or you may see the text:

??? Error using ==> zeros

Out of memory. Type HELP MEMORY for your options.

Once memory fragmentation has occurred, the only way to deal with it is to quit out of MATLAB and then restart it. If you see the memory pressure gauge go red, this is an indication that you may soon run into memory problems. Another symptom of memory fragmentation, short of an actual crash, is that MATLAB starts slowing down to a crawl. Again, the best solution is to just restart MATLAB at this point. It also helps to quit out of other apps, especially browsers. The working set of data will still be present when you restart the Toolkit. It's also not a bad idea to set the Activity Monitor to show the floating window with CPU usage so you can have a better sense of what is going on when you're waiting for the Toolkit to finish a function.

Energy Saver Settings

An issue specific to OS X is that MATLAB does not register as an active program to the Energy Saver controls. Thus, if the Energy Saver is set to put the Mac to sleep after a certain amount of time, it will put it to sleep in the middle of an ongoing MATLAB run. Thus, if you are planning on running long MATLAB sessions, such as eyeblink correction runs, it would be best to at least temporarily set Energy Saver to Never Sleep.

File Preparation

EP Toolkit File Types

The EP Toolkit recognizes three types of files:

1. *continuous* - is a single stretch of unsegmented raw data recording. For many functions continuous data will be treated as a series of one second segments. The EP Toolkit has in-depth functionality for performing segmentation.
2. *single_trial* - is a segmented raw data recording.
3. *average* - is one or more subject averages. The Toolkit's PCA functions are designed to work on combined subject averages in which a single file contains an experiment's entire set of subject averages in the form of separate waveforms. If you have separate subject average files, you can use the "append" function on the subject pane of the Edit function to glue them together into a single combined subject average file or use the single file mode to merge them together with the Read function.

A full EP data file may have all of the following dimensions: channels, time points, cells, subjects, factors, frequencies. This structure can be perused via the Edit function and the

waveforms may be examined directly via the View function. It may also have a seventh dimension of channel relationships, as in coherence, for which the Edit and View functions do not provide separate controls.

The EP file format itself is described in the header of the `ep_readData.m` file. While it is really a .mat file, the suffix for EP files is .ept (to help distinguish it from normal .mat files).

As a side note, while it is true that various software packages can produce .mat files, such files are not useable by the EP Toolkit. The reason is that critical information like whether channels are the rows or the columns are not specified in such files and basic information like the sampling rate will be entirely missing. It's not worth the effort to add support to the Toolkit for them and it's not worth the effort you would have to make to manually add the missing information every time you wanted to use such a file.

Electrode Coordinates

Many EP Toolkit routines, especially the artifact correction function, require 3D electrode coordinates. .ept files maintain the coordinates information internally along with the data itself. When reading in data from another format, the EP Toolkit is able to use the names of the channels to automatically generate the electrode coordinates, using canonical coordinates shared by Robert Oostenveld, if they have standard 10-05 names. If the channel names are non-standard or more channel information is required, then the information can be provided by an EEGLAB .ced file. The EP Toolkit already includes .ced files for a number of montages and they can be accessed as needed.

If a new .ced file is needed, one can edit an existing one using any text or spreadsheet editor as they are actually just .txt files (with a .ced suffix). Alternatively, one may generate one using an EEGlab function by typing the following into the Matlab command window:

- 1) `pop_chanedit([]);`
- 2) click on Read Locations.
- 3) click on Save (as .ced)

Finally, one can read in a file into EP Toolkit, allow it to automatically generate the electrode coordinate information, and then use the Edit function (to be described later) to save this information as a .ced file which can then be modified as needed.

The .ced file will be treated as the definitive statement on what channels should be in the data file and their proper ordering, except for the EGI EGIS and simple binary formats where all channels are always present and in the same fixed order. Except for these two formats, if the data file has channels that are not present in the .ced file, then they will be present as locationless data but will not be utilized for operations that need electrode coordinates like bad channel replacement. If there are EEG channels in the .ced file that are not present in the data file, it will be assumed that the channels were bad channels and

had been dropped (normal practice for Neuroscan files) and they will be added back in as zeroed out channels and they will be internally marked as being bad channels. If all the channel names in the file and in the CED file are different but there are the same number of channels, it will be assumed that they correspond and are in the same order and so the channel names and other information in the CED file will be used.

If you need to edit your own ced file, the following channel types are recognized:

EEG - basic EEG channel

ANS - generic autonomic data

MGM - magnetometer MEG data (not fully supported yet)

MGA - axial MEG magnetometer data (not fully supported yet)

MGP - planar MEG magnetometer data (not fully supported yet)

ECG - electrocardiogram data

pupil - pupil dilation data

ACMx, ACMy, ACMz - accelerometer head position data

TLTx, TLTy, TLTz - head tilt data

BSC - BOSC analysis results

Additionally three labels are recognized during data import:

REF is an EEG channel used as a reference channel during data collection, regardless of whether it has been subsequently rereferenced. It will be relabeled as EEG.

FID stands for “fiducial” and indicates locations that are merely landmarks. They will not have corresponding channels but will be retained for use in co-registration.

BAD is for inactive channels that should be completely deleted (as opposed to channels that were bad for a particular subject but has meaningful data for other subjects and can be dealt with using channel replacement).

Reference Channels

Also, note that data files often have the reference site present only implicitly. Reference channels are by definition zero voltage and so to save disk space many software (not the EP Toolkit) leaves them out entirely. If the reference channels specified in the .ced file are missing from the data file, they will be added as explicit flat channels. Note also that for a mean mastoid reference, if both mastoid channels are missing then they cannot be reconstituted as only their average is zero; they should both have non-zero waveforms that are mirror images of each other that split the difference between the two sites (for more on reference issues, see Dien, 1998 and Dien, 2017). When using software that gives you an option as to whether to include them (such as Neuroscan’s software), you should always choose to do so.

BrainVision's PyCorder program gives the option to use a "no-reference" option. As far as I can tell after discussions with tech support, all this means is that the voltage numbers are not interpretable until the data are rereferenced. It also gives the option to choose more than one channel as being the reference (e.g., mastoids) but then it drops all these

channels and their data are lost so it is best not to do so. Either choose one channel as reference or none and then rereference as needed during analysis.

Net Station Files

If you are using Net Station, you will need to use the mff file format. Unfortunately, the functionality to write files in mff file format is limited (you can't write out average files that includes multiple conditions and subjects). If it does not meet your needs, you can still use Simple Binary, although it has numerous drawbacks. EGIS is no longer an option for NetStation as of version 5 unfortunately.

The EGIS session format cannot represent baseline period information and so that has to be input manually using the Edit function (to be described). EGIS average files can represent baseline period information but when Net Station saves them, it leaves it out. Also, it leaves out the subject number information. Both types of information will need to be added in using the Edit function. EGIS files are only accurate up to about two decimal places but that is normally sufficient. EGIS files, unlike Simple Binary, can support Trial Specific information but Net Station does not support this aspect of the file format. EGIS files also do not preserve the full event information the way Simple Binary files do but as Net Station does not import the event information in Simple Binary files, this is largely moot.

Simple Binary format has the drawback that it does not represent the baseline period information but the FieldTrip import routine (which I wrote) will try to guess what it is. You will therefore need to verify that it deduced it correctly (using the Edit function). Note also that NetStation (as of 4.1.2) loses the trial-specific information when writing out continuous (non-segmented) simple binary files. Also note that the Simple Binary format is not designed to handle epochs of different sizes (although the epoch-marked variant can get around this limitation) so if you try to export such a file from NetStation, the file may be corrupted and cause errors when you try to read it. The easiest way to test for this is to just try to open the file using NetStation. It will fail to open if the file is corrupted in this manner.

Text files can also be used as a data export option from NetStation. A separate file will be generated for each segment (for single trial data) or each average (for averaged files). In this case, use the Single File mode to merge these files back together again.

Another option for data exporting is Matlab file format. Choose the "cell array" option. Depending on the version of NetStation, this will have either a .mat or a .nsf file suffix. Note that the EP Toolkit does not support importing of event data due to problems in how NetStation exports them (if the "export good segments only" option is chosen, the bad segments are dropped only from the data, not the events, making it impossible to determine which events go with which data).

You can also just export the raw data from NetStation (as simple binary or mff) and perform all the processing using the EP Toolkit. This is recommended as the Toolkit will retain a variety of information that will otherwise not be available.

Another issue is the channel locations. Unlike most systems, the channels do not follow the 10-05 system. Some functions like dipole analysis and 3D plotting require first translating coordinates to a standardized space and that requires being able to identify the channel names. In order to enable this translation, the EP Toolkit requires the Montage to be specified when reading an EGI file.

Finally, note that EGI has issued age-normed electrode coordinate files and they are included as ced files. Thus, “9_18AverageNet64_v1.ced” means the 64-channel Hydrocel net for ages 9-18 years old. This age-norming has nothing to do with the Montage selection, which in this case would simply be “Adult Hydrocel 64-channel” as the overall spatial configuration of the electrodes would still be valid.

BrainVision Files

In order to facilitate the use of BrainVision files, I've implemented a method for encoding subject and trial information in the trigger channel that I term a BV header and serves to supplement the normal BrainVision .vhdr and .vmrk files. This information is translated by the EP Toolkit and incorporated into the data file when read and can then be used for segmentation purposes. If the computer is too slow, you might have some problems with consecutive repeats sometimes not being distinguished however (e.g., two ones in a row) even though all the numbers are supposed to be separated by zeros. With current hardware, the process has been quite reliable on my system and I now use the BV header as the primary source of meta-data, with the .txt files serving as a back-up. The EP Toolkit will issue warnings when a header or trial specific information is not decoding properly (and in the latter case the trial will be dropped). If using PyCorder or BV Recorder, make sure to set the events to be saved as S1-S255, although the EP Toolkit can also handle S1-15/R1-15 labeling. An example PsychToolbox script (DY3.m) is provided.

One can also use E-Prime files in this manner. An example E-Prime file (AP2-LD-Assoc-CB1.es2) is included in the Documentation folder to demonstrate how to do so. To implement it, one must copy the code from the User tab of the Script window to provide the needed global variables and subroutines. Also the inline code objects named headerTriggers and trialSpecs. The trial information can take as much as a second to encode so to control the ITI the trialSpecs inline object is also set to time how long it took and to then pad the ITI out to the desired length. In order to run this E-Prime script, the file INIT_IO.txt will also need to be present in the same directory.

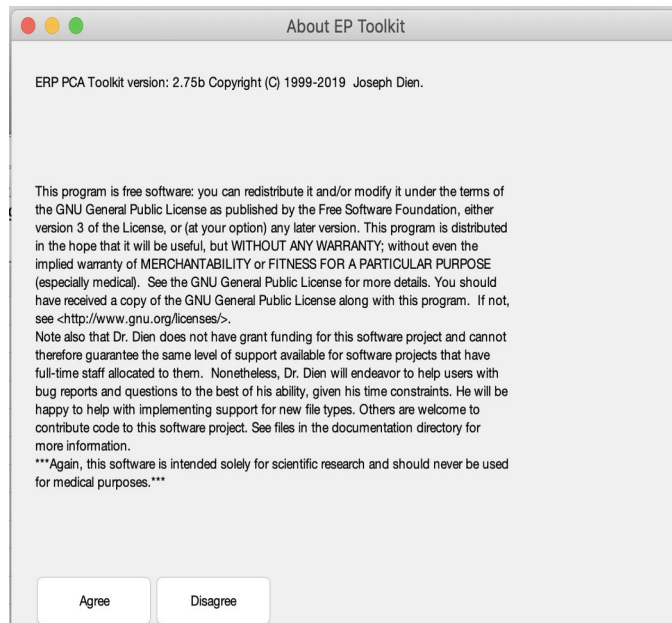
Other Formats

For average files other than EGI EGIS and EGI Simple binary formats, if the cell names all start with "Sub" and are followed by three numbers then the Toolkit will interpret this situation as meaning that this means that this suffix (Sub###) refers to subject numbers and will arrange the data accordingly. Any text following the three numbers will be interpreted as the actual cell name (e.g., "Sub001target" would be interpreted as the cell "target" for subject "Sub001"). This is a workaround for a limitation of the FieldTrip I/O code.

Running the Toolkit

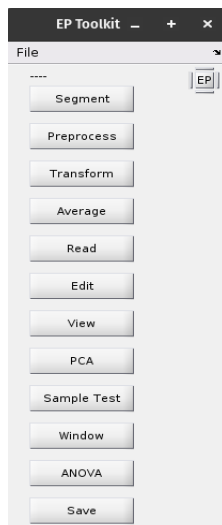
Starting Up

To start the Toolkit, type `ep` at the command line of MATLAB. The following splash page will appear:



Read it and indicate whether you agree with the stipulations. If you disagree then the Toolkit will quit (sorry).

The Main Window should also appear, showing the Main Menu:



Note that some of the options will be grayed out until appropriate data have been loaded. In the upper right corner is a status button. When the EP Toolkit is working on something, it will turn light blue and a black bar will rotate around the edges. It will turn dark blue when it is conducting file operations that should not be interrupted to avoid data corruption. It will turn green when finished. If you wish to abort data processing or otherwise reset the status of the EP Toolkit, press the status button. Also, note that at least under OS X there are persistent glitches in Matlab's drawing of the interface, with buttons and so forth entirely missing. Also, sometimes it becomes impossible to type in text boxes. If you find these issues happening, just click on the Matlab command window and the problems will resolve. I haven't find a solution thus far, except to switch to Linux or Windows.

Preference file & EPwork directory

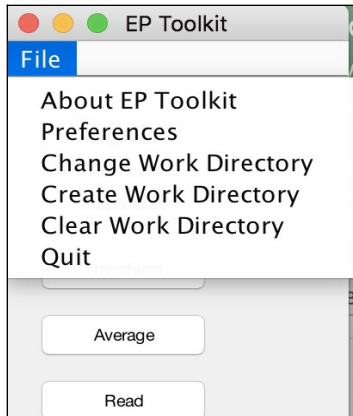
A working directory called EPwork will be placed in Matlab's default user directory. On a Mac, this is in the Documents/Matlab folder. Other operating systems have their own user locations. A permanent preference file called EPprefs will be placed inside the EPwork folder. If you already had an EPwork or EPpreferences folder installed from a prior version of the Toolkit, it will continue to use them.

EPwork contains a working copy of each data file that you have loaded into the Toolkit. DO NOT modify the contents of the EPwork directory. It is meant for the internal workings of the Toolkit.

The EPwork working directory makes it possible for the Toolkit to maintain a working set of data. This set is listed in several of the functions. You can add files or remove files from the working set using the Read function. The resulting data file from PCA is also added to the working set. When using these functions, you just select the data to be processed from this working set. When you are done, you must use the Save function to make a permanent copy of the file in the working set.

For the top four functions (prior to Read) the working set is not used except as templates. For these four functions (Segment, Preprocess, Transform, and Average) you will select the files to be processed directly from the disk and the resulting files will be saved directly to the disk.

You can maintain multiple EPwork directories as long as they are located in different directories. When you start up the EP Toolkit, the EPwork directory in the current directory will be used if present. You can switch to a different EPwork directory using the Change Work Directory command in the File menu of the Toolkit and you can create a new one by using the Create Work Directory command.



Since EPwork contains a working copy of the files, it may take up a large amount of disk space so make sure that the disk has sufficient space to accommodate them. You can use the Clear Work Directory command to clear it out. The EPwork directory and its preferences file will remain. If your main disk does not have enough space, it is possible to specify a directory on a different disk as being your default Matlab directory.

Restarting

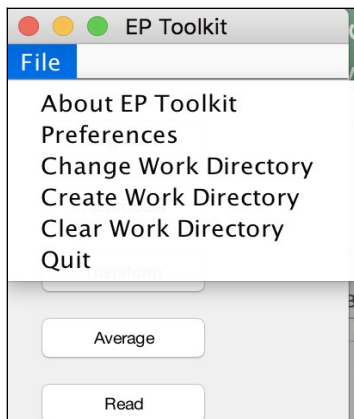
If the EP Toolkit is stopped in mid session (as by memory fragmentation), you can continue where you left off by restarting MATLAB (on a non-Mac, EPwork will need to be in the active directory when you type `ep` into the command line). If you have encountered some other type of error state (such as due to a bug in the Toolkit) it may be sufficient to just click on the status button. If all else fails, just restart the toolkit by using the `ep` command.

Quitting

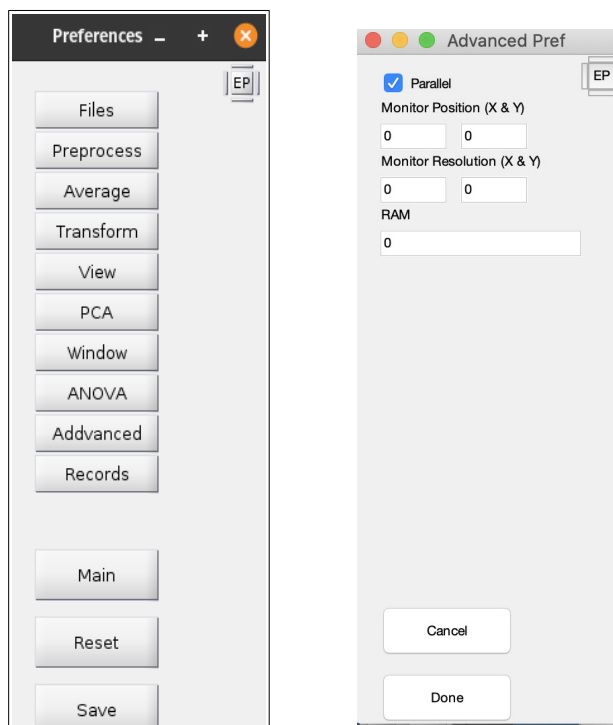
When you Quit out of the Toolkit, you will be asked if you wish to clear the working set. The EPwork directory and its preferences file will remain.

Monitor Preferences

Unfortunately, the information available in Matlab about multiple monitor configurations on the host computer is limited. In order to provide more control over where the EP Toolkit windows are appearing and in what space, there are preference settings to provide this information. First choose Preferences from the File Menu.



Next click the Advanced button.



The Parallel checkbox controls whether the Parallel Toolbox is used for certain functions. Usually doing so can speed up operations but sometimes it does not so it can be useful to be able to turn it off.

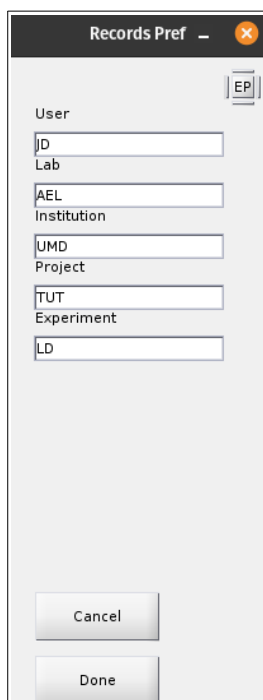
There are four settings for the monitor. The first two is the location of the lower left corner of the desired monitor area in pixels. The second two is the size of the monitor area. If, for example, you have two 1920x1080 monitors set up to be side-by-side areas in the host computer and you want the EP Toolkit to be showing up in the rightmost monitor, then set the position to be X=1921 and Y=1 and the Monitor Resolution to be 1920 and 1080 respectively. Or if you wanted to also avoid a 100 pixel tall Task Bar at the bottom of the screen, you would set the position to be 1921 and 101 and the

Resolution to be 1920 and 980. These settings will only affect the appearance of new windows. None of the existing EP Toolkit windows will be affected by changes to these settings.

Once finished, click the Done button. If you wish these changes to be temporary, click Main and they will only apply until you restart EP Toolkit. If you wish them to persist between sessions, click Save. If you have second thoughts, click Reset and everything (including other preference settings) will be reset back to Default settings.

Records Preferences

You will notice that there is a string of dashes at the top of the Main Menu. This is indicating that the records settings are currently blank. The records settings are intended to be included in the history records for the file (to be described) so that you can keep track of who made changes to the dataset and when for team projects. To set the records settings, again go to the Preferences but this time click on Records.

A screenshot of the 'Records Pref' dialog box. The title bar is dark gray with the text 'Records Pref' and a close button. Below the title bar is a small 'EP' logo. The main area contains several text input fields, each with a label to its left: 'User' (with sub-label 'ID'), 'Lab', 'AEL' (with sub-label 'Institution'), 'UMD' (with sub-label 'Project'), 'TUT' (with sub-label 'Experiment'), and 'LD'. At the bottom of the dialog are two buttons: 'Cancel' and 'Done'.

Enter in values for User, Lab, Institutions, Project, and Experiment. Keep the values short, perhaps no more than three characters, to make sure they will all fit at the top of the main menu. Other than for record keeping, these values have no function.

Toolkit Functions

Tutorial Example Dataset

For ERP analyses, the recommended order of operations is to first low-pass filter and mains correct, then segment, then apply artifact correction. Filtering should be conducted first because filtering algorithms do not handle the edges of recordings well; if you filter the raw data, then the resulting glitches will be in discarded data. Segmentation should then occur, followed by artifact correction. Artifact correction should occur afterwards because: 1) the ICA used to remove blinks benefits if the very noisy periods outside of the experiment blocks are first excluded (although one can also manually trim the files) and 2) the baseline correction that can then be applied to each epoch provides a lossless high-pass filter that further benefits the various artifact correction steps.

An exception is for ERP data collected inside an fMRI scanner as the gradient and the ballistocardiogram correction algorithms benefit from having intact data before and after each artifact. For such data, the fMRI correction option of Artifact Correction should be run first, followed by the low-pass filter, followed by segmentation, and then the rest of the artifact correction procedure.

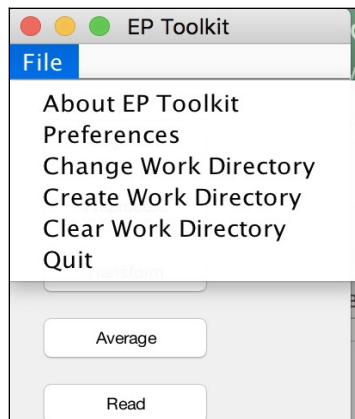
For traditional spectral analyses, segmentation should only be applied if the data are divided into epochs (as in time-frequency analyses). If the data are continuous in nature (as in over several minutes), then segmentation can be skipped entirely if each recording is one condition.

This tutorial will utilize the example dataset described earlier in the Example Dataset section.

Read

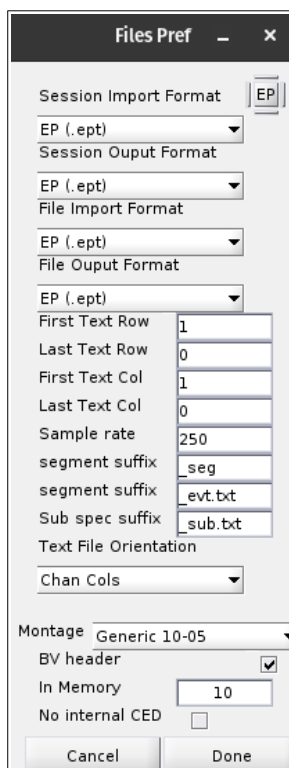
Setting Preferences

Before reading data, you might want to adjust the preferences. Choose Preferences from the File menu.



Then click on Files:

The following window should appear:



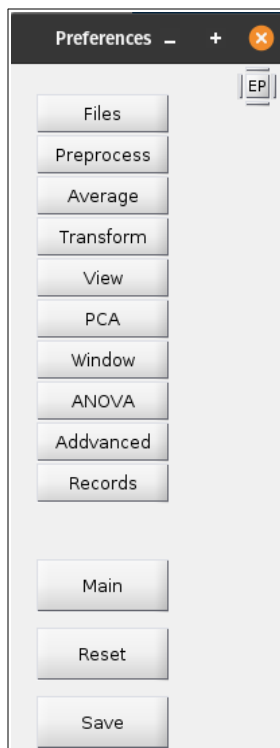
The Files preferences pane allows one to change the default settings for file operations. The first four change the default file type settings for the pop-up menus for read and write functions. The next four control whether any of the rows or columns should be ignored when reading in text data files. The Sample Rate field determines the default sampling rate when reading in data contained in text files. The seg file suffix is for the output of

the Segment function and can be changed if one has different types of segmented files that one wants to automatically differentiate with the suffix (e.g., prime_seg). Event file suffix specifies the suffix for event text files. Subject spec suffix specifies the suffix for subject text files. Text file orientation controls whether rows or columns of text data files are to be understood as being channels (as opposed to time points). The Montage popupmenu specifies the default montage for Montage controls. The BV header setting controls whether the Toolkit will look for EP header information encoded in BrainVision files.

The "in Memory" setting controls how many datasets can be cached in RAM to speed up operations by avoiding the need to read files from the disk. Any changes made to a dataset will be immediately saved to the working set copies kept on the hard drive so these cached copies will never hold any unsaved information and so can be freely cleared. The user will have to determine the ideal setting based on the amount of available RAM and the size of the datasets.

Finally, the "No internal CED" checkbox indicates that the EP Toolkit should ignore the electrode coordinates stored in mff or set files. Instead, the EP Toolkit will ask for a CED file.

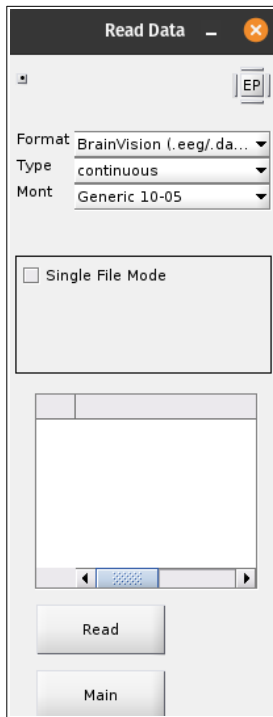
If any changes are made, then click Main, otherwise either Cancel or Main is fine.



Click Main to return to the Main Menu. Click Save to make any changes permanent (applying to future EP Toolkit sessions). Click Reset if you wish to return all the preference settings to the original default values.

Reading Data Files

First, click on Read button in the main menu. This function reads files into the EP Toolkit's working set for use with the Edit, View, PCA, Window, and Save functions. The working set is also used in a limited way for some other functions, as will be seen. The working set is a set of working copies of your files and so any changes you make will not be reflected in your permanent files until you Save your work to the disk.



To use, specify the file format. Make sure the file has the proper suffix (noted after the format name). If the file is not .ept format, it will also be necessary to indicate what kind of file it is (Type). The options are “continuous” (one long recording), “single_trial” (epoched but not averaged), and “average” (averaged epoched data). There are also legacy options for “grand average” (average of subject averages) and “factors.” It is possible to read in a batch of files by selecting multiple files in the dialog, once the Read button is pressed. You also need to specify the Montage (mostly to indicate whether the electrode montage follows the standard 10-05 system or an idiosyncratic one like the EGI systems). If it is an EGI dataset, specify the appropriate montage; otherwise, choose “Generic 10-05.”

If the data take the form of separate files for each subject, cell, trial, session, and/or frequency, then it will be necessary to click on the Single File Mode option and to specify

in the Subject and Cell fields which characters in the file names correspond to the subject and cell labels (as in 4:6 and 7:9 for APP001TAR.txt for subject #001 in the TAR condition). It is also possible to specify trial number if single-trial is chosen as File Type (the trial labels would need to be numbers, as in 001). Frequency can also be chosen if the data is frequency-domain and would also need to be numbers (e.g., 08). Note that it will be necessary for each file to have an appropriate suffix in order to be recognized (listed along with the file format name). By default, the channels will be the columns.

It is possible to construct a separate supplemental event file to contain the event information. By default, such a file would have the same root name as the data file but end in _evt.txt (e.g., test_evt.txt for test.bin). When any data file is read in, if a text event file is present in the same directory, then it will also be read in and its contents will replace the event information in the data file (if any). This functionality is especially helpful for file formats that do not accommodate event information (like text files) and for manually editing the event information. The file should be tab-delimited. The first row has header names. The first column contains the subject number and the second column contains the cell number for each event (for a continuous file, these will all be "1"). The Type, Sample, Value, and Duration numbers follow FieldTrip conventions, wherein Type is a generic category name (e.g., "trigger") and Value is the more specific event information (e.g., "prime"). Sample is which sample (of the epoch for segmented data) and duration is its length in samples. Following EGI mff conventions, it is also possible to have "keys" associated with each event that contain more information. Each key has four fields: code (the label), data (the actual information), datatype (the type of variable, mostly ignored), and description (explanatory comments, mostly ignored). In the event file, keys are named code1, data1, datatype1, and description1 and so forth (see next figure).

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
subject	cell	type	sample	value	duration	code1	data1	datatype1	description1	code2	data2	datatype2	description2	
1	1	1 Stimulus	51	51	1	stim	# Message: C	char						
2	1	1 eye-tracker	90.9215	saccadeET	1	SMlevent	1	short	word	colleagues.	char			
3	1	1 eye-tracker	151.923	saccadeET	1	SMlevent	1	short	word	(AP)	char			
4	1	1 eye-tracker	178.918	saccadeET	1	SMlevent	1	short	word	INDIANAPOL	char			
5	1	1 eye-tracker	51.1445	fixationET	1	SMlevent	1	short	word	colleagues.	char			
6	1	1 eye-tracker	110.92	fixationET	1	SMlevent	1	short	word	(AP)	char			
7	1	1 eye-tracker	161.9178	fixationET	1	SMlevent	1	short	word	INDIANAPOL	char			
8	1	1 eye-tracker	192.9183	fixationET	1	SMlevent	1	short	word	Indiana	char			
9	1	2 eye-tracker	30.9215	saccadeET	1	SMlevent	1	short	word	colleagues.	char			
10	1	2 eye-tracker	91.923	saccadeET	1	SMlevent	1	short	word	(AP)	char			
11	1	2 eye-tracker	118.918	saccadeET	1	SMlevent	1	short	word	INDIANAPOL	char			
12	1	2 eye-tracker	213.9142	saccadeET	1	SMlevent	1	short	word	Indiana	char			
13	1	2 eye-tracker	50.92	fixationET	1	SMlevent	1	short	word	(AP)	char			
14	1	2 eye-tracker	101.9178	fixationET	1	SMlevent	1	short	word	INDIANAPOL	char			
15	1	2 eye-tracker	132.9183	fixationET	1	SMlevent	1	short	word	Indiana	char			
16	1	2 eye-tracker	224.9235	fixationET	1	SMlevent	1	short	word	State	char			
17	1	3 eye-tracker	40.923	saccadeET	1	SMlevent	1	short	word	(AP)	char			
18	1	3 eye-tracker	67.918	saccadeET	1	SMlevent	1	short	word	INDIANAPOL	char			
19	1	3 eye-tracker	162.9142	saccadeET	1	SMlevent	1	short	word	Indiana	char			

There are also other supplementary files that can be read in. Subject spec info can be provided in text files with the suffix _sub.txt in much the same manner as the supplementary event files. Additionally, .edf files generated by EyeLink 1000 software are recognized if they have the same root name but end in ".edf" and will be bundled into the read file. Ditto .asf video files generated by BrainVision systems. Ditto .sfp electrode coordinate files, as in those generated by BrainVision CapTrak systems.

In any case, click on the Read button and then the AP5-005-LA1.eeg file in the tutorial folder. It will then ask for a CED file, displaying the contents of the electrodes directory in the EP Toolkit directory. Choose easycap65EOGbvr.ced, which has the standard 65-channel easycap locations plus four EOG channels, as appropriate for data collected using BrainVision Recorder, and then click Open. For your own data, you would need to construct your own CED file, using the included ones as examples. Keep an eye on the Matlab command window to monitor its progress. You should see something like:

Reading: AP5-005-LA1.eeg

Warning: It appears that the trigger bits setting on Pycorder/Recorder was set incorrectly (see tutorial).

Attempting to fix bad bit setting problem.

Read in 68 channels.

Read in 275600 points.

Read in 1 subject.

Loading the ced file:

/home/jdien/Dropbox/Matlab_Programs/EP_Toolkit/electrodes/easycap65EOGbvr.ced

Adding electrode coordinates from CED file easycap65EOGbvr.ced.

The number of extra channel names in the ced file easycap65EOGbvr.ced match the number of missing channels names.

Will make the assumption that they are the same channels and in the same order.

If this assumption is incorrect, you will need to add the missing channels to the CED file.

Will keep the channel names in the CED file and also add the missing reference channels.

The following ced file channels are not represented in the data file:

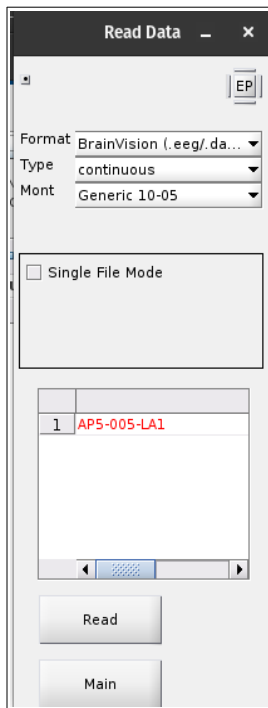
Fp1

The name of the experiment is:

The pre-stimulus period of the data is: 0 msec.

Process took 25.22 seconds.

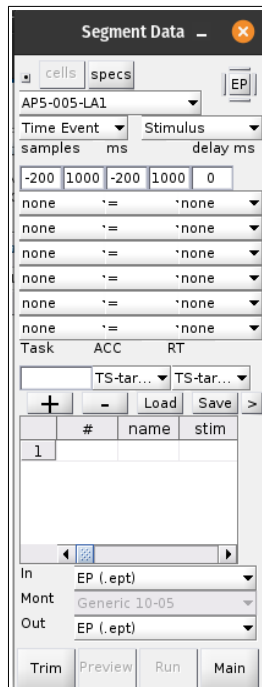
The files that you loaded in will now be listed in red on the EP Toolkit pane. A copy of them have been loaded into the working directory (EPwork) but will not be taking up space in the computer memory. If you need to purge any from the set of work files, click on them and they will be immediately removed (the red is a reminder not to click on them unless you really want to delete your working copy of the file). Also, a star will be displayed in front of every file that contains unsaved changes (changes following the initial file import but not including anything that occurred during the import itself).



Next click on Main to get back to the main menu.

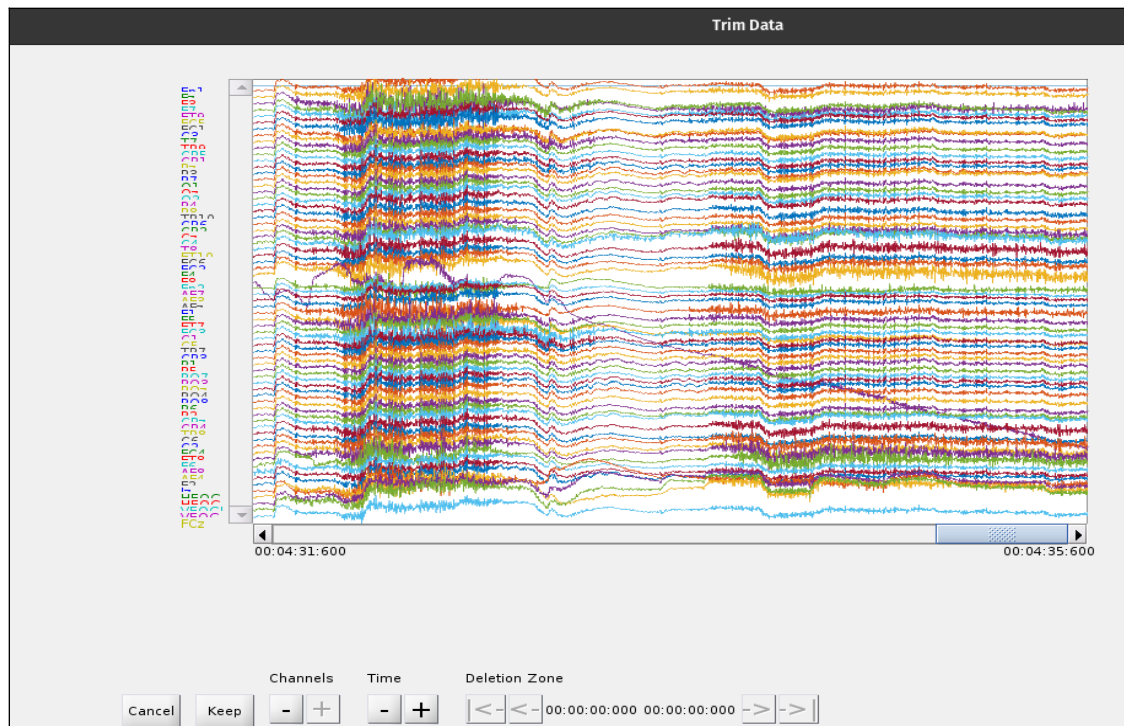
Trimming

The next step is to trim the data to remove the unwanted recordings. This step is usually not necessary, but could be helpful if there was a false start that needs to be removed or the plan is to run a frequency-domain analysis of the unsegmented data. Trim operates on data that has first been imported into the working set, which you have already done. Click on Segment to bring up the Segment pane.

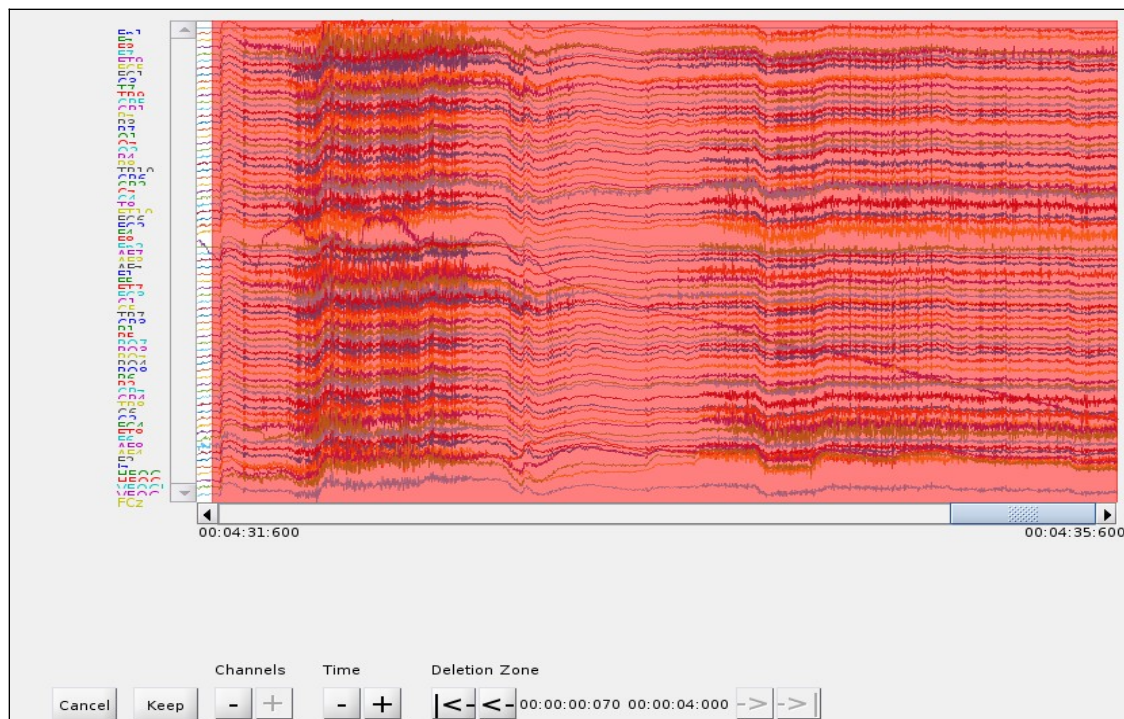


First select AP5-005-LA1 in the first menu (which lists every continuous data file in the working set). If you press the Trim button a new window will open up with each channel occupying a separate line to facilitate a high-level view of the recordings. By default, four seconds of data will be displayed at a time and for all the channels. The - and + buttons below allow one to zoom in and zoom out the channels and the time points (menemonic for the buttons is to think in terms of showing less vs. more channels and time points). Events are shown along the top and marked across the channels with a black line. Discontinuities in the recording (boundary events) are marked with a red line. Moving around the data should be pretty quick and smooth with a decent computer but Matlab sometimes starts bogging down and will need to be restarted.

At the end of the recording, the participant started moving and stretching, resulting in large movement artifacts that might degrade the quality of the artifact correction process. One might therefore decide to delete it if the plan were not to perform segmentation first (in which case this part of the recording would be discarded anyway).



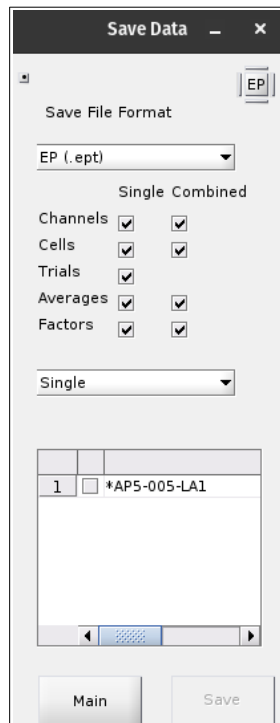
You can delete everything in the window at the end of the recording. Drag the mouse across the entire window. This will highlight this region in red. Then right-click within the red deletion zone and select the word “delete” to delete this portion of the recording.



Next zoom in with the - button (less time points) so you can see both the start of the trimmed recording and the red boundary marker. Drag the mouse from off the left side of the screen up to the red boundary marker. This portion of the recording will be highlighted in red. Then click on the ->| button to extend the right edge of the deletion zone to extend to the sample just prior to the next event, which in this case is the boundary event. Normally this means that the event would not be part of the deletion zone but boundary events are deleted if they end up at the edge of a recording. The -> button would extend the right edge by a single sample. Then right-click within the red deletion zone and select the word “delete” to delete this portion of the recording. Once done, click on the “keep” button to keep all the changes to the recording and click “cancel” to discard them. Then click on Main to return to the main menu.

Save

The next step is to save the trimmed file to the disk. Changes made to the copy in the working set do not affect the original files. The Save function at the bottom of the main menu allows files in the working set to be saved permanently. Just specify the file format and which types of data are to be saved. By default, all adds to the data are saved (that is, all the ERP data plus all the combined channels, cells, subjects, and factors), although some adds may require separate files. Also, if one saves a factor file in EGIS average format then it will be necessary to save a separate file for each subject, in addition to the Grand Average Data. If not needed and one wished to avoid the clutter of all the subject average files, one could just deselect "Averaged Data." It would also be advisable to save a copy of the file in EP format as it will retain information that might otherwise be lost in translation to other file formats. When ready, just click on the name of the dataset to be saved. The asterisk in front of the file name indicates it has unsaved changes. Indicate that the new file should be named: as AP5-005-LA1_trim.ept. It will ask you if you wish to change the name of the copy in the working set, to which you should say Yes. If you have a number of files in the active set that you want to save and you don't want to click on each individually, then you can select the checkboxes and then click the Save button.



If you wish to convert a batch of files directly from one format to another without first loading them in, then change the pop-up menu from "single" to "convert" and it will allow you to select a batch of files on disk and convert them for you without loading them into the working set. When converting from continuous BrainVision format files, there is an option to fix damaged EP headers. When converting from continuous Simple Binary format files, there is an option to also read in events .txt files as a workaround for problems reading EEG files from NetStation 4 where exporting in .mff format is not an option and other formats lose events information. If one chooses BrainVision format and continuous file type, a "Fix damaged EP header" option is unlocked that allows one to fix cases where the RAs started Recorder after they started the new block, resulting in part of the EP header be missing. If checked, the EP Toolkit will allow one to select another file with an intact comparable file header (e.g., AP5-005-LA2) to replace the damaged one. The trial specific information is not affected by this procedure.



Exporting to EEGLAB

One may often wish to save into the EEGLAB format. The .set file format is an option but incompatibilities between the way the two file formats are organized makes it difficult to provide full compatibility. For example, factor results cannot currently be saved to .set file format and likely never can (The EEGLAB .set file format is predicated on the approach of keeping all the single trial data and then generating the averaged data on the fly. Factor matrices are applied to the single trial data and are stored alongside the raw single trial data. There doesn't appear to be a way of having the factor matrices be applied to the averaged data.). If the .set file format is not working for you, then you may wish to try one of the EGI file formats.

EEGLAB calls the EGI simple binary format "EGI .RAW". However, it would be better to use the option "From other formats using FILE-IO" as it will then use the FieldTrip code (which makes an effort to deduce what the baseline period should be since this information is missing from this file format). You find the options under "File> Import Data". You can also import EGI EGIS format files using "File>Import Data>From other formats using FILE-IO". Remember that while this file format keeps track of the baseline period information it does not keep track of events so that information will be missing.

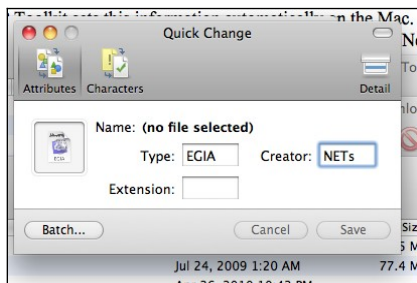
Exporting to Net Station

If you wish to export to NetStation 4, use either EGIS or Simple Binary formats. I recommend using EGIS unless the file is very large (in which case the file format may not be able to accommodate the data and will generate an error message when you try to

save it). The major reason for using EGIS average files instead of Simple Binary is that the latter does not represent the prestimulus period information. EGIS session files don't represent prestimulus period information either. The reason why this is important is that the lack of prestimulus information causes problems with a number of NetStation waveform tools, especially the Combine Files function. If you are using NetStation 5 then you will have to use Simple Binary or MFF despite their limitations.

In order to properly display the contents of an EGIS file, NetStation 4 needs the electrode formats. The montage information from NetStation 4.3 differ from prior versions of NetStation. Since only version 4.3 includes Hydrocel net montages, this means that versions of NetStation prior to 4.3 cannot view such data sets. Also, depending on the version you have, NetStation may not give you the full list of possible electrode nets. The EP Toolkit used to be able to automatically add this information but they no longer work under current versions of OS X. I don't know of any other way to get the necessary montage information into the files.

Regardless of the NetStation version, EGIS and simple binary files will need to have their file type and creator set. You will need to use a utility like Quick Change and use it to change the file type to "EGIA" and the Creator to "NETs" without the quotation marks. If the file were subsequently moved to such a computer then NetStation will not recognize the files and this procedure will need to be repeated. A good program for doing so is Quick Change (<http://www.everydaysoftware.net/quickchange/index.html>). Changing the Creator field to NETs allows the Mac to know that the file belongs to NetStation. Changing the Type field to EGIS for .egis files and eGLY for single_trial and average (segmented) .sbin files and UGLY for continuous (unsegmented) .sbin files allows NetStation to know which type of file it is. Just type in the fields and then drop the file on the box on the leftmost side or onto the program icon itself (including a Dock icon).



Note that NetStation (as of 4.1.2) does not seem to read in event data at all from such files. If you need the event information, you should use the Mark-up tool to add the events back in once the files are converted back to NetStation format.

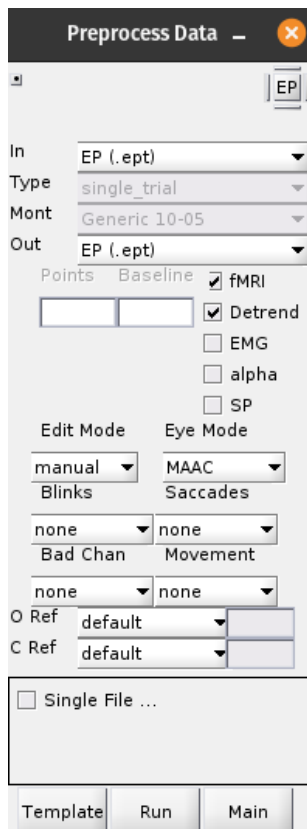
For NetStation 5, you pretty much need to use .mff files. Unfortunately, the only available code for writing to mff format is very limited (there has to be an already

existing mff file with the same structure as the data you wish to write). This functionality seemed not worth the effort to support so the EP Toolkit does not write out to mff at all.

Click on Main to return to the main menu.

Removing fMRI Artifacts

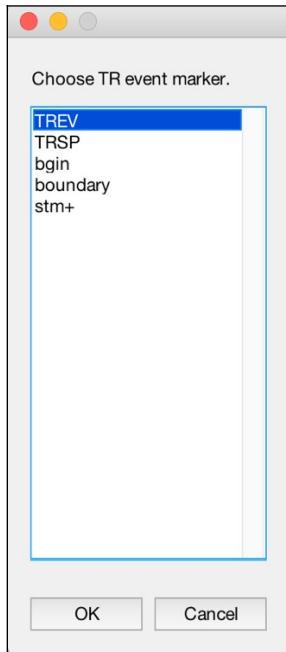
There is great interest in collecting EEG data during fMRI scans, although it produces artifacts which can reach tens of thousands of microvolts. While not part of the tutorial dataset, this section will describe how such artifact correction can be accomplished. Make sure you installed the MRI correction routines as described in the installation section. Click on the Preprocess button on the main menu. Click on the fMRI option to invoke the MRI artifact correction routines (further explained in the Preprocessing section) and Detrend to correct for overall drift. Turn off all the other options (which will be explained in the Preprocessing section). The controls should now look like the following.



Note the small button in the upper left corner. This is a shortcut to the preferences section for the preprocessing function. If you want to check the settings or make a quick change, just click on it. Clicking on either the Cancel or the Done button will take you straight back. Note that any changes you make will only apply to the current session. If

you wish changes to persist, then you should use the Preferences menu to Save the settings, as previously described.

Then click on Run and select the file. If checked and the fMRIb function is specified in the preferences, the EP Toolkit will ask for the TR event marker as it needs to know when each rep began in order to correct gradient artifacts (e.g., TREV).

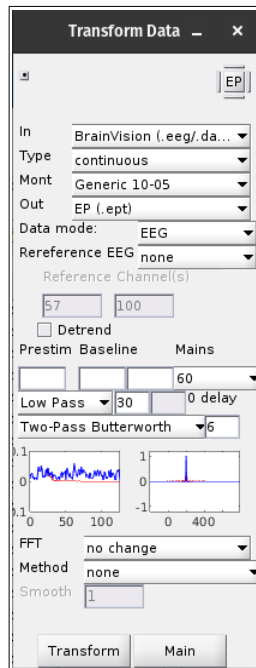


After this point the preprocessing will proceed without further input. It will need an ECG channel in order to correct ballistocardiogram artifacts. It will add 'qrs' events to each point that it detects a heartbeat. Watch Matlab's command window to monitor its progress. Ignore the colorful Artifact Correction figure that pops up for now. When it is finished, the command window will show "Done" and the buttons at the bottom of the Preprocessing pane will stop being grayed out. A new file with the suffix `_bcg.ept` will contain the corrected data. Click on Main to return to the main menu.

Transform

The next step is to low-pass filter the data, because you want to do this prior to segmentation to avoid having edge artifacts at the boundaries of the segments. Click on Transform. The Transform function is used to rereference, baseline correct, filter, and/or convert to spectral measures.

My suggestion is to use a two-pass filter (which preserves the peak latency) for conventional analyses and to use a one-pass filter (which preserves the onset latency) for sample-by-sample analyses. In any case, set up the controls as follows.



The first setting is the file format, which in this case will be .ept (the EP Toolkit's native file format). Since this file format specifies the type of data, the next two settings are not necessary and are grayed out. The fourth setting is the file format of the resulting file and should also be .ept. In general, it is recommended to use the .ept format while using the the EP Toolkit as other file formats will result in loss of EP Toolkit-specific information.

The Data mode setting indicates what type of channels should be transformed. For example, if ECG was selected then filtering would be applied to the ECG channel instead of the EEG channels.

The next control rereferences the data. If reference is set to Traditional then it will be referenced to the channels listed below it. If it is set to Average then those fields will be ignored and it will instead be referenced to the average of the entire dataset (Dien, 1998b). Alternatively, the data can be transformed to CSD (current source density) form (Perrin et al., 1989), which is essentially a high-pass spatial filter which eliminates broadly distributed voltage fields in favor of sharply defined focal features. It has the benefit of being reference-independent and of allowing for a crude source localization but runs the risk of eliminating ERP components of interest that happen to be broadly distributed. Finally, it can be average-referenced using the PARE-correction, which corrects for the lack of electrode coverage on the underside of the head (Junghöfer et al., 1999). For EGI high-density montages (which are designed to maximize head coverage) I'm not seeing a lot of difference from average reference and the EP Toolkit's implementation is quite slow and can take hours. You can use the Topos Rereference right-click option to see kind of effect it has on your data. The PARE correction is influenced by the reference in effect prior to the correction so the EP Toolkit always

applies an average reference first. We will not rereference the data yet as doing so would risk spreading the effects of bad channels to other channels.

Detrend, as for the Preprocessing function, corrects for overall drift across the dataset (or epoch for segmented data) by fitting and removing a linear function to the channels. It is generally not recommended except for continuous data as applying it to segmented ERP data can result in attenuation of ERP components.

The prestimulus period is length of the epoch prior to the onset of the event to which the epoch was locked. The Prestim field allows one to label segmented data with the appropriate prestimulus time but has no other effect. There is no need for this setting until the data has been segmented. Note that some file formats, such as EGIS session, cannot represent the prestimulus period and so this information will be lost if they are used as the output format.

The Baseline period (often the same as the prestimulus period) is the part of the epoch that is used to estimate zero voltage. It need not be the same as the prestimulus period. The two fields specify the baseline period in ms (i.e., "-100" and "0" means the first 100 ms). We will return to this procedure after segmentation.

Mains is a specialized notch filter that removes much of the electromagnetic frequency noise from unshielded electrical equipment. To do so, it uses the CleanLine function that is part of the PREP Pipeline package (Bigdely-Shamlo et al., 2015). You can set the default Hz in the Preferences. It is best to use it now, prior to segmentation.

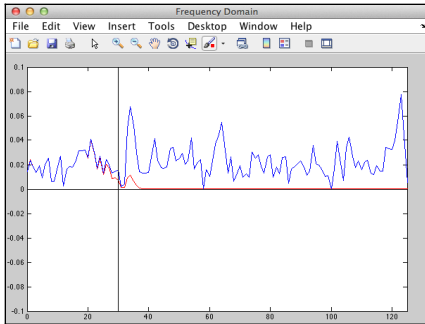
The Delay field is not a control. It indicates how much of a delay (in ms) is being caused by a given filter setting so that it can be corrected for when segmenting the data.

The next setting specifies the filter range. Low Pass means passing through frequencies below the setting. High pass means passing through frequencies above the setting. Band pass means passing through frequencies within a specified range. Band stop means stopping frequencies within the specified range. Notch means stopping only the specified frequency. The filter settings will be grayed out until a number is entered into the first field to denote that filtering is disabled.

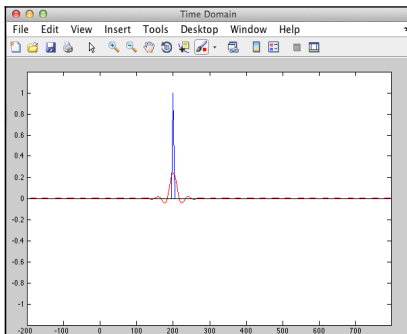
The next row sets the filtering parameters (for an excellent discussion, see Luck, 2014). The first aspect is "one-pass" vs. "two-pass" (also known as "causal" vs. "non-causal"). A one-pass filter starts from the front and goes to the back, with the result that the EEG activity can be smeared a bit towards the back, potentially artifactually delaying the latency of ERP components (the extent of this delay for the sample figure is indicated in the Delay field). A two-pass filter adds a second pass, starting from the back and proceeding to the front. This has the effect of undoing any changes to latency but at the cost of smearing effects towards the future, potentially artifactually advancing the onset latency of ERP components. The second pass also increases the strength of the filtering.

The field following this menu specifies the order of the filtering, which is to say the strength of the filtering.

In general there is a trade-off between accuracy of filtering in the frequency-domain and in the temporal-domain. This can be seen in the two figures on the next row. The first generates some random data in blue and then illustrates the effect of the filter in red. The stronger the order, the steeper the roll-off (the shorter the transition zone between unfiltered and filtered) and the more complete the filtering, but the more smearing occurs in the temporal domain.



This effect on the temporal domain can be seen in the figure in the rightmost figure, which provides a spike in blue and shows in red what the data looks like after filtering. Both figures can be expanded by clicking on them. The frequency domain figure usually shows a vertical line indicating the location of the nominal filter settings. If the filter lines don't appear, close and open the expanded figure a few times.



Butterworth, FIR, and FIRLS are three filter types made available by Matlab. In general the Butterworth is the recommended filtering algorithm. Try playing the settings and see what effect they have on the figures.

Finally, one can convert the data into the frequency domain using FFT (fast Fourier transform). "no change" means staying with voltage measurements over time. The "Frequency" and "Time-Frequency" are for performing frequency domain analyses and will be explained in the Spectral Analysis section.

This step is also performed upon files on the disk rather than on the working set (which is why the preparatory functions of Segment, Preprocess, Transform, and Average, are listed above Read on the Main Menu).

For the tutorial, click on the Transform button and select the all the .eeg files (clicking on the first and last desired entries with the shift key held down to select all the ones in between and control-clicking to select/deselect any further entries). It can be helpful to click on the file requestor options to have the files listed as detailed text entries (rather than icons) and to have them sorted by file size (as the .eeg files are all much larger than the other files). It will also ask you again for the CED file to apply to the entire batch, which will be easycap65EOGbvr.ced. It will generate new files with the ending _mf.ept and place them in the same directory as the input files. This step does not use the working set in any way. Moving files into and out of the working set would simply slow things down. Once the files are output in .ept format, it will no longer be necessary to specify a CED file in future steps as the information is incorporated into the .ept file format. The suffixes are “b” for baseline corrected, “c” for CSD, “f” for filtered, “m” for mains corrected, “r” for rereferenced, “s” for spectral frequency transform, and “w” for wavelet time-frequency transform (even if the wavelet algorithm wasn't used for the time-frequency transform).

Segmentation

First, switch to the Read pane to read in one of the new _mf files by selecting the ept format, and then clicking on the Read button to select the file (AP5-001-LA1_mf.ept). It will take substantial time to read in since it is a large file. The reason this data needed to be loaded into the working set is that in order to segment it helps to have a representative continuous data file to serve as the template (to provide the types of events present in the experiment). Once at least one set of continuous data has been read into the working set, the Segment button will no longer be grayed out. Switch to the main menu by clicking on the Main button and then click on the Segment button to reveal the Segment Data function.

The top popup menu is the list of continuous and single-trial files in the working set. Select the dataset that will serve as the template for the segmenting specifications (e.g., AP5-001-LA1_mf.ept). The top portion helps make it easier to construct the segmentation table in the lower half but can be bypassed directly in favor of typing the criteria directly into the segmentation table.

Each row of the segmentation table will define a condition. There is no limit to the number of conditions that may be defined. Each condition may have up to six criteria stipulated to define the condition. The criteria on a given row have an AND relationship to each other, so for a potential segment to be included in the condition it must pass all the stipulated criteria. In order to define criteria with an OR relationship, provide more than one row where the condition name is the same. A segment that meets either set of stipulations will be included in the condition.

In the upper half of the pane, the “Time event” menu will then be populated with the event names in the template file. Only the contents of the Type field (the general event label) will be listed. Then select the event type that will serve as the time-locking event, which in this case will be “Stimulus”. “New Segment” is the event used to mark the beginning of the data recording.

Then set the epoch period. In this case, enter in -200 samples in the first box and 1000 samples in the second box to specify a 1200 ms epoch starting 200 ms prior to the event. If prior filtering resulted in a delay (as due to the use of a one-way low-pass Butterworth filter or EGI GSN300 and GSN400 amplifiers) then enter the delay into the "delay ms" field so that the epochs are offset accordingly.

The following six lines of popupmenus allow one to define the nature of the segments to be generated. First, one must indicate which events will be used to define the segments. In this case we wish to analyze the brain response to the target words appearing on the screen (200 ms prior and the 1000 ms after).

In the first row, change the first popup menu to “value”. The second column provides a number of options for how the spec value might relate to the criterion value. Comparisons are evaluated numerically for numbers, otherwise alphabetically. For now, leave it at “=”. The third popupmenu has been populated with the contents of the “value” field, which in this example dataset consists of “S 1” (presentation of the target words) and “S 4” (button press). Set the popupmenu to “S 1”.

The popup menus will also contain the fields associated with Trial Specs (TRSP) events, if present. Trial Specs are special event structures that are used by some data collection software, including BrainVision with the EP header, that provides trial-specific information like reaction time. They are generated at the end of each trial and are associated with an epoch if the epoch includes the time point when the Trial Spec was generated. The criterion menu includes the fields included in any Trial Spec present in the data file but will only be considered for segmentation if a Trial Spec event fell within the defined epoch. If multiple Trial Specs are present in an epoch, only the first will be used. Fields derived from the Trial Specs will be prefixed with a “TS-“ to differentiate them from those derived from the time-lock event. If there is a trial-spec named EPoffset, the epoch will be delayed by that number in ms (negative for earlier), allowing for trial-level control of segmentation offsets.

Now that one has specified which events will be used for the segmentation, one now needs to specify how one will tell apart the different types of segments to be analyzed. In the second row, set the first popup menu to "TS-cond". Select LANF (LANF is a filler cell, which will not be analyzed, but will contribute to the signal detection statistic). These are all the criteria needed for this example dataset.

Below the criteria popup menus are some other helpful information. Enter the name of the task in the Task field with a + if this cell is one of the target cells (words in this case) and a - if it is one of the non-target cells (non-words in this case) or else leave blank. This information is used during averaging to generate signal detection statistics. In this case, we will type LD (for lexical decision) and a - since the N in LANF indicates they were non-words. The Toolkit should have already automatically determined that TS-ACC is the trial specific with the accuracy information and TS-RT is the trial specific with the reaction time information, so you don't need to change these popup menus.

Then click on the + button. This will take the settings you have chosen and insert it into the segmentation table. Change the criterion value to LANU and click on + again to add the second condition. Go through all the possible values of TS-cond. You should end up with six rows in the segmentation table. Make sure to change the Task field to LD+ for the word conditions (LAWF, LAWR, LAWU). If you forgot to change it, you can make corrections directly in the segmentation table. If you'd like more space to see the segmentation settings, click on the small arrow button just above the table, next to the "Save" button.

You should then enter the names of the conditions into the second column, which in this case is the same as the cell labels (e.g., LANF). It is best if the names of the conditions are in alphabetical order. Also, the EP Toolkit works especially well when the cell names consist of equal length letter strings where each position codes for a level of an ANOVA factor with a unique letter. You can then edit the contents of the segmentation table as desired. For example, if the criterion value you wanted to choose wasn't in the menu, you can change it at this time. Also, for the ACC and RT columns, if you enter in numbers then the resulting segments will bear those values rather than obtaining them from a specified field. For ACC, 0=incorrect, 1=correct, 2=timeout (these codes can be changed in Preferences). For RT, numbers are the reaction time in ms. If left blank, then these two fields will just be ignored.

Segment Data

cells specs

AP5-001-LA1_mf

Time Event Stimulus

samples ms delay ms

-200 1000 -200 1000 0

value 'S 1

TS-cond 'LAWU

none 'none

none 'none

none 'none

none 'none

Task ACC RT

LD+ TS-ACC TS-RT

+ - Load Save

#	name	stim	prest...	post...	delay	task	ACC	RT	spec1	rel1	value1	spec2	rel2	value2	spec3	rel3
1	LANF	Stimu...	-200	1000	0	LD-	TS-ACC	TS-RT	value	=	S 1	TS-cond	=	LANF	none	=
2	LANR	Stimu...	-200	1000	0	LD-	TS-ACC	TS-RT	value	=	S 1	TS-cond	=	LANR	none	=
3	LANU	Stimu...	-200	1000	0	LD-	TS-ACC	TS-RT	value	=	S 1	TS-cond	=	LANU	none	=
4	LAWF	Stimu...	-200	1000	0	LD+	TS-ACC	TS-RT	value	=	S 1	TS-cond	=	LAWF	none	=
5	LAWR	Stimu...	-200	1000	0	LD+	TS-ACC	TS-RT	value	=	S 1	TS-cond	=	LAWR	none	=

In EP (.ept)

Mont Generic 10-05

Out EP (.ept)

Trim Preview Run Main

There is also a flexible segment option available for the case where one wishes to average together segments with different lengths (most likely TFT data). In this case one clicks on the "time event" popupmenu to change it to "flex event". As before one sets the event that the segments are locked to but now instead of specifying a set duration one specifies the event that ends the segment in the row below. In the line below one sets how many samples will be in the flexible segments. All the segments will be interpolated to yield segments the same length regardless of the original length. The default setting of 20 results in samples representing 5% increments of the original segment length. This procedure is likely meaningless in the time domain (except possibly for slow potentials) but can be useful for TFT data.

To delete the last condition in the segmentation table, click on the “-“ button. You may save the segmentation for later use by clicking on Save. To access this saved segmentation table, click on Load.

To verify that the criteria have been correctly specified, click on the Preview button. It will determine how many segments would be generated if the segmentation table was applied to the template file and provide the number in the # column. The button will not be enabled till all the conditions have a name (the second column of the segmentation table). Remember - only the contents of the segmentation table affect the actual segmentation. All the controls above the table are just for facilitating the construction of the table. One could actually bypass these controls entirely and just construct the segmentation manually by typing directly into the table. In any case, at this point, the function should look like:

Segment Data

cells specs

AP5-001-LA1_mf

Time Event Stimulus

samples ms delay ms

-200 1000 -200 1000 0

value '= 'S 1

TS-cond '= 'LAWU

none '= 'none

none '= 'none

none '= 'none

none '= 'none

Task ACC RT

LD+ TS-ACC TS-RT

+ - Load Save <

#	name	stim	prest...	post...	delay	task	ACC	RT	spec1	rel1	value1	spec2	rel2	value2	spec3	rel3
1	11 LANF	Stimu...	-200	1000	0	LD-	TS-ACC	TS-RT	value	=	S 1	TS-cond	=	LANF	none	=
2	18 LANR	Stimu...	-200	1000	0	LD-	TS-ACC	TS-RT	value	=	S 1	TS-cond	=	LANR	none	=
3	10 LANU	Stimu...	-200	1000	0	LD-	TS-ACC	TS-RT	value	=	S 1	TS-cond	=	LANU	none	=
4	8 LAWF	Stimu...	-200	1000	0	LD+	TS-ACC	TS-RT	value	=	S 1	TS-cond	=	LAWF	none	=
5	13 LAWR	Stimu...	-200	1000	0	LD+	TS-ACC	TS-RT	value	=	S 1	TS-cond	=	LAWR	none	=

In EP (.ept)

Mont Generic 10-05

Out EP (.ept)

Trim Preview Run Main

The first column criteria columns also provide a “-follows-“ option. If this is chosen, it means that a time-lock event should only be used to generate a segment if it follows (at any distance in time and with any number of intervening events) another event. When “-follows-“ is chosen, the third column menu is populated with the list of events present in the dataset. Once one chooses the event to follow, the next row down will be populated to contain the fields for that event (but no Trial Spec fields). Thus, one might specify that only a response event (i.e., “S 4”) that follows a stimulus event (i.e., “S 1”) event from LAWR would meet these criteria (i.e., -follows- = S 1; TS-cond = LAWR). This following row will always be dedicated to the -follows- event, even if one leaves it blank. One may also specify a time window within which the followed event must appear. To do so, set the second row after the -follows- to -rangesamps-. For example, to specify that the event to be followed must be one second or less prior to the time-lock event, set it to <= 250 (if sample rate was 250 Hz). This second row after -follows- is not dedicated to the -follows- criterion and could be any other type of criteria instead.

Or one could specify that a resp time-lock event immediately preceded by any another other event would meet criteria (i.e., -follows- ~= S 1; none = none). One could also string together two “-follows-“ criteria. In this case, the first “-follows-“ acts as before. The second “-follows-“ criterion specifies an event that is interposed between the “-follows-“ event and the time-locked event. Thus, one might specify the criteria that the response event must follow first a stimulus event of LAWU and then a stimulus event of LAWR (i.e., -follows- = S 1; TS-cond = LAWU; -follows- = S 1; TS-cond = LAWR). Only two such “-follows-“ events may be specified. It is possible to specify unrelated criterions between the two “-follows-“ criteria. It will not affect the interaction between the two “-follows-“ criteria. A “-follows-“ criteria cannot be used in the sixth slot because there is not a further slot to complete the specification. A corresponding “-

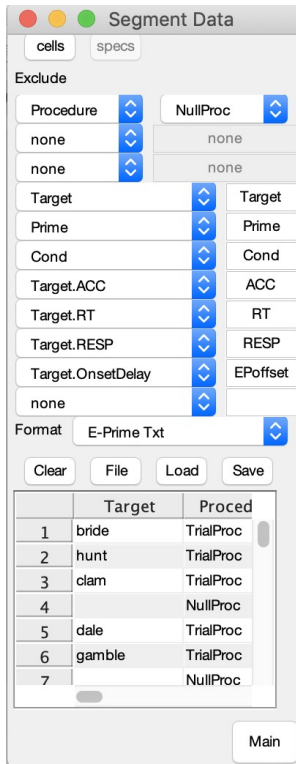
precedes-" option is also available. It operates in the same manner as -follows- but refers to events happening after the time-lock event.

Another pair of special options in the first column criteria popup menus are -responseCorr- and -responseErr-. They are for datasets where responses are marked with an event and where correct and incorrect responses have different labels. They work like a specialized -precedes- option where the resulting segment is marked as being a Correct or Error trial respectively.

If everything is ready, save the segmentation specs by click on Save. Set the file format of the data files to be read in and the file format to save the resulting single-trial data files and then press Run. It will ask you to indicate which files (on the disk drive, not in the working set) to segment. If you click on "modification date" in the file requestor, the newest ones (e.g., _mf.ept) will be grouped together, facilitating selecting them all. The copy of the data loaded into the working set will not be affected by the segmentation process. The Segment function will generate new files with a _seg suffix (e.g., AP5-001-LA1_mf_seg.ept).

It is also possible to apply the segmentation function to single-trial data that has already been segmented. In this case the existing segments will be reassigned to new conditions if they meet the stipulations or otherwise dropped. This can be a useful approach if, for example, one has a complicated dataset that can be potentially segmented in different ways, one wishes to run segmentation prior to Preprocessing (as recommended since baseline correction enhances the process), and one does not wish to rerun the Preprocessing function after each and every use of the Segment function.

For data with behavioral data (reaction time and accuracy), one should make sure to retain the error and the bad RT trials during segmentation as they will be handled during averaging.



Although not relevant for this example data, the segmentation function is also capable of making use of information from E-Prime .txt files (the ones that are automatically saved during a session and can be used to generate edat files using the E-Recovery program). Click on the “specs” button to switch to the specs subpane. From here, one can load in the E-Prime .txt file using the File button and it will be decoded.

This example file will be used to populate the menus so you can set up the settings, just like on the Cells tab. Also, this example file will be used for the Preview mode. If you switch back to the Cells tab and change the dataset, this specs file will be cleared.

The top two rows specify what trials, if any, should be excluded (as in practice or null trials). The remaining rows specify which fields should be added to the resulting segmented file. For each one, the text field allows one to specify a more concise name for the new trial spec. Any spec fields chosen in this manner will also be added to the segmentation menus for convenience. These settings can then be saved and loaded for later use. The clear button resets everything.

Once the spec settings are finalized, they will be automatically used during segmentation if E-Prime .txt files are present that have the same name as the EEG files (other than the file suffix). The number of segments will need to match the number of trials in the E-Prime .txt file (not including excluded trials). Note that this system is primarily intended for datasets where a single code (e.g., S 1) is used to denote the segmentation event and the cells are then distinguished based on the trial specs.

The trial specs “RT”, “ACC”, and “EPoffset” are reserved labels that have special meaning to the EP Toolkit. They refer to reaction time, accuracy, and timing error of the event. The former two will be used for behavioral information during averaging and plotting (these two spec names can be changed in Preferences). The latter will be used, if available, to adjust the segmentation. This information can be obtained from the .onsetdelay field for the stimulus event.

Press Main to return to the starting menu when finished.

Artifact Correction

The next step is to perform artifact correction. The Toolkit can automatically eliminate blink, saccade, movement, and alpha artifacts from session files as well as detecting and correcting bad data. It does so via the Multiple Algorithm Artifact Correction (MAAC) procedure, which applies an algorithm that is especially well-suited to the characteristics of each of the artifacts (Dien, 2024).

It is recommended for the data to have already have been segmented into discrete trials if it is meant for ERP analysis as using the full continuous data takes more time and disk space than is needed. Also, my observation is that applying the artifact correction (especially the ICA-based eyeblink and saccade corrections) work best on baseline-corrected segments as it provides a loss-less high-pass filter.

Chunking

The artifact correction function reads in a session file and segments it into manageable chunks which are temporarily stored (since an ICA of an entire session file can often exceed the available RAM). Since the final chunk (the remainder) would often be too small for an effective ICA run, it is appended to the previous chunk and the total chunk is divided into two chunks that are smaller than the others but large enough to process. The chunking is not allowed to divide any trials in two. Then it processes each chunk individually and stores the result. Then it glues them back together into a new session file with a “_e” appended.

Note: If the data are large enough to be chunked and you have more than 2GB of RAM, you should go to the preferences menu and see if you can increase the chunking size without running into out of memory errors. About 100,000 per GB of available RAM seems to generally work well. Basically, if the preprocessing routine is chunking your data, try upping the preference setting and see if it'll still run. Unfortunately Matlab does not provide a way to adjust this setting automatically.

Default Procedures

Preprocessing parameters can be changed in the Preprocessing Preferences, but the default procedures of the Toolkit are as follows:

If the data was collected inside an MRI scanner, first run the data through Preprocessing as a continuous file with just the fMRI correction option chosen. An electrocardiogram channel (with the channel type designated as ECG) will be required to detect heart beats for ballistocardiogram artifact correction. The fMRI correction is invoked simply by checking the fMRI box. The option is determined by the Preference setting (see #19 below). Both options provide both MRI pulse sequence and ballistocardiogram artifact corrections. Then, whether or not fMRI correction was performed, either segment the data and run it through Preprocessing as single-trial data or just run it through as continuous data, depending on analysis needs. All steps are independent of referencing scheme. The full Multiple Algorithm Artifact Correction or MAAC procedure is presented elsewhere (Dien, 2024).

- 1) *fMRI noise correction* - always takes place first since the noise is so large it will swamp anything else in the recordings.
- 2) *Correct for baseline shifts* - The second step is to prepare the data by controlling for the wild DC shifts sometimes present. Baseline correction and/or detrending can address much of them. Detrending fits a linear trend line to a waveform that is then subtracted from it. For segmented ERP data only baseline correction should be used as detrending can attenuate ERP components. Detrending can be useful for continuous data.
- 3) *Bad channel identification* - The first step is to check for any channels whose best absolute correlation with neighboring channels falls below .4 (across all the timepoints, irrespective of bad trials). In order to detect channels that went bad partway through the session, it will divide the session into five sections (if the badTrials setting is at the default of .20) and test for bad channels in each. If a channel is bad in at least one such section, it is considered to be globally bad and is excluded from further analyses. Furthermore, if a channel has more than 20% (if the badTrials setting is at the default of .20) timepoints over +/-500 microvolts from the median in a section, it will also be marked bad.

If the session has more than ten percent bad channels then it will be considered a bad subject and it will not be further processed. A warning will be provided if any of the globally bad channels are neighbors. Additionally, flat channels are identified as being bad channels (unless it was identified as being the reference channel). Additionally, channels that correlate perfectly with a reference channel will be marked bad (this happens when a file with a flat bad channel is rereferenced). Channels that are shorted together (perfectly correlated) will generate a warning in the log but will not be touched. Points that are deemed to be bad (due to exceeding the saturation values, which are +/- 1000 by default) are set to NaN as such extreme outliers can otherwise distort the results. The Edit Mode option controls whether this automatic routine is performed.

- 4) *Saccade correction* - For the MAAC, the next step is to correct two artifacts associated with saccades: saccadic spike potentials or SPs (Blinn, 1955) and/or corneo-retinal dipoles or CRDs (Lins et al., 1993). SPs are produced each time the eyes initiate a

saccade and are problematic because they resemble a P300 and they also mimic gamma band activity in spectral analyses (Yuval-Greenberg et al., 2008). First a blink detection is run (if the blink correction option has been chosen) to identify putative blink time points so they can be excluded from the SP correction process. To further complicate things, SPs can distort the blink detection procedure so prior to the blink detection a preliminary SP detection run is conducted just to exclude those time points from the blink detection run. Needless to say, these putative SP time points are not excluded from the SP correction process itself. Once the putative blink time points have been identified the actual SP correction procedure is conducted using a vector-filter algorithm. A SacPot event is recorded at each such point to facilitate analysis and segmentation of saccades. Also, a SacPot channel is added that describes the time course of the data that has been removed.

After this is done, then the CRD artifact is corrected using a regression algorithm. The CRD is a persistent voltage field that reflects the current gaze direction so it is not a saccade artifact so much as it is an eye position artifact. The CRD produces positivity in the direction of eye gaze. XEY and YEY channels are added to the data indicating the gaze direction inferred from this procedure. This estimate is roughly accurate for horizontal gaze direction and less so for vertical. For continuous data, the putative eye movement activity can then be observed using the View Scan function. Unfortunately, head turning complicates interpretation unless it is restrained.

The routines for all three artifacts (blink, SP, CRD) use templates to identify the artifacts. These templates can either be generated automatically or can be manually produced using the Template button on the Preprocessing pane. High quality manual templates are also provided in the templates folder. Automatic template generation requires four VEOG and two HEOG channels. Normally, the EP Toolkit is able to automatically determine which channels are the EOG channels (or the next closest EEG channels) if it is given the electrode coordinates information. If the automatic determination process is not working, it is also possible to set the EOG channels manually via the preferences settings. Just in general, it is best to check what channels are being used for the blink correction (reported by the preprocessing log) and confirm that they are the correct ones. If one of these channels is a bad channel, the EP Toolkit will first generate a best estimate of it via interpolation, which may take some time.

5) *Blink correction* - For the MAAC, the next step is to perform the blink correction process. The artifact detection routine uses EEGLab's runica routine to decompose the data into basic scalp topography components. These components are then compared to a blink template which specifies the topography of a typical blink (that can be generated either automatically or via a manual template constructed via the Template button of the Preprocessing pane). Components which correlate highly (.9 by default) with this blink topography are deleted from the data. The removal of the blink topography will apply to blink recoveries as well (which have the inverse scalp topography and are caused by the amplifiers recovering from the sudden blink amplitudes). Event markers are added to

indicate the following for each blink: blink_start, blink_peak, and blink_end. Also, a blink channel is added to describe the timecourse of the data that has been removed.

For the one-step ICA and the EMCP options, steps 5 and 6 are combined into a single step.

6) *Movement artifact removal* - Using a new approach (Dien, 2024), movement artifacts can then be removed. Whereas ICA is highly effective at removing eye blinks, it is not suited for movement artifacts because they are typically highly variable between trials so ICA can't be applied to them at the level of a dataset (there may not even be enough ICA factors available to account for all of them, let alone the ERPs themselves, since there cannot be more factors than there are variables, which in this case is the number of channels). ICA can't be applied to isolated trials because it needs more observations (i.e., time points) than are available from a typical trial. In contrast, PCA (using the Promax rotation) works even with small numbers of observations and seems to provide adequate results for individual trials. For this reason, a temporal PCA (using the Promax rotation) is used to identify factors which account for activity with an amplitude difference between minimum and maximum values greater than 200 μv and this activity is removed.

7) *EMG artifact removal* - Using a newly developed method (De Vos et al., 2010) based on blind source separation canonical correlation analysis (BSS-CCA), the toolkit will use code kindly made available by the developer to attempt to remove EMG activity. Essentially it capitalizes on the characteristic of EMG that it is more discrete than EEG in both spatial and temporal domains by preferentially removing activity that is not correlated between adjacent time points across multiple channels. I have not had the opportunity to evaluate its efficacy so I cannot make a recommendation but it does seem like a promising approach. My impression is that it works best if the sampling rate is on the order of 1000 Hz so it will not be used with the tutorial datasets (which are 250 Hz). Note that EMG correction requires the Signal Processing Toolbox.

8) *Alpha Correction* - If a participant is sleepy or otherwise zoning out, alpha activity can arise that is so large that it swamps ERPs even after averaging. This experimental option removes some of the alpha activity using a spectral-spatial two-step PCA to remove the major activity in the alpha band. This is very experimental and I cannot guarantee that it won't remove part of the ERPs as well. It should not be used on data that will be analyzed using spectral measures. It can require a great deal of RAM and has a tendency to crash even my 32GB Mac unless I make sure to free up all the memory that I can first.

9) *Bad channel and trial identification* - Bad channels are then detected on a trialwise basis. They are defined as having a difference of more than 100 μv from the minimum and the maximum values in that trial (so the trial contains substantial swings in the voltage). They are also declared bad when the maximum divergence from all the other channels is computed and even the least different is 30+ μv (so all the other channels are quite different). Flat channels are also marked as being bad, unless they were identified as being a reference channel AND the channel is flat over the entire dataset. If more than

10% of the channels (including globally bad channels) are marked as bad then the entire trial is marked bad. If a channel is marked as bad on more than 20% of the good trials, then it is marked as globally bad. The Edit Mode option controls whether this automatic routine is used and/or manual editing marks from before the automatic preprocessing routine was set in motion.

10) *Bad channel replacement* - If the "replace" option was chosen, channels that were marked bad are replaced via interpolation from the good channels of the scalp voltage field (the entire scalp is interpolated and then the missing channel voltage is obtained from this interpolated scalp surface) and bad trials are zeroed out. The quality of interpolation for channels on the edge or with neighboring bad channels is likely to be reduced.

11) *Log file* - A log file is generated to summarize the results of the session. Also, a template file is generated that provides the scalp topographies of the templates used (whether automatic or manual) and of the overall data that were actually removed.

If you wish to change the Preprocessing preferences setting, go to the preferences as before.

Parameter	Value
No Figure	☐
Moving Window	80
Chan Min-Max	100
% Bad Channel	10
# Neigh Chans	6
Neigh Diff μv	30
Bad Chan Corr	0.400
Blink Corr	0.900
% Bad Trial	20
Size of Chunks	200000
Warn Trials/Cell	15
Bad Neighbors	☐
μv Move Fac	200
Move Corr Facs	20
EMG Ratio	9
EMG Threshold	15
EOG channels	
Spike Pot	2.000
Saturation	1000
fMRI correct	fMRib O...
Blink Rotation	Infomax
Saccade Method	regression
SP Rotation	vector

- 1) *No Figure* is an option to not provide a summary figure for the artifact correction process. While a very useful figure, it requires substantial memory and so dropping it can be helpful when encountering recalcitrant memory problems.
- 2) *Moving Window* is the number of milliseconds over which the artifact correction routines average the data in a form of low pass filtering. The larger the number, the less sensitive it is to high frequency spikes.
- 3) *Chan Min-Max μv* is the maximum allowed change in voltage levels for a channel during a trial before it is deemed to be a bad channel for that trial.
- 4) *% Bad Channel* is the maximum percentage of channels allowed to be bad in a trial before it is deemed to be a bad trial.
- 5) *# Neigh Chans* is the number of channels considered to be a neighbor for purposes of the artifact correction algorithms. The electrode coordinates are then used to figure out which channels are to be used.
- 6) *Neigh Diff μv* is the maximum voltage difference allowed between a channel and its neighbors before it is deemed to be a bad channel.

- 7) *Bad Chan Corr* is the multiple-R regression criterion for determining whether a channel is a globally bad channel over the entire session.
- 8) *Blink Corr* is the correlation criterion for determining whether an ICA factor matches the blink template and should therefore be subtracted from the data. This setting is also used for the saccade and saccadic spike potential corrections.
- 9) % *Bad Trial* is the maximum number of trials a channel is allowed to be judged bad before it is deemed to be globally bad.
- 10) *Size of Chunks* is the number of time points that are read into each chunk (about 100,000 per GB of available RAM seems to generally work). If there is sufficient memory, it is best to process a datafile as a single chunk so be sure to set this number as high as is possible. Unfortunately, on the Mac it is not possible to have MATLAB set this parameter automatically so the user has to set this. If the data files are already being processed as a single chunk then raising this number will have no effect.
- 11) *Warn Trials/Cell* is the minimum number of good trials that is considered to be sufficient for a cell. Any cells dropping below this number will trigger a warning in the artifact correction log. There is no other effect of this setting.
- 12) *Bad Neighbors* is an option where if two neighboring channels are marked as being locally bad then the trial is also marked bad (because this typically means that a movement artifact of some sort is present, as opposed to isolated bad channels). This is a very stringent criterion and should not be used for developmental data.
- 13) μ v *Move Fac* is the maximum voltage difference (maximum-minimum) allowed by a factor by the movement artifact correction step. Factors exceeding this limit are deemed to reflect artifacts and are subtracted from the data.
- 14) *Move Corr Facs* is the number of factors to be retained by the movement artifact correction routine. A larger number results in a more accurate but slower process.
- 15) *EMG Ratio* is the minimum ratio of signal power to EMG noise to retain during EMG correction. In other words, factors smaller than this number are discarded as likely being EMG.
- 16) *EMG Threshold* is the Hz threshold considered to be the lower bound of possible EMG frequencies during EMG correction.
- 17) *EOG channels* are the EOG channels. If left blank, the Toolkit will try to automatically determine which are the EOG channels based on electrode coordinates. If it fails to properly identify them, then fill in this field to specify them manually: [LUV RUV LLV RLV LH RH]. Specify missing EOG channels with a -1.

18) *Saccade Potential* is the minimum microvolt threshold for detecting saccadic spike potentials.

19) *Saturation* is the voltage level at which the data should be considered to be bad and excluded from the blink and saccade correction steps. These points are also set to NaN during the blink correction procedure. Although the setting is labeled "saturation", really it means outlier points as modern amplifiers have much more range than this.

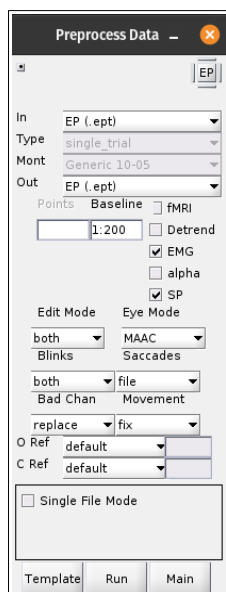
20) This pop-up menu is a list of available fMRI artifact correction routines. At present the two available are the fMRIB EEGLab plug-in which uses the Optimal Basis Set method (Niazy et al., 2005) and the AMRI EEG fMRI Toolbox which uses an ICA-based method (Liu et al., 2012) for removing ballistocardiogram artifacts. They also both remove gradient artifacts. I've gotten better results with the OBS method in my initial efforts so I've made it the default but your mileage may vary so it would be best to try both. The two software packages have many options but I don't yet see any reason to implement a user interface to access them.

21) This pop-up menu specifies which rotation is used for blink corrections, so you could choose SOBI rather than Infomax for example.

22) This pop-up menu specifies which rotation is used for CRD saccade corrections, so you could choose Infomax rather than Regression for example.

23) This pop-up menu specifies which rotation is used for saccadic spike potential corrections, so you could choose Infomax rather than Vector for example.

Setting the Preprocessing Controls



In (Input File Format) should be set to EP (.ept). It is recommended to set Out (Output File Format) to EP (.ept) since this is the native file format for the EP Toolkit program and will preserve information that might otherwise be lost should another file format be used. At the end of the process the data can be saved into a different format as needed.

The next field is the Points field. Oftentimes only a portion of the epoch is needed for the analysis. If so, it may be desirable to specify the timepoints in the Points field (will not segment continuous data). Not only would doing so save time and disk space, it would also avoid losing trials due to artifacts outside the parts of the epoch of interest. They should be described in terms of samples. Colon means that a range of numbers is being described. For example, "1:250" would indicate that the first 250 time points should be retained. If each sample is 4 ms (i.e., 250 Hz sampling rate), then these samples would correspond to the first second. If the field is left blank then by default the entire trial will be retained. The label of the field is initially grayed out to denote that this control is inactive but if timepoints are typed into the field then the label will no longer be grayed out.

The Baseline field specifies samples to use for baseline correction (ignoring the baseline field of the data file). If the baseline parameter is set, the function will baseline correct each trial using the time points specified (e.g., 1:25 means use the first twenty-five time points). This ensures that random swings in the baseline do not degrade the quality of the automatic artifact detection process. The use of this option is strongly recommended. Note, though, that the output data will then be baseline corrected. For this tutorial dataset, type in "1:200".

There is then a column of checkbox controls. The first, fMRI, indicates when to apply fMRI artifact correction routines for EEG collected inside an MRI scanner. Generally for ERP data it makes most sense to run this routine as a separate step on the continuous data, segment, and then run the rest of the artifact correction routine on the segmented data.

The next checkbox, Detrend, indicates whether to detrend each trial. In general, this option is not recommended for segmented ERP data as it may attenuate the ERP components but can be of use for continuous data prior to filtering and for segmented data that will be analyzed in the frequency domain.

The third checkbox, EMG, uses blind source separation canonical correlation analysis or BSS-CCA (De Vos et al., 2010) to remove EMG activity. My informal observation is that it seems to work well but needs a high sampling rate (e.g., 1000Hz). Click it.

The fourth checkbox, alpha, uses frequency-spatial two-step PCA to attenuate unwanted alpha activity.

The fifth checkbox, SP, removes saccadic spike potentials, using the method specified in the preferences setting and the saccades pop-up menu. Click it.

The Edit Mode popup menu indicates whether the results of previously performed manual editing should be utilized to determine bad channels and trials, either in combination with or instead of the automatic routines. Manual scanning for bad data will be explained in a later section. Leave it at "both."

The Eye Mode popup menu controls the eye artifact correction method. MAAC (Dien, 2024) is the default stepwise method described earlier and should be used for this example. ICA is the use of a single-step ICA, from which each of the eye-artifact factors are removed. EMCP (Gratton et al., 1983) is the classic regression-based method for correcting blinks and saccadic CRD artifacts.

The next popupmenu, Blinks, sets whether the blink correction step should use a manually generated blink template ("file"), an automatically generated template ("auto"), or both ("both"). Leave it on "both". There is also an "eye-track" option that is not yet documented. For EMCP, this setting only affects how the data are marked up with blink events and does not affect the actual EEG data. For the File option, the EP Toolkit comes with a canonical blink template file for greater convenience.

The next popupmenu, Saccades, sets whether the saccadic CRD and the saccadic spike potential correction steps should use a manually generated blink template ("file"), an automatically generated template ("auto"), or both ("both"). There is also an "eye-track" option that is not yet documented. For EMCP, this setting only affects how the data are marked up with saccade events and does not affect the actual EEG data. For the File option, the EP Toolkit comes with a canonical saccadic CRD and saccadic spike potential template file for greater convenience. In general, I recommend using the "file" option as it allows for a very rough eye gaze estimate, but only if the montage of the available file template is reasonably compatible with the dataset. If making a manual template for this purpose, it would be necessary to include a calibration phase to provide pure examples of vertical and horizontal saccades.

The Bad Channel Correction control indicates whether to "replace" bad channels (via interpolation) and to zero out bad trials, to "mark" the bad channels and bad trials with a flat line and a large spike (when saving to non ept file formats), or to do nothing except to record their existence in the log (and in the file if ept format). For the tutorial, choose "replace."

The Movement Correction control indicates whether to use a PCA procedure (Dien, 2024) to "fix" movement artifacts or not. For the tutorial, choose to use it (quite helpful for noisy datasets).

The next set of controls is for indicating which, if any, of the channels are explicit reference channels, meaning that their voltage numbers are included in the data. The "O" line is for the original reference channel(s) during data collection. The "C" line is for the current reference channel(s), for cases where the data has been rereferenced.

A single reference channel will by definition be a flat line of zeros. Usually, in order to save on disk space, such a channel is simply left out of the file and is considered to be present implicitly. Mean mastoids reference channels will have a mean of zero and will split the difference, becoming mirror images of each other. Sometimes one of them will be left out of the file as it is redundant with the other one but will still be considered to be present implicitly. If both mean mastoid channels are missing, then nothing can be done and they should be treated as being missing channels.

The reference channel(s) needs to be indicated when: 1) a single channel reference is explicitly present so that the EP Toolkit will know that it is flat because it is a reference channel rather than because it is a bad channel. 2) when mean mastoids reference is utilized and both channels are explicitly included because the ICA process used to extract eye blinks can be derailed by their presence (since they are mirror images of each other, they will have a perfect correlation of -1, which can end up causing computational problems). "ave ref" means average referenced. "1-2 refs" means one or two explicit reference channels. "default" means that the reference channels indicated by the CED file (if any) should be used. These values provided by the CED may not be correct if they indicate the original reference electrode and the data were subsequently rereferenced to some other scheme.

In this tutorial dataset Fp1 was used as the recording reference and was present only implicitly; however, since the CED information was provided when the data was read in, it was automatically added back into the dataset. For this reason, "1-2 refs" and "1" would be appropriate settings (but only if BV Recorder was used for data collection, as Pycorder generates an ambiguous reference scheme that needs to be rereferenced). Since the reference channel was already provided by the CED file, it is sufficient to use "default" for such files.

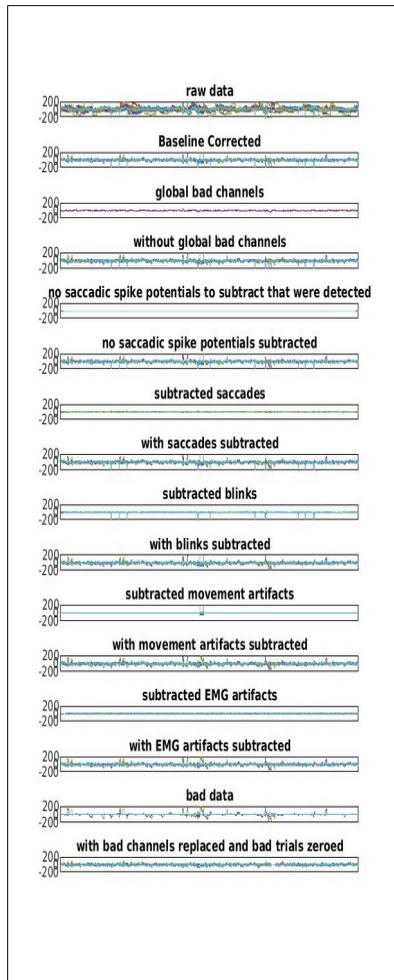
The Single File Mode controls operate in the same fashion described under "Read Data".

When done selecting the proper settings, click on the Run button. It will ask for the three templates (blink, saccade, and spike potential). Choose the ones in the templates directory of the EP Toolkit (just click on Open as each requestor appears). If you have template files in the active directory (if you made customized ones for the study), then it will show you them and you'll need to navigate to the templates directory to use the canonical ones there.

Processing can take anywhere from five minutes to three hours per data file, depending on its size and the speed of the computer. The channel replacement can take up half the total period. The tutorial files take an average of 7 minutes each on a 3.8 GHz 24-core 128 GB 3960x Ryzen Threadripper Pop!_OS 22.04 Linux computer running MATLAB 2024b. It's best to just set up an entire batch and let it run overnight. Remember to set the computer to not sleep if it is a Mac.

During the artifact correction process, a new Artifact Correction figure will be presented illustrating the progression of the process. For each chunk (there will be multiple chunks if the size of the data file is larger than the maximum chunk size that you set earlier on in the preferences), it will chart the state of the data in a butterfly plot, which consists of all the trials laid end to end, with one line corresponding to each channel. Artifacts can be easily detected as high amplitude divergences from the central mass of lines. Several such plots are provided:

- 1) the original data.
- 2) the data after baseline correction.
- 3) the data identified as being global bad channels.
- 4) the data with the global bad channels subtracted.
- 5) the portion of the data identified as saccadic spike potentials. They should be small and distinct.
- 6) the data with the saccadic spike potentials subtracted.
- 7) the portion of the data identified as corneo-retinal dipole artifact. They may be quite large and continuous, depending on the nature of the experiment.
- 8) the data with the corneo-retinal dipole artifacts subtracted.
- 9) the portion of the data identified as blinks. They should be sharp spikes.
- 10) the data with the blinks subtracted.
- 11) the portion of the data identified as movement artifacts.
- 12) the data with the movement artifacts removed.
- 13) the portion of the data identified as bad channels or bad trials.
- 14) the data with bad channels interpolated and bad trials zeroed out.



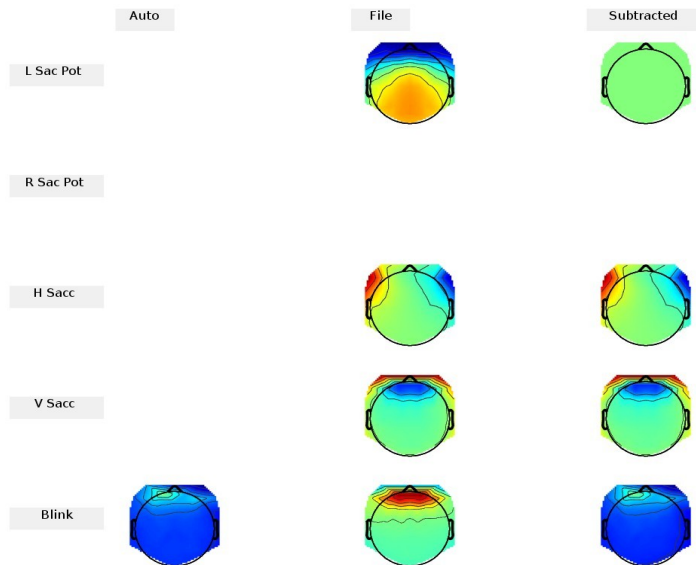
The subplots for options that were not chosen will not be included in the summary figure. If a combined subject average file is being preprocessed, then each subject will be laid end to end just like trials are for single-trial data, resulting in just a single summary figure.

The following files will be generated:

- 1) an eyeblink corrected file with a "_e" suffix added. A new regional channel will be added that contains the timecourse of the removed blink activity (upper VEOG-lower VEOG) and another for saccades (if these two types of corrections were made). This can be helpful for determining whether any residual blink activity might be responsible for apparent experimental effects. Also, blink start/peak/end events will be added to mark the maximum point of the blink if it exceeded 100 microvolts at the peak channel and is separated by at least 300 ms from the last blink peak. Saccade events will be added to mark the point at which the differential between the HEOG channels first exceeded 2 microvolts prior to the saccade peak.

2) a .jpg file of the butterfly plots. Each one should be examined, principally to determine if the blinks were in fact removed. If not, it may be necessary to rerun the file using a customized blink template formed using blinks just from the one file.

3) a .jpg "templates" file of the artifact and template topographies. It is best to check this file to confirm that the algorithms did a reasonable job at characterizing the artifacts, especially when using automatically generated templates. The following figure is for the case where "auto" was used for the saccades and "file" was used for the blinks.



4) A file titled Artifact_Correction_Log containing the record of the preprocessing session. It is highly recommended that it be examined carefully for signs of problems.

For the first session file, the log file output is:

ARTIFACT CORRECTION SUMMARY

The blink channels are: LUVEOG(1) RUVEOG(32) LLVEOG(67) RLVEOG(68).

The saccade channels are: LHEOG(65) RHEOG(66).

Minimum number of good trials per cell to avoid warning message is: 15.

Moving average window for smoothing during bad channel detection: 80 ms.

Difference from minimum to maximum for bad channel: 100 μ v.

Percent of bad channels exceeded to declare bad trial, rounding down: 10%.

Number of electrodes considered to be neighbors: 6.

Minimum predictability from neighbors to not be considered globally bad: 0.4.

Maximum difference from most similar neighboring electrode to be considered bad: 30 μ v.

Blink template option is: bothTemplate.

CRD saccade template option is: fileTemplate.

spike potential template option is: fileTemplate.

Using MAAC stepwise procedure.

Threshold correlation with blink template: 0.9.

For blinks, using EEGLab function runica to perform Infomax rotation.

Threshold correlation with saccade template: 0.9.
 For saccades, using regression method.
 Saccade Potential Threshold: 2.
 For spike potentials, using vector method.
 Detrend data: OFF.
 Percentage of good trials chan is bad to declare a channel globally bad: 20%.
 Number of factors to retain when correcting movement artifacts: 20.
 Maximum voltage difference allowed when correcting movement artifacts: 200.
 Minimum ratio of signal power to EMG noise to retain during EMG correction: 9.
 Hz threshold considered to be the lower bound of possible EMG frequencies during EMG correction:
 15.
 Bad channels and trials: replace.
 Samples used to baseline epochs: 1:200.
 Editing mode option is: both.

 Working on: /media/jdien/Data1/workshop/tutorial/AP5-001-LA1_mf_seg.ept.
 The channel 1 is marked as being the original reference.
 The channel 1 is marked as being the current reference.
 Baseline Correcting.
 Warning: shorted channels: FC5-FT7; TP9-TP7; P3-P7; P3-P5; P7-P5;
 No out-of-range time points set to NaN.
 Global bad channels: 32
 Temporarily interpolating right upper VEOG channel to enable artifact correction.
 Warning: interpolating may not work sufficiently well. It may be necessary to redo this session using
 manual templates.
 Starting preliminary spike potential scan to exclude them from the preliminary blink detection routine.
 Nothing matches saccadic spike potential template so no detection performed.
 Starting preliminary blink scan to identify which time points to exclude from the saccade and the
 spike potential corrections.
 Starting eyeblinks routine.
 The rank of the blink data was: 68
 7 blink trials used for template generation.
 The highest automatic blink fac correlation (of 1) was: 0.9775
 No manual blink facs met criteria. The highest correlation was: 0.7797
 No blink periods detected.
 Starting final spike potential correction, excluding blink time points.
 Nothing matches saccadic spike potential template so no correction performed.
 Starting saccade correction, interpolating blink time points.
 Starting final blink correction.
 Starting eyeblinks routine.
 The rank of the blink data was: 68
 10 blink trials used for template generation.
 The highest automatic blink fac correlation (of 1) was: 0.99179
 No manual blink facs met criteria. The highest correlation was: 0.81113
 No blink peaks detected but blink-related activity corrected.
 1 trial corrected for movement activity.
 Starting EMG correction.
 There was 1 bad trial.
 Originally there were 0 bad trials.
 There were 1 bad trials due to too many bad channels.
 There were 0 bad trials due to neighboring bad channels.
 For good trials, there was an average of 0.52703 bad channels per trial.

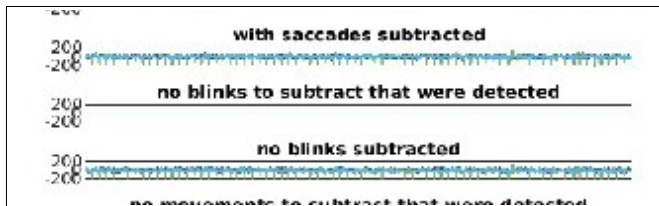
Warning: Cell LANF has only 11 good trials.
Warning: Cell LANU has only 10 good trials.
Warning: Cell LAWF has only 8 good trials.
Warning: Cell LAWR has only 12 good trials.
Replacing bad channels.
The file took 4 minutes to process.

The header gives information about the settings used for the preprocessing run. Then each section after that provides a summary of each file processed. In this case, it is providing a warning that some pairs of channels appear to be shorted together as they are nearly identical. The EP Toolkit does not take any action when it detects possible shorts aside from issuing the warnings. They can help warn the experimenter that extra attention to the electrode applications may be needed. Such shorts reduce spatial resolution but should not invalidate results.

For maximum rigor, one could perform an initial artifact correction run with just blink and/or saccade correction where Bad Channel Correction and Movement Correction are set to None, followed by manual scanning and adjusting of edit marks, followed by another automatic artifact correction run with blink and saccade correction turned off and Bad Channel Correction and Movement Correction turned on.

Manually Forming an Blink Template

The automatic artifact correction procedure does not always work. It is important to always inspect the results after each run. For example, in the example below (not part of the tutorial), the blinks were not detected properly. This is especially evident in the summary figure. In this kind of butterfly plot with fP1 reference, blinks appear as spikes below the overall line. While they are quite apparent in the “with saccades subtracted” line, no blinks were detected and so they are still there in the next line “no blinks subtracted.”



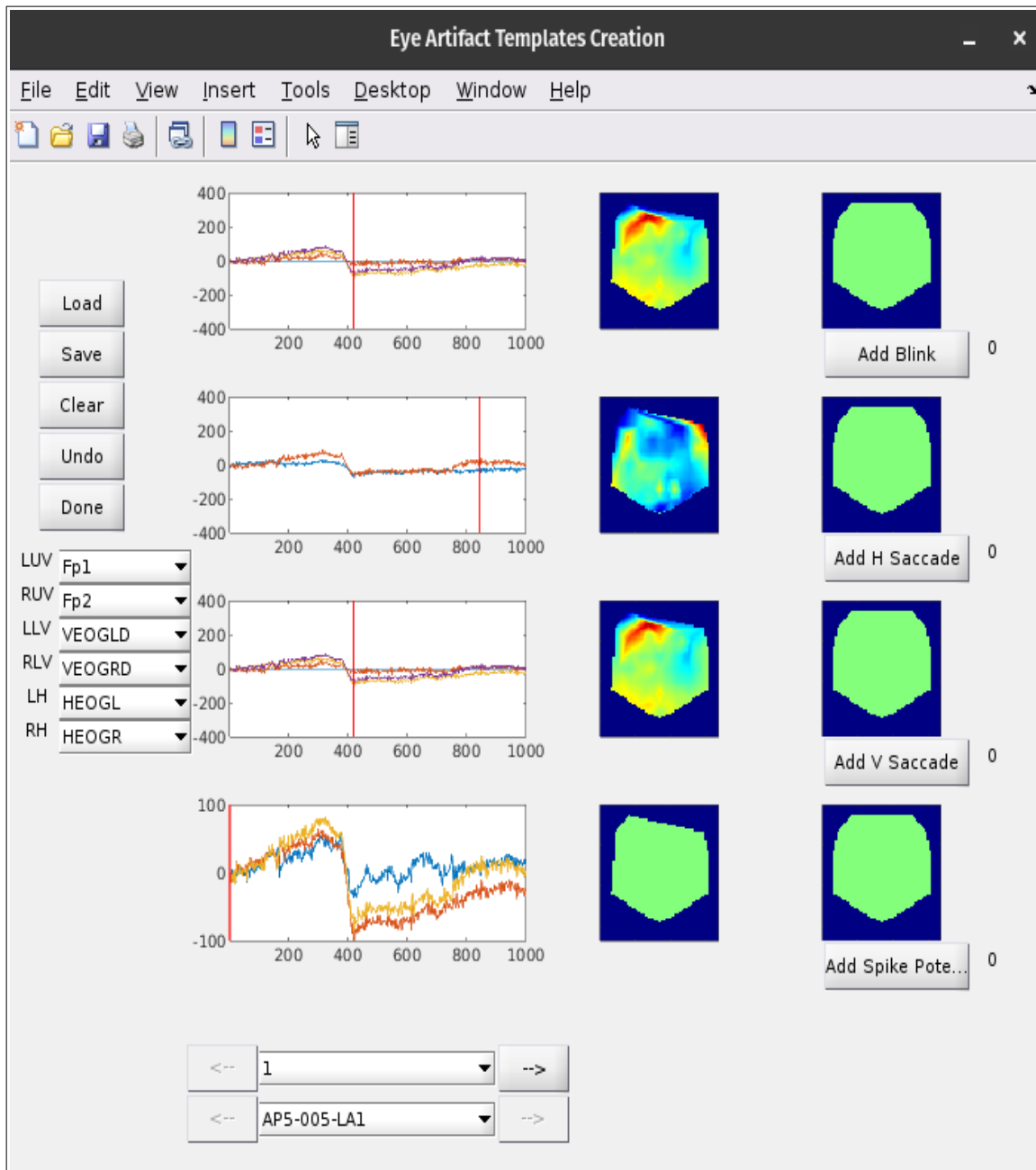
Moving to the Artifact Correction Log, the initial blink scan for the spike potential correction was fine, but for the actual blink correction, the output read:

*No blinks auto-detected to form the blink template.
No manual blink facts met criteria. The highest correlation was: 0.89009
No components match blink template so no correction performed.*

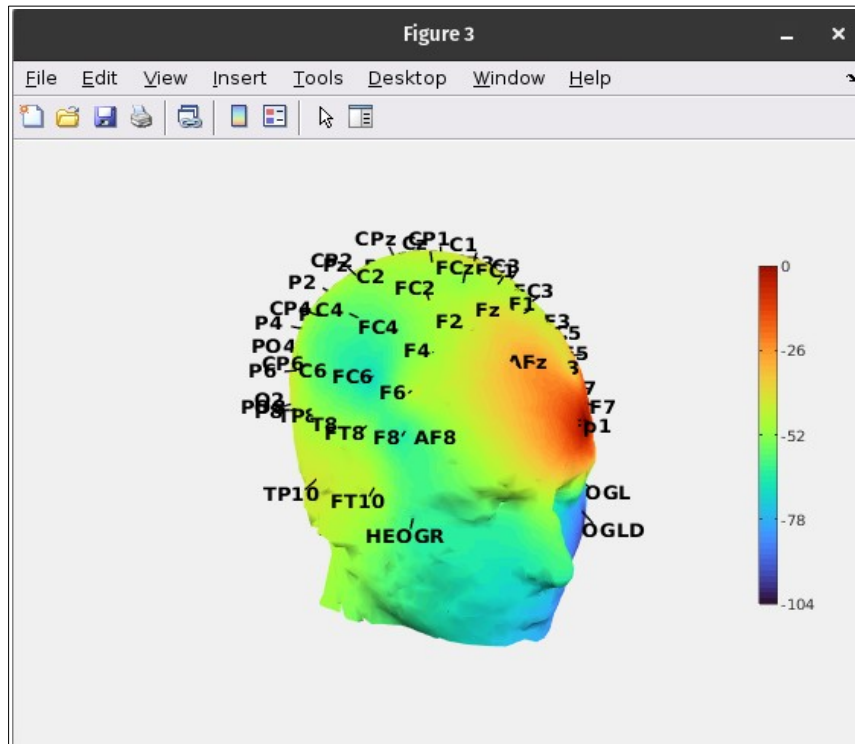
Since the criterion threshold was set at .90, it only just missed being used. One simple solution is to just lower the threshold to, say, .88 and rerun the files. The drawback to lowering the threshold is that it increases the chance of falsely identifying a non-blink ICA factor as a blink and eliminating it. In this case, the relaxation of the threshold would be very modest, so it should not be an issue. In other cases, a bespoke blink template might be necessary. For example, if one's EEG montage did not include lower VEOG channels, then it would differ significantly from the canonical template included with the EP Toolkit.

An alternative approach is therefore to generate artifact templates manually. This next section will describe how to do so. First click on Read in the Main Menu to load in the file. This will bring up the Read Data Panel. Specify EP (.ept) file format. You don't need to specify the Type and Montage since the segmented files are .ept file format, which provide that information already. Load in the segmented files. Doing so will place them in the working set, from where you will be able to construct the manual templates.

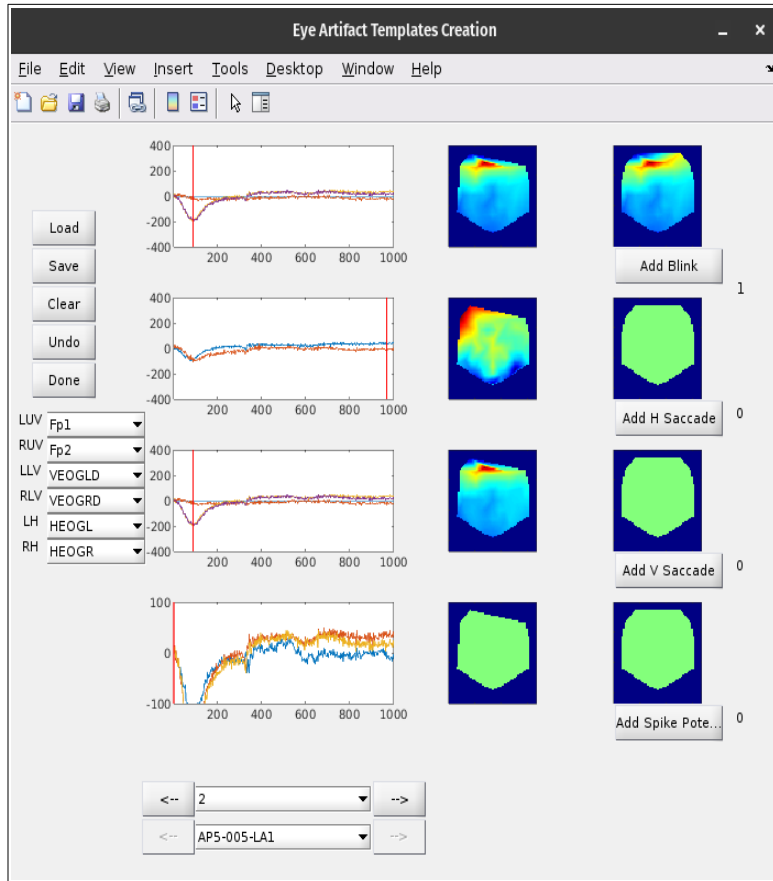
Now click on Main to go back to the Main Menu. Next, click on Preprocess. Click on Template to bring up the Template Creation Window. In the first row (blinks), it shows the VEOG waveforms of the first data file, one second epoch at a time. It also marks the time point with the maximum voltage (across all the channels, including the ones not plotted so it may not always be apparent why it's chosen the time point) with a red line. To the right of the waveforms is the scalp topography at that point of maximum voltage (across all the channels). To the right of that is the scalp topography of the new blink template. Since nothing has been added to it yet, it is a blank green. The second row (horizontal saccades) shows the two HEOG channels and the scalp topography at the point of maximal divergence between the two HEOG channels. The third row (vertical saccades) shows the VEOG channels and the scalp topography at the point of maximal divergence between just the VEOG channels. The fourth row (Spike Potentials) should show Pz and the lower VEOG channels. The first column is a blank green because no credible spike potential was detected in this trial. The topos are a bit lopsided at the top because the right lower VEOG has registered as a bad channel in this function's quick scan, which is likely why the artifact correction procedure had trouble with this participant, and so has been excluded from the topos.



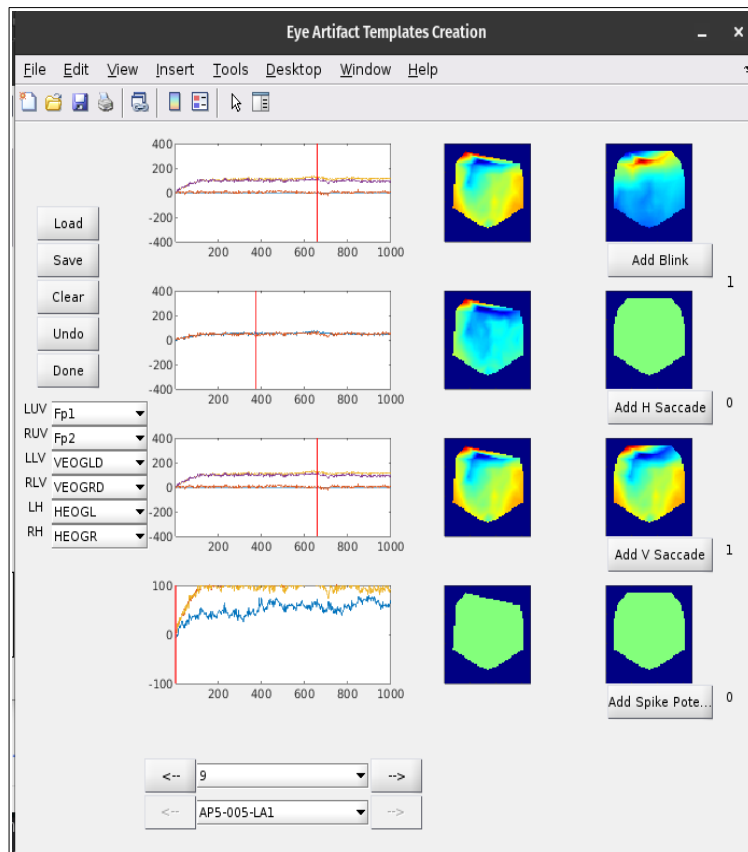
One should first confirm that the correct channels have been automatically identified as EOG channels. The channels are listed on the left side of the window and can be manually changed if need be. These settings only affect which channels are used to automatically identify artifact candidates. One could also visually confirm electrode locations by clicking on any of the 2D plots (the first row first column candidate blink topo in this example). The electrodes were not mapped to the 3D head perfectly but close enough for visualization sake. The important thing is that there were not any major distortions, such as being rotated 90 degrees to the right.



Clicking on right arrow or the pop-up menu below the waveforms brings one to the first blink (in the 2nd one second epoch). It can be recognized as such because of the smooth, high amplitude bell-shaped curve which the scalp topography image shows to be tightly centered around the eyes. Since the data are still referenced to Fp1, the upper VEOG channels are flat and the lower ones are negative. Click on the Add Blink button to add this blink to the template. The counter below the blink template image increases to one. If a mistake was made, one can Undo the last addition. One can also Clear the template formed thus far. If the red line is not at the right time point, you can drag it by left-clicking on it and then moving the cursor.

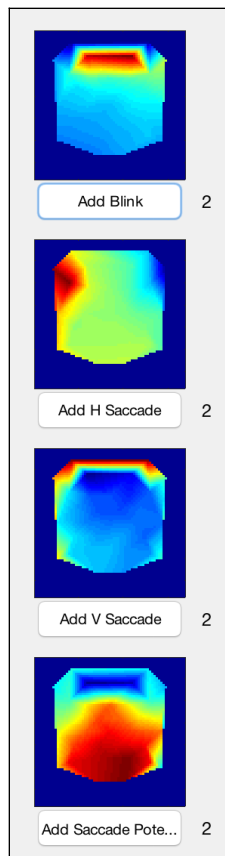


There is no hard and fast rule on how many blinks to use in a template but I find that having ten of them (evenly sampled from the files) tends to yield acceptable results. The template doesn't have to be exact, though, since the eyeblink correction process only uses it to determine which ICA factors correspond to blinks and the match to an individual's blink does not need to be exact (the default setting in the preferences is to correlate .9).



The first clear vertical saccade shows up on the 9th trial. It involves a clear and smooth divergence between the four VEOG channels depicted on the second row and the topography shows a clear asymmetry of blue above and red below, and just in the front of the head by the eyes. Click on Add V Saccade. Horizontal saccades are not really present in this recording, but have a similar inversion laterally. Note that the template image are sometimes reversed from the chosen example. The EP Toolkit automatically flips these examples to be consistent with left-sided horizontal saccades, otherwise the left and right saccades would cancel out. We'll be using the sample template files for the rest of the tutorial, but this is a good exercise for getting familiar with the artifacts. Spike potentials are also not really present.

Keep clicking through until you have added two each of blinks and horizontal saccades from each of five files (total of ten each, not necessarily from the same ones). Not every file will have clear examples of these artifacts so don't worry if you need to skip one. Avoid examples of artifacts that are noisy and contaminated with other artifacts as they will degrade the effectiveness of the template. Ignore the horizontal and spike potentials. Following are examples of the artifacts from another dataset where they were all clearly present.



Once finished, click on Save and give the file a name. It will ask to do so thrice, once each for blinks, saccades, and spike potentials (if such a template has been constructed). One can also Load a template back for later inspection or to add more examples to it. Click on Done to go back to the Preprocess Data Window.

Once the template has been formed, it is no longer necessary to have the files in the working set. The preprocessing step operates directly upon the files on the disk (since reading in a large batch of files to the working set can take some time, it's more efficient to just read them in one at a time during the batch session rather than having to wait around). To avoid confusion, one may wish to purge the working set at this point. To do so, click on Main to return to the Main Menu and then click on Read. From the Read Window, one can purge the files by clicking on their names (hence the red color to remind you that they will be deleted). Alternatively, one can choose Clear Work Directory from the File menu of the Main pane, at which point it will ask if you want to clear the working set. Once done, return to the Preprocess Window (by clicking on Main and then click on Preprocess).

What you do next depends on what kind of analysis you are planning on conducting. If you are planning on a spectral analysis then you should use the Transform function followed by the Average function (next section). If you are planning on an ERP analysis

then you should use the Average function followed by the Transform function. For the purposes of this tutorial, we will proceed with an ERP analysis.

Averaging

To perform the Average step, click on Main to return to the Main Menu. Next, click on Average. Set the file formats and types to the appropriate values. It is recommended to use the EP .ept format. In the figure, File Type and Montage are grayed out because the .ept format includes the information about what type of file it is. The Model popupmenu specifies which dataset in the working set, if any, to use as an example dataset for some of the controls to be described later. Make sure to import one of the preprocessed session files into the working set using the Read function if you haven't already.

The Drop Events control instructs the averaging procedure to drop the events contained in the session files in order to reduce the size of the resulting file. The box can be unchecked so that the distribution of the events can be inspected in the View function to determine whether there is the possibility of confounds.

The drop noise box likewise instructs the function to not retain the noise estimate produced by the +/- reference (Schimmel, 1967) in which every other trial is inverted during averaging, resulting in an average in which the ERP has been cancelled out, leaving only noise. By default this information is retained and can be visualized via the View function, examined via the Edit function's QC pane, or analyzed via the meanNoise option of the Windowing function.

Average Data

In: EP (.ept)

Type: single_trial

Mont: Generic 10-05

Out: EP (.ept)

Model: AP5-001-LA1_mf_seg_e

☒ Drop events ☐ Drop noise

Trial SD: none

Type: subject

Subject: S:7 Session:

Cell:

Proc: Average

Method: Mean

☒ Drop Bad Trials

ACC: Error ☒ Time ☒

RT: Median ☒

Min RT: 100 Max RT SD: 2

Average Main

The Trial SD popup menu instructs the function whether to retain the trial-wise standard deviation information generated during the averaging process. This information can be used as another method of estimating noise levels and also point-by-point standard error bands. In order to recalculate the SD information during such operations as rereferencing and combining of channels, the covariance between the channels must also be retained. This information can result in a substantial memory and time overhead during subsequent operations. The default setting is therefore to drop this information. One can also drop just the covariances, in which case the SDs will simply be dropped during operations where the covariance information would have been needed to update the SDs. If the data are filtered during the Transform step, the SD information will be dropped as no longer corresponding to the data; if both are desired, then the filtering should be performed on the preprocessed single-trial data prior to averaging.

The standard averaging type is the subject average. One can also choose to generate an item average (Dien, Frishkoff, et al., 2003) wherein the averages are grand averages of each item. If choosing this option, it is necessary to first load an example file into the working set, then to choose it as the Model from the popupmenu, and then to specify which trial-specific field provides the item labels. If, for example, the spec chosen is that of the stimuli, then each subject slot will be one of the stimuli. Stimuli from different cells would continue to be segregated.

The Subject box is used when the original file format did not include an identifier for the subject and so it needs to be extracted from the file name, otherwise the Toolkit has no way of knowing when data from files should be combined into one subject and when they should be kept separate. If one includes the subject label in the file name (e.g., AP5-001-LA1_mf_seg_e.pt), then one indicates in the box which characters in the file name represent the subject label (e.g., 5:7 to denote 5th through 7th characters). When you select the entire batch of files to be averaged, the EP Toolkit will use this information to identify which files belong to a given session and will merge them together. This field is colored red as a reminder as it is easy to forget. Following it is a Session box that can be similarly used to specify session information (e.g., "pre_" and "post") and Cell box for files where there are separate files for each cell.

The settings at the bottom of the pane control the handling of bad trials. The checkbox indicates that trials judged to be bad by the Preprocessing function should be dropped. Trials can also be dropped on the basis of behavioral data if available. If a Model dataset has been selected, the next two popupmenus will allow specification of which trial specs provide the accuracy information (upper one) and reaction time information (lower one). If the Model dataset does not have any trial specs named "ACC" or "RT", then by default these two popupmenus will be set to "none" and the associated controls grayed out. If this information is present but under another name, then one can manually choose the trial specs on the popupmenus. By default, the accuracy data is coded as: 0=error and 1=correct and 2=timeout. These values can be changed in the Preferences settings.

If the settings are used, then the average function will compute the reaction time and accuracy data for the resulting averages. The checkboxes on the ACC row indicate whether error trials and timeout trials should be dropped from the averages. The RT method popupmenu indicates whether the mean, median, or trimmed mean (using the trimming parameter from the preferences) should be used for the reaction time data, independent of the method used for the EEG data (median is the default as it is more robust than the mean against outliers). The min RT field indicates that RTs smaller than this number should be dropped as being false start trials. The Max RT field indicates that RTs with a standard deviation larger than this figure (computed within cell) should be dropped as being an outlier. When computing cell accuracy, only error trials will be counted as errors. All other dropped trials will be excluded from the accuracy computation.

If one or more tasks were specified during segmentation, then signal detection measures (D-Prime and c) will also be generated for each task for each subject. Signal detection theory measures (Green & Swets, 1966) are designed to improve on accuracy measures by separating decision criteria from true sensitivity (e.g., one can obtain 100% target accuracy by responding to every trial). Although b has been the most commonly used decision criteria measure, c has become preferred as it is independent of the value of D-Prime (Stanislaw & Todorov, 1999). These values can be inspected using the Edit function's Subject pane and will generate text files suitable for ANOVAs by the ANOVA function's "behavioral" option.

There are several possible procedures for the EEG averaging, each with associated methods:

1) Average - which is the standard averaging process. Single-trial files will be averaged into subject averages and subject averages will be averaged into grand averages. If multiple single-trial files are chosen, a single combined subject average file will be generated in which each of the separate subject averages are present. This is the recommended approach as it greatly facilitates analysis and is the method that works best for the EP Toolkit.

The averaging methods are:

- a) mean - which is the conventional averaging procedure.
- b) median - which uses the median instead of the mean.
- c) trimmed mean - which uses the mean but first trims off the most extreme values at each time point. The default value (which can be changed under Preferences) is 25% so the highest and lowest 25% will be dropped. This results in a more robust estimate of the central tendency of the event related potentials. The trimmed mean is recommended over the median (Leonowicz et al., 2005).

2) Latency-Lock - is averaging in which a trial spec such as reaction time is used to time-lock the averages. So although a stimulus event may have been used to generate the epochs, one can then use this option to realign the epochs in terms of the ms measure

provided by the trial spec (usually reaction time). The new time-lock point will serve as the new time zero. One should also specify a range of values that will be accepted as latencies (e.g., 300-700 ms). If no range is specified, the procedure will default to 100 ms after the start of the epoch and 100 ms prior to the end of the epoch. Due to the need to accommodate latencies at the extremes of this range, the length of the resulting epoch will be equal to the length of the epoch before and after the latency range. For example, for a latency range of 300-700 and an epoch of -200-800 ms, the resulting epoch would be $(-200-300)$ to $(800-700) = -500$ to $+100$ ms or 600 ms long. The new zero will be the latency lock event. When choosing this procedure, a menu with all the single-trial files in the working set will be presented as potential model files. A menu populated with all the trial specs present in the chosen model file is available below it. Thus, one could select the reaction time trial spec if it was so desired. The model file's sole purpose is to generate the list of possible trial spec names and is not otherwise used.

3) Jitter-Correct - is much like latency-lock except that one does not need to generate the latencies in a prior step. Instead, a rough peak-picking algorithm is employed in which the point of minimum or maximum voltage within the latency range is treated as the peak of the ERP component of interest. One must also specify the channel. This can be especially helpful when one is trying to measure the amplitude of an ERP component with a variable latency (like the P300) to control for the effects of jittering on the amplitude measure, but only if the signal-to-noise is good enough. A full list of the datasets in the working set is available under Channel Model. One chooses one of them and then a list of channels from this model file will populate a Channel menu below it. One should also specify whether the positive or negative peak in the window is to be used. Thus, one could specify that the averages are to be aligned according to the positive peak in Pz within the 200-500 ms range if one wished to jitter-correct the P300.

4) Frequency-coherence - generates a coherence measure, using either multi-taper (with the indicated frequency smoothing) or Hanning methods.

5) Time-Frequency-Coherence - does the same but producing a TFT equivalent (that is, with a timecourse as well), using the phase-lock wavelet method.

6) RIDE – is a hybrid procedure (Ouyang et al., 2011, 2015) in which the epoch is divided into an initial S period with ERP components time-locked to the stimulus, a late R period with ERP components time-locked to the response, and an intermediate c period with an ERP component (like the N400) which is somewhere in between. One specifies the periods expected (S, R, and one or two c). Then one indicates the latency and the window of the periods (with the R period defined in terms of a window about the response event). One can also experiment with the C cut-off (low-pass Hz filter) and lag settings (ms range within the C window). The procedure will then generate S and R averages and then a jitter-correction to fit the remaining ERP activity after the other two are removed. The resulting averages are then recombined into a single averaged waveform. The RIDE code is included with the EP Toolkit with the permission of Dr. Ouyang and the full RIDE manual is included in the Documentation folder. The

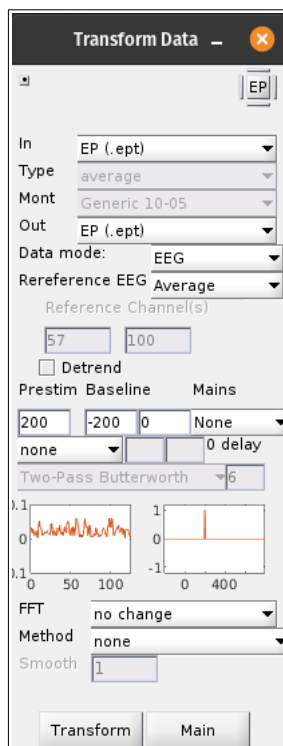
examples directory has been excluded for space reasons but can be obtained from the RIDE website. The procedure is quite interesting, although I am not in a position to make a recommendation at this point.

7) ReSync - is another experimental procedure (Ouyang, submitted) in which one or more windows can be defined. For each one a peak channel may be specified or else it may be based on all the channels using the "global" option. To provide the list of channels, a model dataset in the working set must be chosen. Within each window (which should be non-overlapping) at the specified channel(s) the waveform is corrected for jitter. The details are provided in the documentation. Again, I find this procedure interesting but cannot vouch for it at this point.

When ready, click on Average. Select the desired session files (sorting by size makes this easier) and they will be combined into a single combined subject average file (name it "tutorial_avg.ept" when it asks for the name) with separate averages for each subject. The files will be selected from those on the disk and not from those in the working set. This process will take a while. The command window will announce which file is being worked on. When it is finished, the Average and Main buttons will no longer be grayed out. If the session files were generated by the Toolkit's preprocessing step and are in EP format, then trials that were marked as bad will be dropped from the average. Likewise, uncorrected bad channels will be dropped. If the prestimulus duration information hasn't been added yet to the file (which it has not yet for the tutorial dataset) then a warning message that the covariance information needed for the MNE software won't be calculated correctly. This is not a problem unless one is planning on exporting the data to FIFF file format and submitting it to MGH's MNE software.

Click Main once it is no longer grayed out to get back to the Main Menu.

Now go back to the Transform panel to rereference the data to average reference, provide the information that the data have a 200 ms prestimulus period, and baseline correct using that 200 ms period. Note that specifying the prestimulus period will not by itself baseline correct the data. The prestimulus period is usually used for baseline correction but not necessarily. This setting is just to provide the information on the duration of the prestimulus period. Once you have run this function, it will generate a file with an _rb suffix (standing for "rereference" and "baseline correction" and "filtered"). The suffix is not necessary for the purposes of the EP Toolkit and are included only for your convenience. Note that updating the SD information will be quite time-consuming, if that option was chosen.



Edit

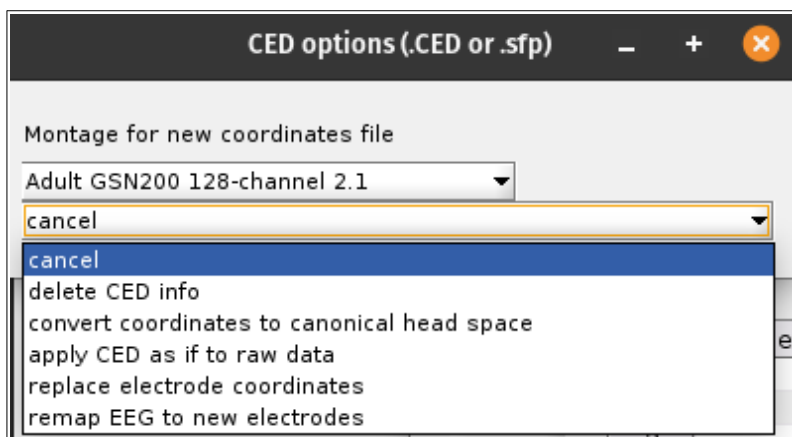
It is now time to enter the newly processed average file into the working set of data. From this point onwards, the analyses will be applied to a copy of the data maintained in the working set. Click on Main to go back to the Main Menu and then Read. Read in the tutorial_avg_rb.ept file described earlier. Click Done and then click on the Edit button. This will bring up the Edit Window, which allows for editing the data file. If there is only one data file in the working set then it will automatically open, otherwise it will be necessary to click on the name of the one that is to be edited. If you had previously clicked on it and it is still highlighted, it will be necessary first to command-click it to deselect it so that you can then select it again. While in the Edit function, a temporary copy of the dataset is created, so changes will not affect the working set copy until one is finished.

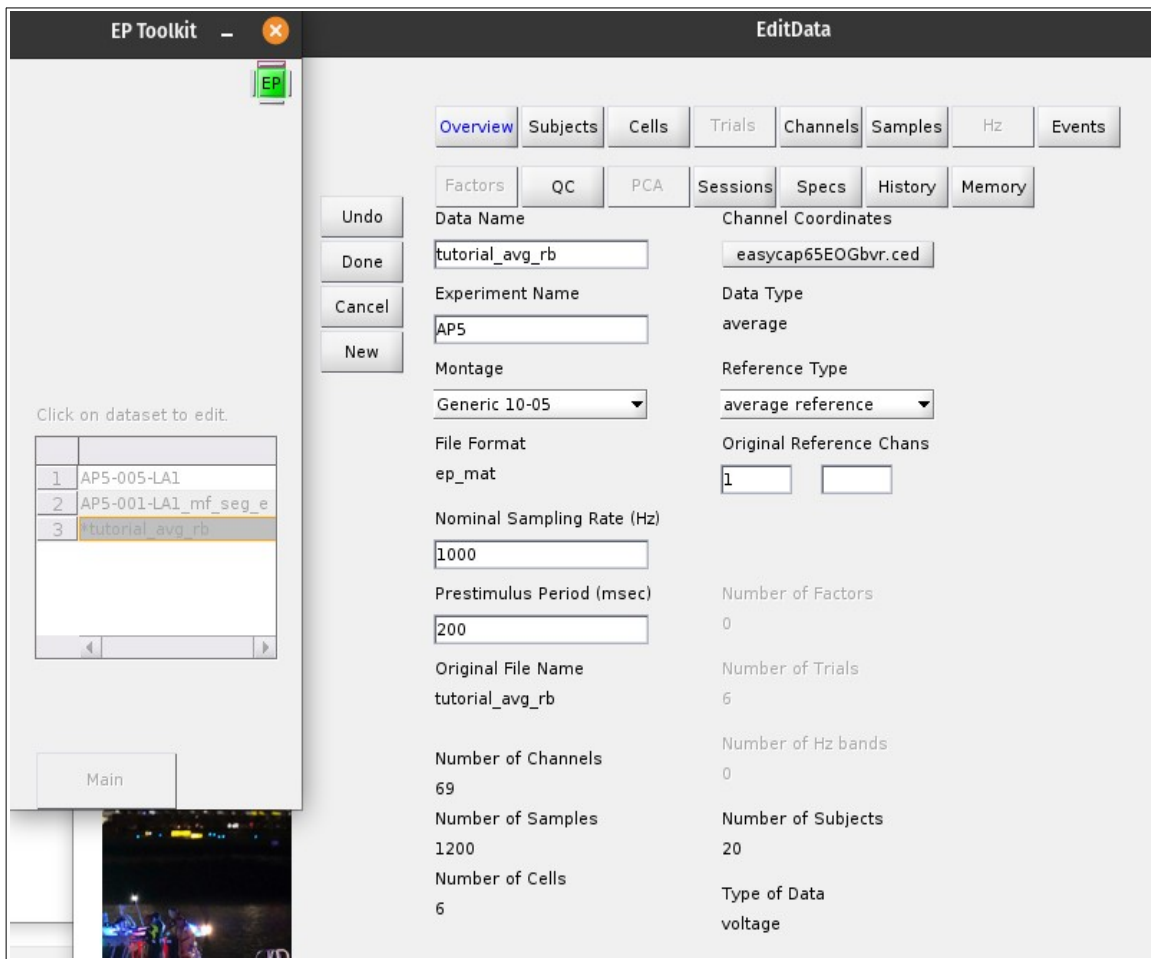
Every pane of the Edit function has the following buttons in the upper left corner. Undo allows one to undo the last change (but no more). Clicking it again undoes the undo. Done closes up the Edit window and saves any changes to the copy in the working set (which must still be saved via the Save function to have a permanent copy). Cancel closes up the Edit window and discards any changes made. New saves the changes made to a new copy in the working set, leaving the original one unchanged.

The Overview pane of the Edit Window provides the general information about the file. The Data Name is the label by which it is identified in the EP Toolkit. Each data file in the working set needs to have a unique Data Name. The Experiment Name is the name of the overall experiment, rather than the name of the specific file. Take this opportunity to provide an Experiment Name by typing it into the appropriate field ("AP5" in this case).

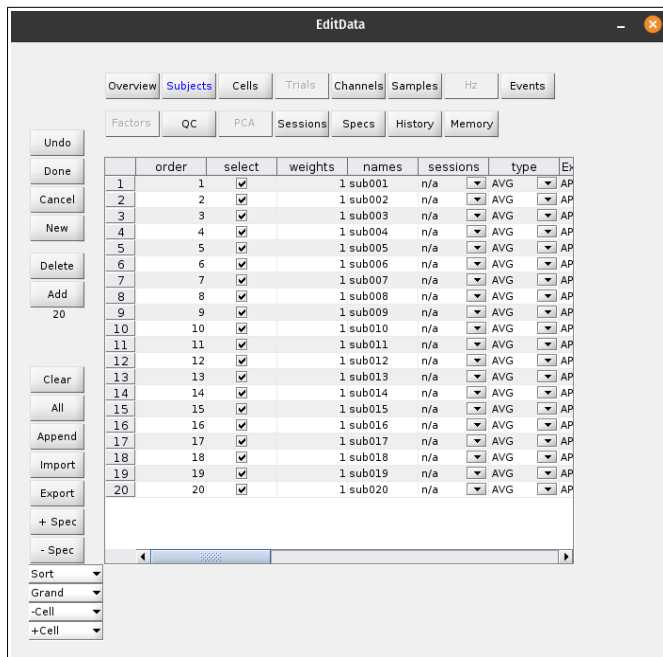
If one clicks on the "channel coordinates" field, one can change all of the coordinates at once. Doing so yields a popupmenu with the available montages, set at the current one. This popupmenu can be used to change the current montage setting, which may be necessary depending on what is done with the electrode coordinates. The second popupmenu provides several options for changing the electrode coordinates:

- 1) cancel - just cancels this current operation.
- 2) delete CED info - deletes all the electrode coordinate information, leaving it blank.
- 3) convert coordinates to canonical head space - transforms the current electrode coordinates to achieve a best fit to a standardized canonical head space (Robert Oostenveld's Standard-10-5-Cap385).
- 4) apply CED as if to raw data - will request a CED or sfp file and apply them as if to raw data. For example, if a CED file and it has any channel types like "BAD", they will be applied to the file. Likewise, implicit references will be added as appropriate.
- 5) replace electrode coordinates - will request a CED or sfp file and use them to replace existing coordinates. Any electrode names that are present in the file but not in the data or vice versa will instead simply be dropped.
- 6) remap EEG to new electrodes - will request a CED or sfp file and use them to interpolate the existing data to the new electrode locations, using a best fit transform for the two head spaces. It will be assumed that all the electrodes in the file are new locations even if the names (e.g., "E1") match existing ones. The existing channels will then be dropped.

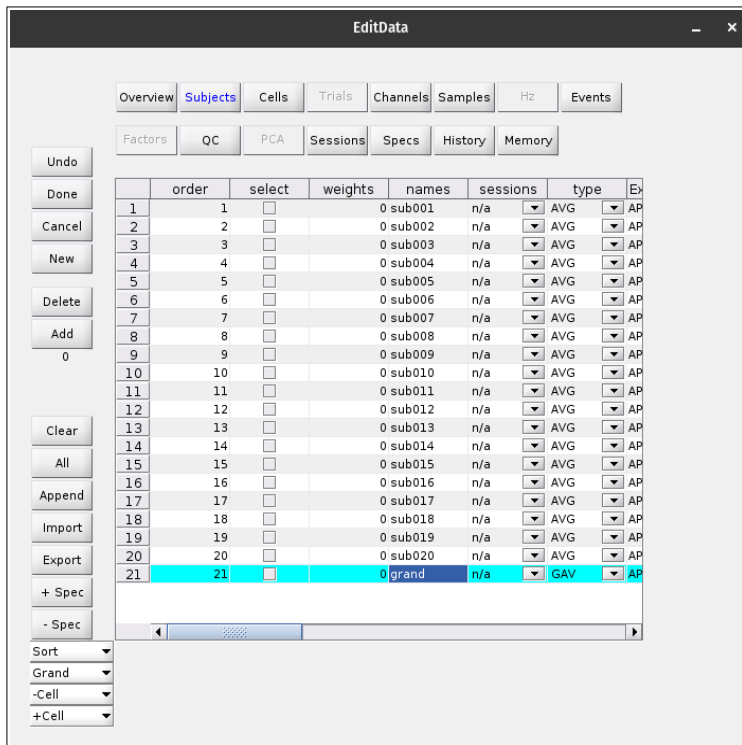




Next, we are going to generate a grand average after clicking on Subjects. Click on All. This will give all the subjects a weight of one. Note that the number below the “Add” button now says 20, to help you confirm that you have selected the intended number of subjects.



Then click on Add. The new entry is marked as a GAV type to keep it distinct from the subject averages which are AVG type and has automatically been named “gave”. Click on “gave” and change it to “grand”. Note also that the age column contains the average age of the sample and the gender column contains a summary count of the females and males. The new row is highlighted in blue to indicate it is a virtual grand average, meaning that it is dynamically updated from its constituents. If a change is made in one of the contributing subjects averages, this grand average will immediately reflect the change. It will not reflect additions (as it would not know which to add it to), just deletions. If you check the select box for the virtual grand average, the weights for all its constituent averages will be filled in, allowing you to check which ones are contributing to this grand average. Not all GAVs are virtual and will not be highlighted if just a normal grand average. If you delete all of the subjects for a virtual grand average, it will convert to a normal grand average. Also, virtual grand averages must always be at the end of the subjects list, although they can be reordered amongst themselves.



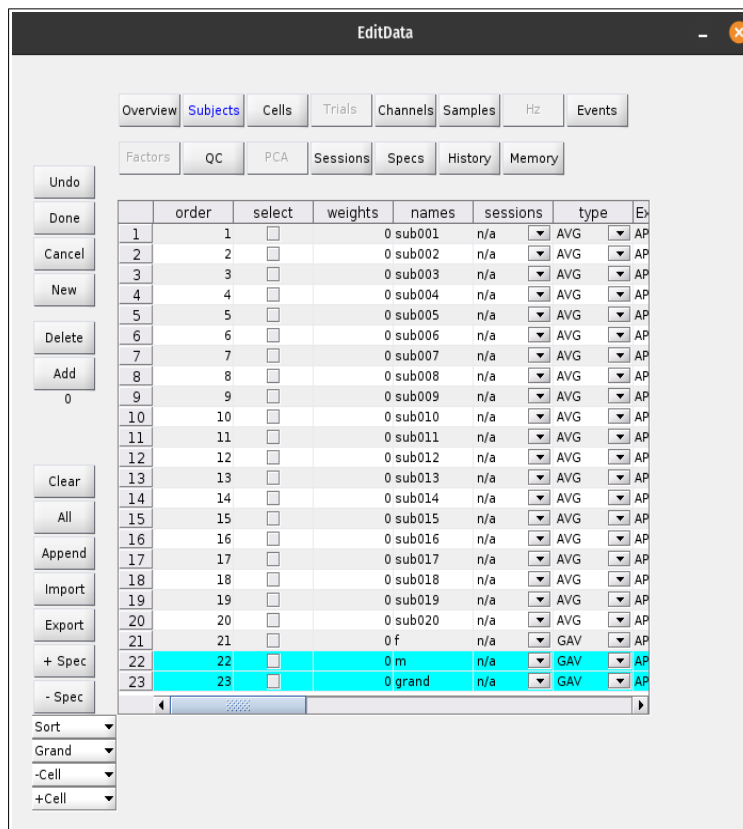
The fourth to last column in the Subjects pane contains the original file names of each subject. You might want to confirm that the file names correspond to the subject names. Expand the width of the file names column so you can see the entire names by moving your cursor to the boundary between the column names until the cursor changes to a double-headed arrow, then click and drag. If need be you can shrink the size of the other columns so that the “names” column and the “fileName” column can be seen in the window at the same time. You can also drag the “fileName” column so it is next to the “names” column. Doing so will not change the file itself and if you later return to this file the fields will be back in the original order. The subjects should be in alphabetical order of the file names. It would be simplest if the subjects are in order so you should reorder them if necessary. To do so, you can type in the new number in the “order” column and the subject will immediately be moved to just below that number (so typing in “.5” will move it to be the first). Even easier, you can just select Names from the Sort pop-up menu in the left-hand column of controls. If you don’t like something about the result, just click on “Undo” to put it back the way it was (but it will only undo one action!).

This pane can also be used to append more subjects, exporting the data from this pane to a text file, or adding/deleting columns (specs). One of the main uses for such specs is to specify groups for between-group ANOVA factors. If using the EP Toolkit for the ANOVAs, the first letter of the entries need to differ (e.g., grp1 vs. grp2 would not work).

To append more subjects, click on Append and then indicate which file is to be added to the present file. It will be assumed that appended file has the same format as the present

file. An error message will be generated if the appended file differs too much from the present file.

To add a grand average of a subset of the subjects, indicate in the weights column how much to weigh each subject and then click Add. If you just want equal weighting, you can just click in the checkbox and the value "1" will automatically be entered into the weights column. When using the weights to generate combinations, a running total of the weights will be maintained on the left side of the pane. **Note that any weights input into the pane will not take effect until one has clicked elsewhere or pressed enter.** Keep an eye on the running total to verify that each weight value you enter has been recognized. If the weights sum to zero then they will be weighted exactly as entered (e.g., a -2 and a 1 and a 1 one would result in the first subject being doubled and then subtracted from the sum of the other two). If the weights do not sum to zero (e.g., a 1 and a 1 and a 1) then they will be divided by the total weights to provide a mean value (e.g., the total in this case would be 1+1+1=3 and so the three subjects' data would be added together and then divided by 3). For ease of use, if you click on the "Grand" popupmenu and then choose a column name (e.g., gender), it will automatically add grand averages corresponding to each value in that column (e.g., one for male and one for female). Go ahead and do so, which should add two more virtual grand averages, one for "f" and one for "m". Then enter 23.5 as the order for "grand" to move it to the end.



Normally if one is generating a difference wave, each cell from the one set would be subtracted from the respective cell in the other set. The -Cell and +Cell popup menus allow one to specify a single cell in the negative and/or the positive subset that are applied to all of the cells in the other subset, as when one set of subjects is providing a baseline cell that is subtracted from all of the cells in the other set of subjects.

Finally, there is a sessions column to indicate the session for multi-session designs where a participant had recordings on more than one occasion (e.g., "pre_", "post"). The session data will already be present if the Session field was used in the Average function. For non multi-session data, this all can be ignored.

In any case, let's shift to the Cells pane. The Cells pane lists the cells in the data. One can delete, reorder, rename, add combined cells, append more cells, or export the information.

Generate a LAWU-LAWR difference wave by entering a weight of 1 by the first and a weight of -1 to the second.

	order	select	weights	names	types	target	
1	1	☐	0	LANF	SGL	0	LANF
2	2	☐	0	LANR	SGL	0	LANR
3	3	☐	0	LANU	SGL	0	LANU
4	4	☐	0	LAWF	SGL	0	LAWF
5	5	☐	-1	LAWR	SGL	0	LAWR
6	6	☐	1	LAWU	SGL	0	LAWU

Then click Add. Note that the new condition is marked as having the type CMB to denote that it is a combined cell and is therefore not to be treated as a separate cell in the succeeding analyses (it will be ignored when doing PCAs or windowing data for example). Note also that it is a virtual cell and functions much like the virtual subjects.

	order	select	weights	names	types	target	
1	1	☐	0	LANF	SGL	NaN	
2	2	☐	0	LANR	SGL	NaN	
3	3	☐	0	LANU	SGL	NaN	
4	4	☐	0	LAWF	SGL	NaN	
5	5	☐	0	LAWR	SGL	NaN	
6	6	☐	0	LAWU	SGL	NaN	
7	7	☐	0	+LAWU-LAWR	CMB	0	

Finally, rename the resulting new condition if you wish. Note that conditions must have unique names. In a session file, renaming a cell to have the same name as another cell results in their being combined. The LANU and LANR conditions are both non-words, differing only in which condition to which their sublexical characteristics are matched, so it makes sense to use a waveform collapsed across these two conditions when plotting them, so generate one now by giving them both a weight of one and then clicking on the Add button. Rename it LAN.

	order	select	weights	names	types	target	
1	1	☐	0	LANF	SGL	NaN	
2	2	☐	0	LANR	SGL	NaN	
3	3	☐	0	LANU	SGL	NaN	
4	4	☐	0	LAWF	SGL	NaN	
5	5	☐	0	LAWR	SGL	NaN	
6	6	☐	0	LAWU	SGL	NaN	
7	7	☐	0	+LAWU-LAWR	CMB	0	
8	8	☐	0	LAN	CMB	0	

A Grand popupmenu is also available for the Cells pane for average data. If the SGL values for the chosen columns are all the same length and they are fully crossed (e.g., AS, AT, VS, VT), then it will be assumed that they represent the values of an experimental design and average waveforms will be automatically generated for each main and interaction cell (e.g., A, V, S, and T). If some of the cells are selected using the Select column, then only that subset will be considered. The Trial Weights setting is respected. There is no need for this at present, but could be useful later on if you get a lot of effects and want to plot their corresponding waveforms. Choose “names” from the Grand popupmenu to see what this looks like (wait for it to complete process), then click Undo as all the extra cells would be cumbersome for now.

The Trials pane is available only for session files and so its button is grayed out. Aside from listing all the trials, their edit codes, and their trial specific data (like reaction time) if any, it provides an option to "Import" all the cell names from a text file, useful for epoch-marked Simple Binary files and a simple averaging function useful for taking a quick look at a session file's averaged waveforms.

The Channels pane provides similar functionality for channels. One can, for example, generate regional channel averages (Type “REG”) in the same manner. Sometimes there are also fiducial channels listed (marked “FID”). Fiducial channels are not data channels but rather standardized locations used for co-registration purposes. For BrainVision files, the impedance values will also be listed. You can resize the pane to allow all the columns to be shown.

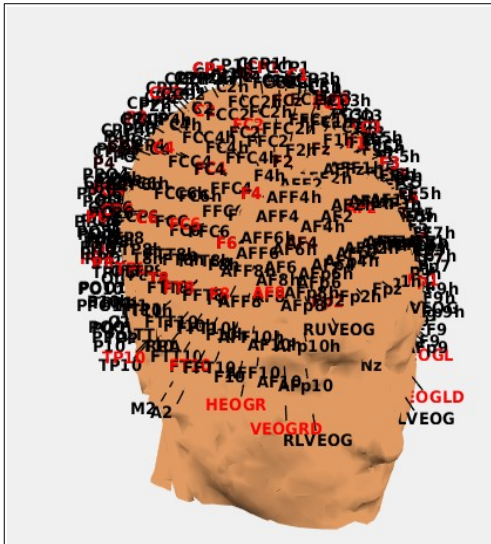
	order	select	weights	Name	Type	elocX
1	1	☐		0 Fp1	EEG	8.5960
2	2	☐		0 Fz	EEG	6.3807
3	3	☐		0 F3	EEG	6.0713
4	4	☐		0 F7	EEG	5.2972
5	5	☐		0 FT9	EEG	2.5427
6	6	☐		0 FC5	EEG	3.1354
7	7	☐		0 FC1	EEG	3.3240
8	8	☐		0 C3	EEG	-0.0404
9	9	☐		0 T7	EEG	-0.0404
10	10	☐		0 TP9	EEG	-2.6234
11	11	☐		0 CP5	EEG	-3.0785
12	12	☐		0 CP1	EEG	-3.4047
13	13	☐		0 Pz	EEG	-6.4615
14	14	☐		0 P3	EEG	-6.1521
15	15	☐		0 P7	EEG	-5.3780
16	16	☐		0 O1	EEG	-8.6768
17	17	☐		0 O2	EEG	-9.1212
18	18	☐		0 O2	EEG	-8.6768
19	19	☐		0 P4	EEG	-6.1521
20	20	☐		0 P8	EEG	-5.3780
21	21	☐		0 TP10	EEG	-2.6234
22	22	☐		0 CP6	EEG	-3.0785
23	23	☐		0 CP2	EEG	-3.4047

The Swap button provides a fix for the common error of swapping two electrodes during data collection. Simply select the two channels involved and then click Swap. The data will be swapped between the two channels, fixing the error.

This pane can also be used to specify electrode coordinates. The 3D coordinates elecX, elecY, and elecZ are listed and may be edited. The 10-05 column provides 10-05 channel names that can be chosen to replace the coordinates with canonical ones (transformed to be consistent with the rest of the existing coordinates for this dataset).

The “New CED” button has the same effect as the “Channel Coordinates” button on the Overview pane.

You can then click on the "Head" button to see the coordinates (if any) plotted on a 3D head, with the canonical locations labeled in black and the EEG coordinates in red. Fiducial locations will be displayed in green. Once satisfied, one can save the electrode coordinates in a new .ced (or .sfp for BESA) file using the "Save CED" button.



The Samples Pane provides some control over the temporal aspects of the data. You can trim the data by specifying a smaller time range. You can also change the nominal Sampling Rate or prestimulus time (doesn't change the data, just the labeling information). You can also resample the data to actually change the sampling rate via interpolation of the data. There is a button (1/2 Rate) to halve the sampling rate and another to double the rate or you can use the Resample Data field to do so with more control. The interpolation uses just the neighboring samples. For samples on the edge of the segment, it uses the edge sample as an estimate of the absent sample beyond the segment edge. Finally, there is an Append button if one needs to attach one set of data to another set of data with adjoining time points (e.g., 1-1000 ms and 1001-2000 ms). Click on 1/2 Rate twice to bring the sampling rate down to 250. Now that the artifact

correction is complete, the high sampling rate is no longer needed and the smaller dataset will be easier to work with.

The screenshot shows the 'Samples' pane of the EP Toolkit. At the top, there are tabs for Overview, Subjects, Cells, Trials, Channels, Samples (selected), Hz, and Events. Below these are sub-tabs: Factors, QC, PCA, Sessions, Specs, History, and Memory. The 'Specs' sub-tab is active, showing a table with columns 'Ms' and 'Samples'. The 'Ms' column has a value of -200, and the 'Samples' column has a value of 1. To the right of this table are two input fields: '1000' and '300'. On the left side of the pane, there are buttons for 'Undo', 'Done', 'Cancel', and 'New'. Below these are two buttons: '1/2 Rate' and '2x Rate'. At the bottom left is an 'Append' button. On the right side, there are three input fields labeled 'Nominal Sampling Rate (Hz)' (with value 250), 'Prestimulus period (msec)' (with value 200), and 'Resample Data (Hz)' (with value 250).

The Hz Pane provides similar functionality for frequency domain data. In this case it is grayed out.

The Events Pane provides a listing of the events in the file. The events have all been dropped due to the Drop Events option of the Averaging function as keeping all the events would make the dataset cumbersome. If events were listed, they would be seen to generally follow the FieldTrip convention in which a .type field provides a general level of information (such as whether the event represents a stimulus event or something else) and a .value field that provides more specific information such as which stimulus. In order to support the mff file format, keys fields may also be present. The keys field of an event may contain any number of subfields holding additional parameters and information.

The Factors and the PCA buttons are only for factor data and so are grayed out.

The next step is to determine if there are any bad sessions. The QC Pane provides quality control information by which problematic data can be detected. This information is generally only available if the EP Toolkit was used for the averaging process and the EP file format was used to store the results. Two controls in the lower left corner allow for numeric cells to be highlighted in red and/or green according to the specified comparisons.

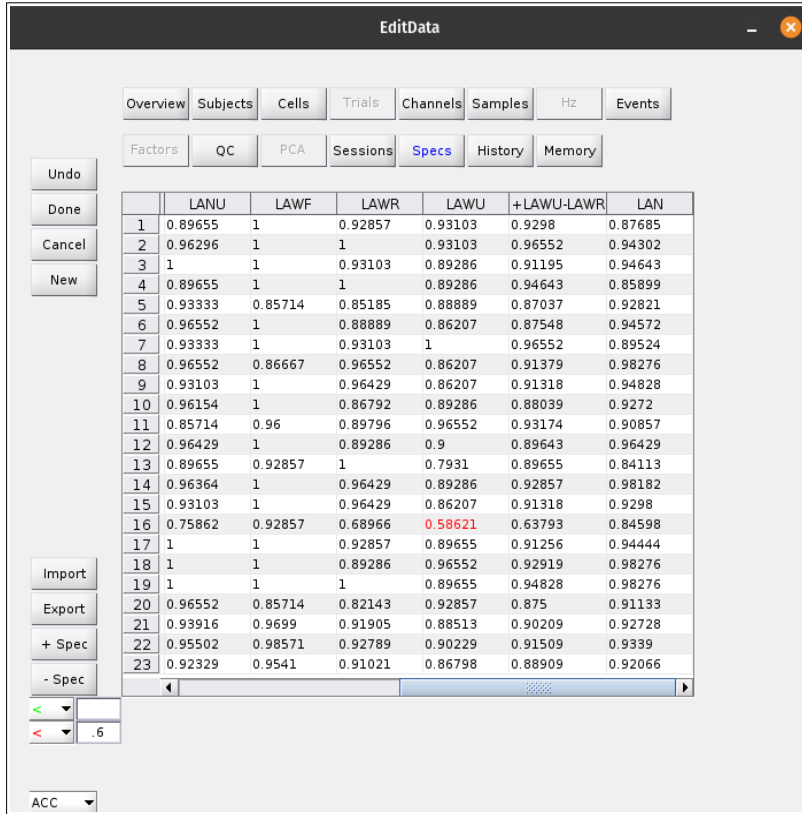
The first subpane (accessed by the buttons at the bottom of the pane) provides the number of trials going into each cell. Type 15 into the red control to indicate that we want to find any cells with less than 15 trials. Only the filler trials (LANF and LAWF), which are not analyzed, miss the threshold, so all is fine.

	select	names	type	LANF	LANR	LANU	
1	☐	sub001	AVG	15	24	25	14
2	☐	sub002	AVG	14	27	28	14
3	☐	sub003	AVG	12	24	26	14
4	☐	sub004	AVG	13	24	26	13
5	☐	sub005	AVG	12	27	27	13
6	☐	sub006	AVG	13	25	27	13
7	☐	sub007	AVG	12	24	27	13
8	☐	sub008	AVG	14	30	26	9
9	☐	sub009	AVG	15	25	25	13
10	☐	sub010	AVG	8	15	19	8
11	☐	sub011	AVG	15	27	26	13
12	☐	sub012	AVG	14	26	25	12
13	☐	sub013	AVG	13	22	25	13
14	☐	sub014	AVG	11	28	22	14
15	☐	sub015	AVG	14	26	28	14
16	☐	sub016	AVG	15	28	23	14
17	☐	sub017	AVG	14	27	29	15
18	☐	sub018	AVG	15	27	29	15
19	☐	sub019	AVG	15	27	27	15
20	☐	sub020	AVG	15	26	29	13
21	☐	f	GAV	137	256	267	13
22	☐	m	GAV	132	253	252	12
23	☐	grand	GAV	269	509	519	26

The Subs Subpane provides the number of subjects going into each average. The Blinks Subpane indicates the proportion of good trials that contained blinks, according to the artifact correction step. The Saccades Subpane does the same for saccades. The Sacc Ms Subpane provides the estimate of the latency of the saccades.

The Move Subpane provides the proportion of good trials with movement artifacts. The BadTrials Subpane provides the proportion of bad trials in the cell. The BadChans Subpane provides the same information on bad channels (either the number that are bad if any are present, or the number that were bad and were dropped during the averaging process, or otherwise the number that were replaced during the preprocessing. A label below the button will indicate which of the three is the case.). The Noise Subpane provides the magnitude of the noise in the cell, as estimated by the +/- reference (Schimmel, 1967) in which every other trial is inverted during averaging, resulting in an average in which the ERP has been cancelled out, leaving only noise. The numbers in the table are the root mean square (RMS) measures across all the channels and time points in those cells (meaning that the numbers were squared to eliminate negative signs, the mean was taken of the resulting squared numbers, and then the square root was taken of this final number in order to get back to the original metric, without any minus signs - basically just an elaborate way to get the average of the absolute values). Thus, larger numbers are higher levels of noise. This procedure is meaningless for frequency data. Outliers on this measure should be examined closely to determine if they are problematic. The numbers provided by these QC measures can allow for formal empirical criteria to be formulated and reported if it proves necessary to exclude bad subject data.

Next, click on the button at the top for the Specs pane, which allows one to examine the trial specifics of the data, such as the reaction time and accuracy information, if such is available. The popup menu in the lower left corner determines which spec is presented, with the rows being subjects and the columns being cells. Choose ACC to bring up the accuracy numbers. Type in .6 into the red field in the lower left corner to indicate that we wish to find any cells with an accuracy of less than 60%. One such (#16) is found. Click on its checkbox.



	LANU	LAWF	LAWR	LAWU	+LAWU-LAWR	LAN
1	0.89655	1	0.92857	0.93103	0.9298	0.87685
2	0.96296	1	1	0.93103	0.96552	0.94302
3	1	1	0.93103	0.89286	0.91195	0.94643
4	0.89655	1	1	0.89286	0.94643	0.85899
5	0.93333	0.85714	0.85185	0.88889	0.87037	0.92821
6	0.96552	1	0.88889	0.86207	0.87548	0.94572
7	0.93333	1	0.93103	1	0.96552	0.89524
8	0.96552	0.86667	0.96552	0.86207	0.91379	0.98276
9	0.93103	1	0.96429	0.86207	0.91318	0.94828
10	0.96154	1	0.86792	0.89286	0.88039	0.9272
11	0.85714	0.96	0.89796	0.96552	0.93174	0.90857
12	0.96429	1	0.89286	0.9	0.89643	0.96429
13	0.89655	0.92857	1	0.7931	0.89655	0.84113
14	0.96364	1	0.96429	0.89286	0.92857	0.98182
15	0.93103	1	0.96429	0.86207	0.91318	0.9298
16	0.75862	0.92857	0.68966	0.58621	0.63793	0.84598
17	1	1	0.92857	0.89655	0.91256	0.94444
18	1	1	0.89286	0.96552	0.92919	0.98276
19	1	1	1	0.89655	0.94828	0.98276
20	0.96552	0.85714	0.82143	0.92857	0.875	0.91133
21	0.93916	0.9699	0.91905	0.88513	0.90209	0.92728
22	0.95502	0.98571	0.92789	0.90229	0.91509	0.9339
23	0.92329	0.9541	0.91021	0.86798	0.88909	0.92066

Now, go to the Subjects Pane. The one that has been selected is marked here as well. Also, scroll to the last columns, where the signal detection statistics are listed. Make sure that all the good subject D-Prime values are at least 2 (they are). Click on the Delete button to remove the bad subject. Note how the virtual average entries update.

EditData

Overview **Subjects** Cells Trials Channels Samples Hz Events

Factors QC PCA **Sessions** Specs History Memory

Undo Done Cancel New Delete Add 1 Clear All Append Import Export + Spec - Spec Sort Grand -Cell +Cell

	order	select	weights	names	sessions	type	Ex
1	1	☐		0 sub001	n/a	AVG	AP
2	2	☐		0 sub002	n/a	AVG	AP
3	3	☐		0 sub003	n/a	AVG	AP
4	4	☐		0 sub004	n/a	AVG	AP
5	5	☐		0 sub005	n/a	AVG	AP
6	6	☐		0 sub006	n/a	AVG	AP
7	7	☐		0 sub007	n/a	AVG	AP
8	8	☐		0 sub008	n/a	AVG	AP
9	9	☐		0 sub009	n/a	AVG	AP
10	10	☐		0 sub010	n/a	AVG	AP
11	11	☐		0 sub011	n/a	AVG	AP
12	12	☐		0 sub012	n/a	AVG	AP
13	13	☐		0 sub013	n/a	AVG	AP
14	14	☐		0 sub014	n/a	AVG	AP
15	15	☐		0 sub015	n/a	AVG	AP
16	16	☒		1 sub016	n/a	AVG	AP
17	17	☐		0 sub017	n/a	AVG	AP
18	18	☐		0 sub018	n/a	AVG	AP
19	19	☐		0 sub019	n/a	AVG	AP
20	20	☐		0 sub020	n/a	AVG	AP
21	21	☐		0 f	n/a	GAV	AP
22	22	☐		0 m	n/a	GAV	AP
23	23	☐		0 grand	n/a	GAV	AP

The Sessions pane is used for multi-session data. By default, there will always be an n/a row that will list all the subjects. Use the Sessions pane to first define the available sessions by clicking on the Add button and editing their names. Then go to the Subjects pane and use the sessions column to indicate their sessions. Multi-session data should be the only time that a subject name is allowed to be present more than once (for each session). Below is what it might look like, if this had been multi-session data.

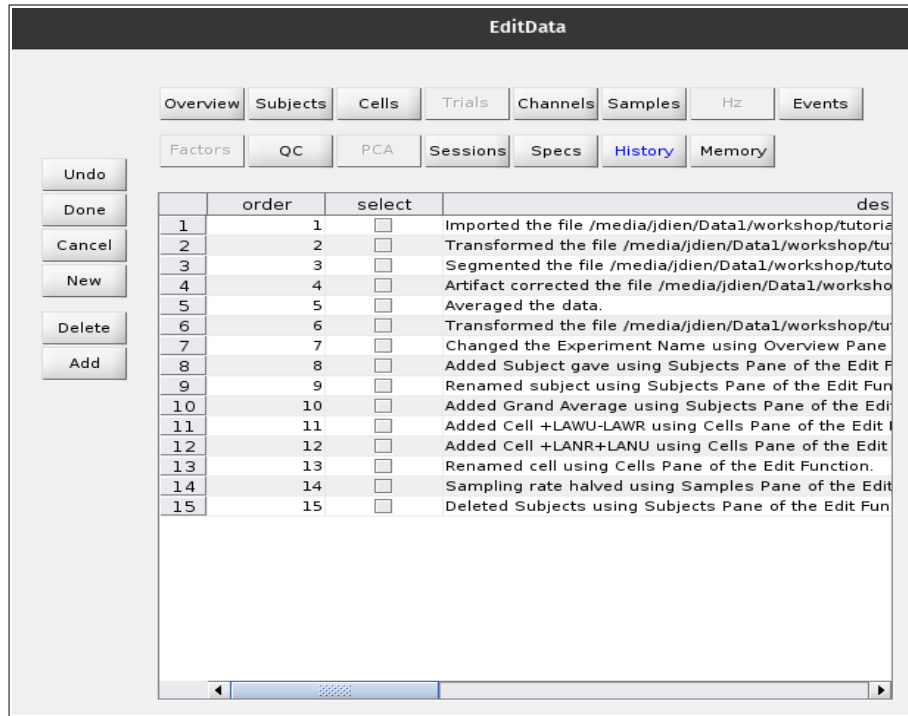
Overview **Subjects** Cells Trials Channels Samples Hz Events

Factors QC PCA **Sessions** Specs History Memory

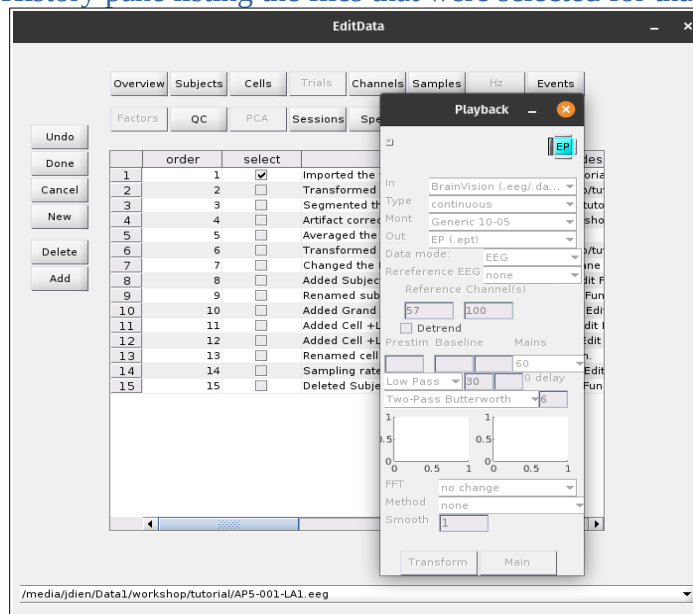
Undo Done Cancel New Delete Add Clear All

	order	select	sessions				
1	1	☐	n/a	sub001	sub002	sub003	sub004
2	2	☐	session 1	sub001			
3	3	☐	session 2	sub002	sub003	sub004	

The History pane provides an ongoing change history for the dataset. Here you can see what changes have been made, when they were made, and by whom (if the Records preferences have been set).



Furthermore, if you click on a checkbox, it will show you the exact settings used for that step. Try clicking on the first one. Note that this image is not functional as it is just a depiction of the settings. Additionally, a popup menu will appear at the bottom of the History pane listing the files that were selected for that operation, if any.



Finally, the Memory pane provides information about the size of various components of a dataset, which can be helpful if one is suffering from memory problems. It also allows one to delete some optional components to save space. As can be seen, while the noise information (one of the options of the Average function) can be helpful, it takes up as much space as the data itself.

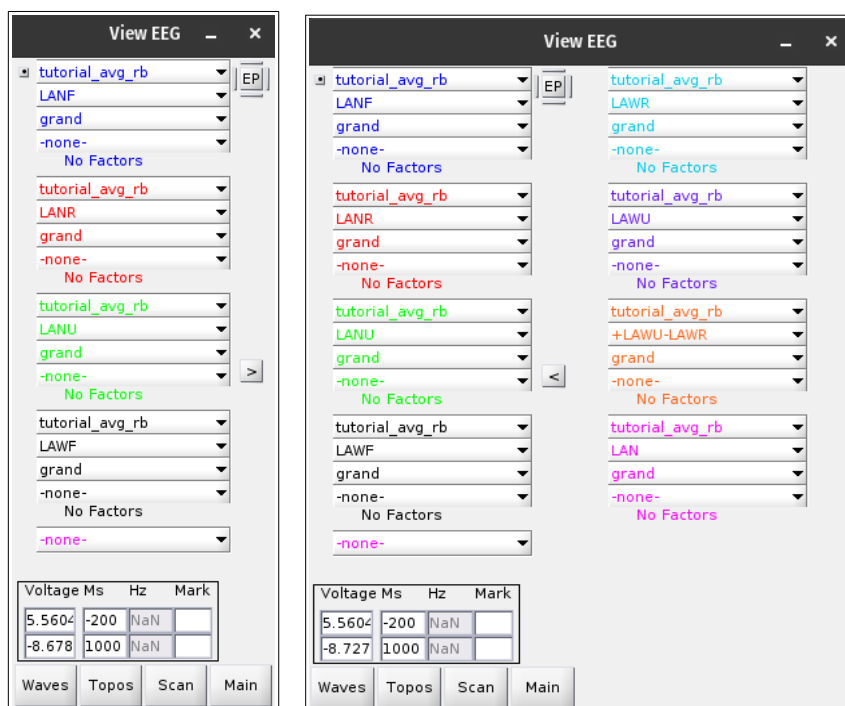


View Data

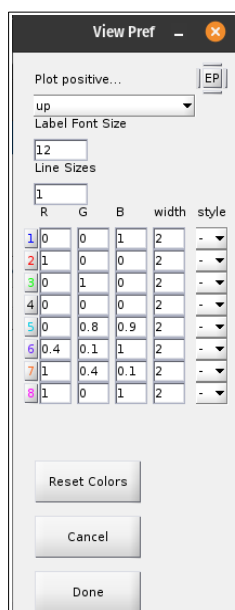
We can now proceed to view the data. Click on Done to return to the Read Pane of the Main Window (and to keep the changes you've made). If you changed the Data name, then it will ask you if you wish to make this be a new dataset in the working set or if you wish to rename and update the original dataset in the working set. Then click on Done to return to the Main Menu.

You may first want to go to the Preferences menu to set whether positive is up or down. The default is up. You can also set the line widths and font size of the channel labels (set to zero to omit entirely), which is helpful for making figures suitable for publication, but otherwise you probably want to leave the settings as is for now. The Topo Map Electrodes setting controls how and whether electrodes are plotted in the View Topos function.

Click on View to shift to the View EEG panel. This function allows one to view the waveforms directly. Up to four colors are available corresponding to four waves in the plots. For each color, there are controls indicating the data in the working set, the cell, the subject, the trial (or for averaged data various data range metrics), and the factor (where appropriate). An additional four colors can be made available if one clicks on the arrow button on the right edge of the pane.



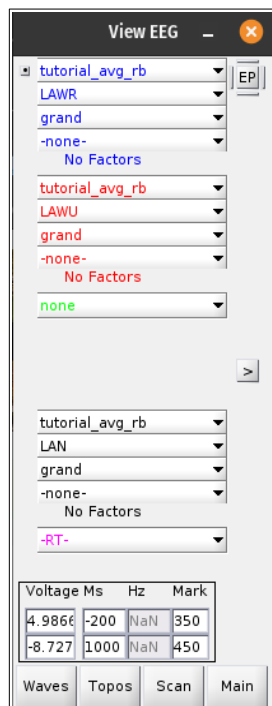
One can change the colors by using the shortcut to the View preferences in the upper left corner. Clicking on the numbers brings up the color pickers. These preferences specify RGB color, line width, and line type. The Reset Colors button changes them all back to the default settings. Once again, any changes you make are only temporary. To make them permanent, you would need to access these same controls via the Preferences menu from the Main pane and then use the Save button.



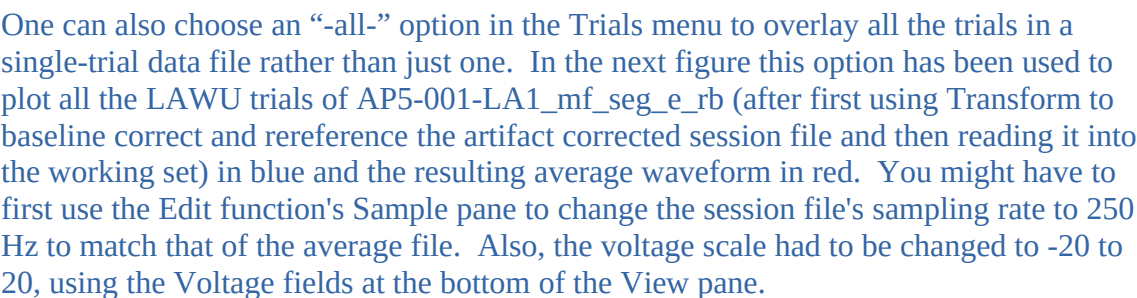
For datasets where there is event data like reaction time, one can also choose one or all of the event types to be marked on the figures using the magenta control. The event markings in this averaged dataset were dropped due to the Drop Events option in the Average function. If they were available and one was chosen, thick event bars would be added to the bottom of each waveform box, overlaying each other, color coded to correspond to the condition. If all events were chosen, they would not be distinguishable, so it is usually best to choose one of the event types (At each sample where an event of that type occurs, the thick event bar will spike up. They are scaled so that the event with the greatest incidence will have a height equal to 25% of the waveform box (even if that event type was not chosen for display). For single-trial data, there will usually just be one of each event type at the most so all the event spikes will be of this height. For averaged data, all the events from the contributing trials have been retained (if the EP Toolkit was used to conduct the averaging and the box was not checked) so the event spikes will provide a histogram of the distribution of the events. Since the event used for epoching will usually be the event type with the greatest incidence (one per trial), it will be scaled to be 25% of the waveform box and the other events will be proportional to it. If there is an RT field in the trial specs, then this can also be chosen as an event marking. There are three options for viewing the data, Waves, Topos, and Scan. Note that the maximum and minimum voltages across all four colors is listed at the bottom under Voltages and these numbers are updated as the settings are changed. These numbers will be used to scale the data plots.

One can also change these settings manually if desired. The Epoch numbers show the ms to be displayed (dropping non-overlapping points if different datasets with different sized epochs are to be displayed together). If the Voltage, Mx, or Hz settings are manually set, their value will be pinned (and will change to blue) and will not change until a dataset is changed or the field is cleared, in which case they will revert to automatic values again.

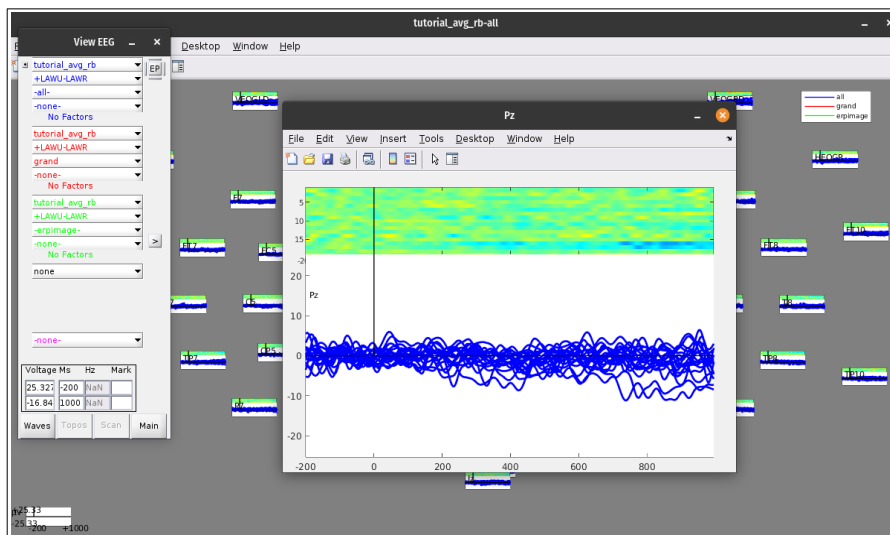
When one shifts to actually viewing the data by pressing the Waves or Topos buttons the scaling will be updated to include the effect of the Ms and Hz controls to provide a better fit (this is especially important for spectral data as the range of values can be quite different between frequency bands). If the Voltage controls were manually set then the scaling will not be changed.



The marker fields allow for optional marker lines to be included on the waveforms. Enter 350 and 450 to indicate that they should be marked at these times. Then click outside the box so that the numbers register. Also, choose LAWR for blue, LAWU for red, none for green, and LAN for black. Also -RT- for magenta. Then click Waves. After a pause the waveforms should be displayed in a new window. A legend box will appear in the upper right corner and a scale will appear in the lower left corner. If you click on the legend, it will let you edit the labels for publication purposes. An expanded view of a waveform can be obtained by clicking on it. Right-clicking on the expanded waveform figure allows one to add a legend. Just as with the main Waves figure, one can click on it to edit the legend labels.



An erpimage (pioneered by Scott Makeig's group) option is also available. Every erpimage will be placed at the top of the channel window with any waveforms below them. Here is the grand average +LAWU-LAWR difference wave in red, with the individual subject average difference waves in blue, and an erpimage (after sorting by subject D-Prime value using the Edit function). In this erpimage, each pixel line is a different subject and the voltages are color coded. Thus, you can see the blue and red streaks corresponding to the subjects that have sharp dips and peaks respectively. The -all- and -erpimage- options are also available for displaying cells, trials, and factors. This figure suggests that the N400 effect in this dataset was strongly reliant on two subjects with especially high D-Prime values; it may be that the effect is sensitive to the quality of the sample, and also that outliers will be a concern during statistical analysis.



For averaged voltage data, the trial menu instead provides several options for better understanding the variability of the data. -GFP- presents the global field power (Lehmann & Skrandies, 1984), basically the standard deviation of the voltage across all the channels at that time point, which provides a way of assessing the degree of ERP activity all over the head in a single waveform. This GFP waveform is present in every channel. I'm not a fan of the GFP because it encourages an overly serial view of ERPs but it can be helpful and is widely used.

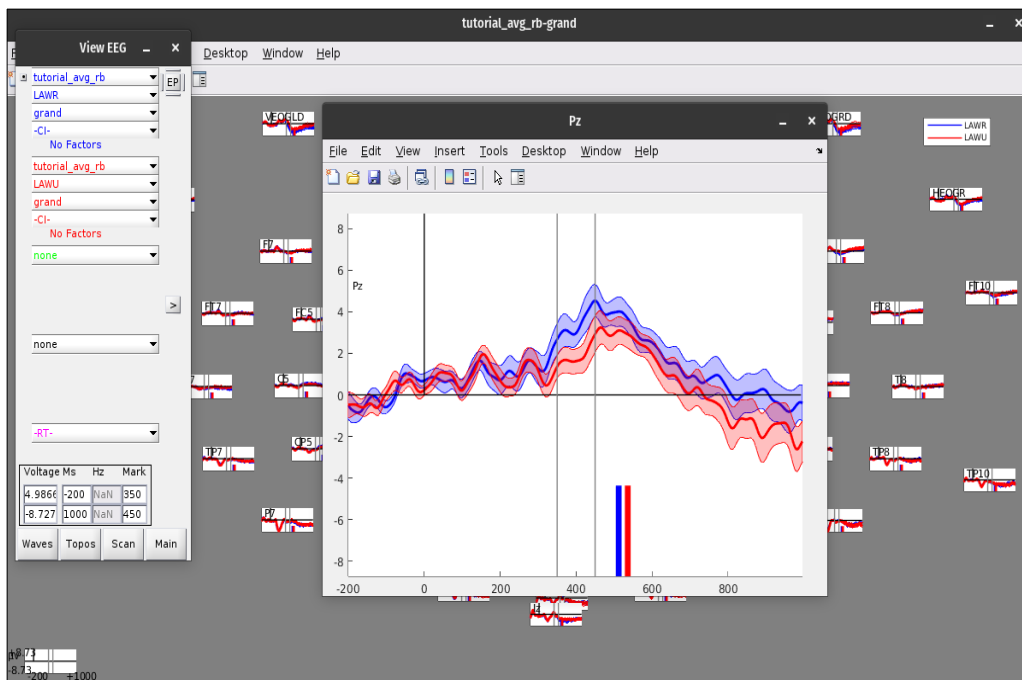
There are also three band options, each of which is presented as a band about the waveform. -noise- is generated during the Average function (unless Drop Noise is checked) using the +/- reference (Schimmel, 1967) and is computed by inverting every other trial, cancelling out the ERP components, and thus providing an estimate of the residual noise level. -Trl StDev- does the same with the trialwise standard deviation (i.e., the variability of the noise about the ERPs) computed during averaging (if a Trial SD option was chosen and only if the SD data were not subsequently lost by the use of an operation such as filtering). For grand average waveforms, the average across the subjects of this trialwise SD will likewise be available. '-Sub StDev-' is the subjectwise standard deviations (i.e., the variability of the subject ERPs around the grand average) is always available for averaged data unless the data were subsequently filtered.

-CI- does the same with an 84% confidence interval. If 95% CI bands from two conditions do not overlap then that time point is statistically significant, ignoring the multiple comparison problem, but this approach is overly conservative (Schenker & Gentleman, 2001). To achieve a .05 alpha from this test, assuming equal standard errors, 84% CI is needed (Payton et al., 2003). If the Statistics toolbox is not installed, the the EP Toolkit provides an estimate by using the coefficient of 1.44, which is appropriate for a sample size of $n=30$ when standard errors are equal.

In order to deal with repeated measures for grand average data, the Cousineau-Morey correction (Baguley, 2012; Cousineau, 2005; Morey, 2008) is applied. This correction is required because a basic CI would not reflect the benefit provided by a repeated measures test. A complication of using this correction is that it is reliant on the factor structure of the test. So for example, the correction would be different for two vs. four levels and for a single four-level factor versus two two-level factors (consider the degrees of freedom). Since specifying the factor structure is too complicated for the View>Waves interface, it simply assumes the case where there is one factor with a number of levels equal to all the visible waves that are not CMB type cells (e.g., difference waves) and that are from the same grand average. Thus, if there are four waves, two each from a male sample and a female sample, then the CIs would be appropriate for contrasting the two male cells with each other and for contrasting the two female cells with each other, but not for contrasting the male versus the female waves.

Also, keep in mind that these error bands merely provide a rough estimate (they also do not take multiple comparisons into account) and they assume circularity (Franz & Loftus, 2012), so formal testing is required for conclusions. In general, best to use the CI bands when only two waves are being plotted as there is no ambiguity about factor structure and also no assumption of circularity. Also, these bands are also not computed using robust statistics for speed reasons. The CI bands for subject average data are based on the averaged trials. The CI bands for grand average data are based on the averaged subject averages.

Compare the LAWR and LAWU conditions of the grand averages using the -CI- band options. Set marks at 350 and 450 to demark the range of the expected N400 effect. Choose the -RT- option for the events. Here we see that within the N400 window, the confidence interval bands of the averaged waveforms do not overlap, suggesting that they are significantly different. The reaction time bars are sufficiently similar that it is unlikely that a delayed P300 latency in the red LAWU condition could account for this difference.



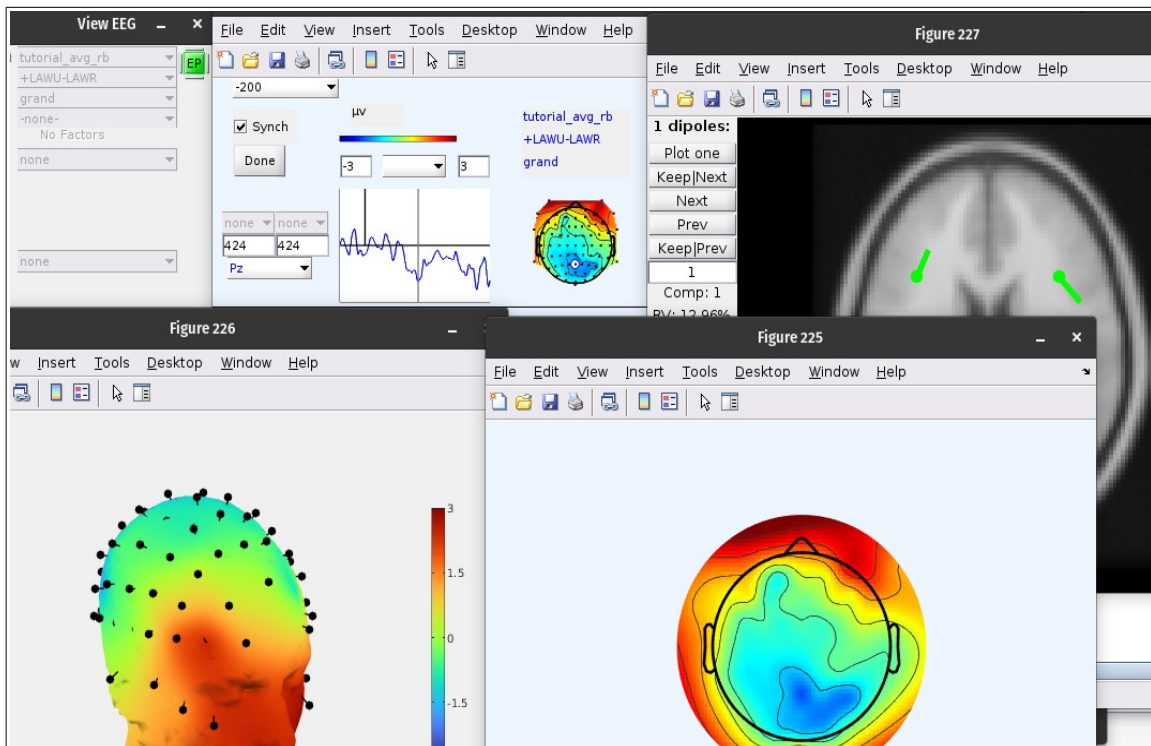
The Topos option provides an alternative viewing format in which the waveforms from one representative channel plus the scalp topography maps are provided (but the button will be grayed out if either the -all- or the -erpimage- options are chosen). Whatever dataset is listed first is used as the reference dataset. Set all but the blue color to “none”. For blue, set it to the tutorial average, choose the difference wave “+LAWU-LAWR” as the cell and “grand” for the subject. Delete the marks. The bands options like CI will not have an effect. Then click Topos. Doing so will allow us to examine the topography of the priming effect over time.

The waveforms for each line will be from the channel with the largest absolute voltages (in the reference dataset) and each topography plot will be from the time point with the largest absolute voltages (in the reference dataset). The gray line in the waveform figure indicates the time point corresponding to the topographical plots and the white dot in the topographical plots indicates the electrode corresponding to the waveform figure. You can change which channel and time point is being displayed using the controls on the left side. Every ERP component is a dipole, with both a negative and a positive pole, and so sometimes the “wrong” pole will be chosen. The channel popup menu for the row will have the channel with the most positive voltage highlighted in red and the channel with the most negative voltage in blue, to make it easy to switch the display to the other pole.

Each row presents a time point, starting with the first (-200 ms). The top row, set off with a black line, is the time point with the highest amplitude effect. One can edit the two boxes with the latency (424 ms in this case) to show other times. If the two numbers are different, say 350 and 450 respectively, then the row will display the mean voltage within this range. This can be done only with the first row.

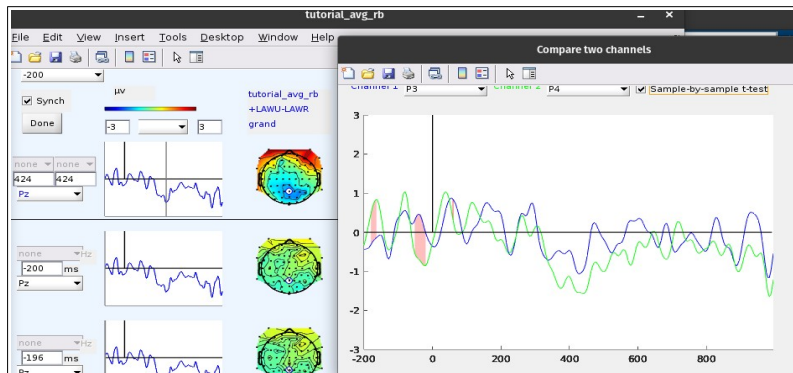
You may also change the channel by clicking directly on a scalp topography map. If you first click on the Synch checkbox then when you change the time or channel for one row, all of the rows will be changed to be the same. Black dots indicate the locations of the other channels but your click doesn't have to land precisely on the desired electrode location (the View Preferences pane has a setting for how the electrodes are mapped, if at all). You may change the voltage range of both the waveform figure and of the scalp topography maps by changing the numbers in the two boxes above the waveform figures. If you wish to make them symmetrical (e.g., +/-2) then you will find it easiest to use the popupmenu in between the two number boxes. Note that you will need to change the number in a box, press enter or click outside the box, and then wait until the page is updated (you can use the Activity Monitor/Task Manager utility to see when it is done as well).

If you choose -all- in the View pane for any of the datasets, then whatever you chose (e.g., a factor) will be listed on each row. If you did not choose anything, then if the datasets are ERP or TFT data then each row will be time points spaced by one sample. If the reference dataset is FFT data then each row will be frequency bands spaced by 1 Hz. If the frequency data is in raw form and hence complex, then every other row will be the imaginary component of the data. In these latter two cases, the first line will correspond to the peak latency/hz and will be set off with a gray line. To navigate to the other pages of plots, use the controls in the upper left corner (a set of buttons if there are a few or a popupmenu if there are many). They will be labeled according to the first row on each page.

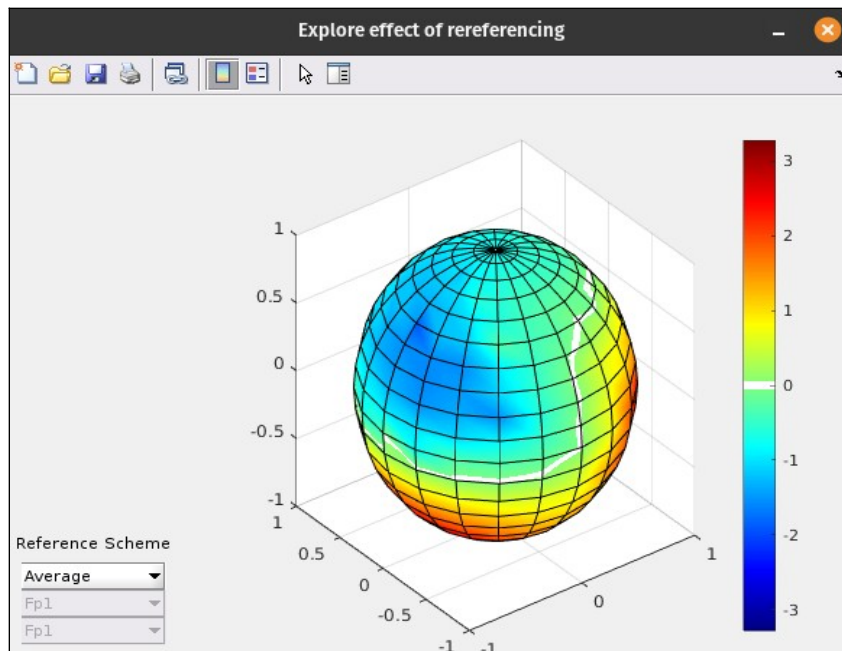


You may also right-click on the scalp topography plots to obtain: 1) an expanded 2D figure, 2) an expanded 3D head that may be rotated as needed, 3) an expanded 2D figure with the electrodes marked and named, 4) rescaling of the topo plots based on the minimum and maximum of the clicked topo plot, 5) 2-Channels, which provides overplotting two channels from the 2D plot chosen and which can provide a sample-by-sample non-parametric t-test if the option box is checked (based on trials for single-trial data and on subjects for averaged data), 6) Topo Grid, which presents the set of topos in the selected column starting with the selected row and extending to the end of the column, in a figure that can be dynamically resized to the desired configuration, 7) simple dipole analysis, 8) simple dipole analysis of jack-knife PCA results (for PCA results only). 9) illustration of effects of rereferencing. The former two are courtesy of EEGLab functions and the dipole analyses are courtesy of a FieldTrip function (using an EEGLab function to display the results).

While the priming effect looks quite lateralized, a quick use of the 2-Channels option suggests it is not statistically reliable.



Choosing the Rereference option will not change the dataset but will help you evaluate the effects of different rereferencing schemes on the effects of interest. Here we can see the initial head topography (approximated as a sphere).



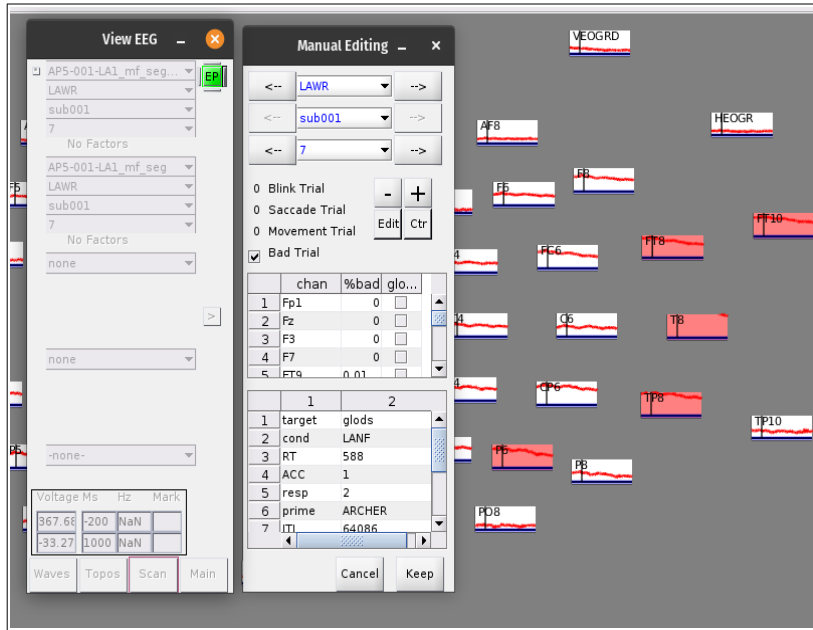
Clicking on the  symbol adds the color scale (white marks the zero voltage region). Since the data is average referenced, the reference menu starts with this option. Note how the sphere is roughly divided into half positive and half negative voltage regions, consistent with the biophysics of voltage fields in the head. In contrast, changing the settings to mean mastoids reference (roughly channels TP9 and TP10 in the tutorial dataset) results in a very distorted voltage distribution. The PARE average reference does make a notable difference for this vertically oriented dipolar field, suggesting that it might be worth considering further as an improvement on the conventional average reference, although it takes longer to compute. The “unknown” reference is for ones that are not recognized by this function and the option should be ignored.

The first time you use the 3D head, EEGLab will need to generate a "spline file" which it will automatically place in the same location as the .ced file. You will need to wait while this is done. It only needs to be done once per .ced file that you are using. The dipole analysis is a bit trickier. In order to allow the function to work on multiple platforms, FieldTrip compiles a binary file on the fly. To do so, it needs a gcc compiler to be available (see installation instructions). If this was not done, then this function will not work. The dipole analysis program will attempt to fit two symmetric hemispheric dipoles. If they end up running into each other then it will try again with only one midline dipole. Note that this is a very simple implementation of dipole source analysis and does not seem to provide results as good as that provided by the more refined BESA program. In particular, it seems to have problems with the two hemispheric dipoles running into each other and then exploding into arbitrarily high amplitudes as they interact with each other. BESA provides parameters which help keep the dipoles from doing this. Nonetheless, this implementation of source analysis can be suggestive,

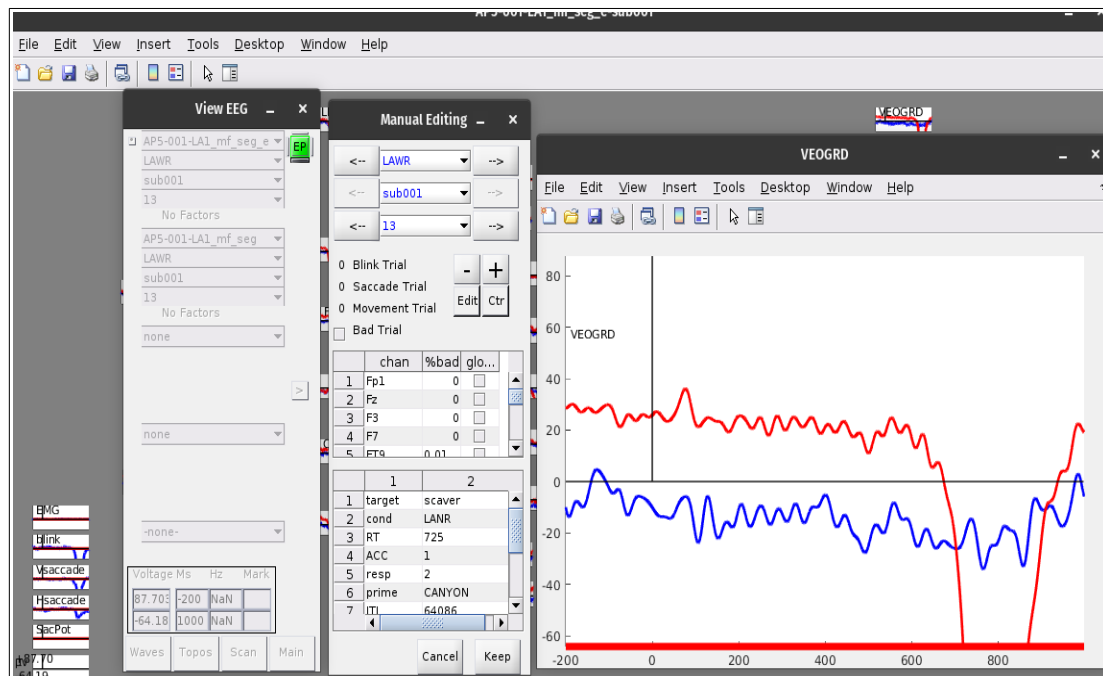
especially when the dipoles have not collided. Press the Done button to return control to the View pane.

The Scan button allows one to scan through data to perform manual editing of bad channels and trials. The interface will differ depending on whether the data are continuous or segmented. In the following screenshot the segmented blue dataset had undergone automatic bad channel/trial detection but no corrections were applied. The red dataset is the same data but after having undergone full automatic artifact correction (note, bad trials do not undergo channel replacement). Channels that were marked as being bad (but not replaced) in the first (blue) dataset are highlighted in red. One can turn this bad channel marking on and off by clicking on the channel if the Edit button is depressed, otherwise doing so will produce an expanded channel window. The Ctr button centers the display of each waveform on zero. The – and + buttons decrease and increase the scaling of the displayed waveforms by half. Bad Trials are indicated by a check mark in the Manual Editing window, and this too can be clicked on or off as desired. There is a table of all the channels that indicates the %age bad across all the good epochs for that subject and a checkbox that allows one to mark it as globally good or bad (will clear all bad channel markings if unchecked). Finally, there is a table that lists the trial specific information for this trial.

One can scan through the data by clicking on the appropriate arrows or choosing from the popup menu, for cell, subject, or trial. Once scanning is completed, one can either keep the manual edits by clicking on “keep” or discard them by clicking on “cancel”. Even if one keeps the manual edits, one will still need to Save the data to disk (see later section). These manual edits will then be corrected if one runs the file through the Preprocessing function with the editMode set to either Manual or Both. Here, we can see a trial where several channels were automatically marked bad, resulting in the entire trial being marked bad (check in Bad Trial box).

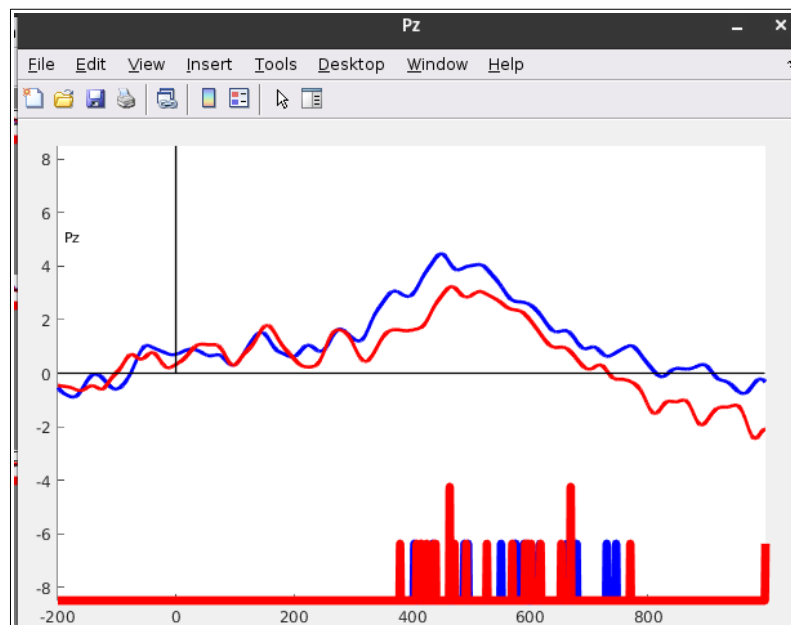


In the following screenshot of the AP5-001 single-trial data, it can be seen that the ICA blink correction has eliminated most of the blink for this trial, but a notable amount still remains, so it is still possible that if there is a differential distribution of blinks between the conditions, it could result in an apparent experimental effect.

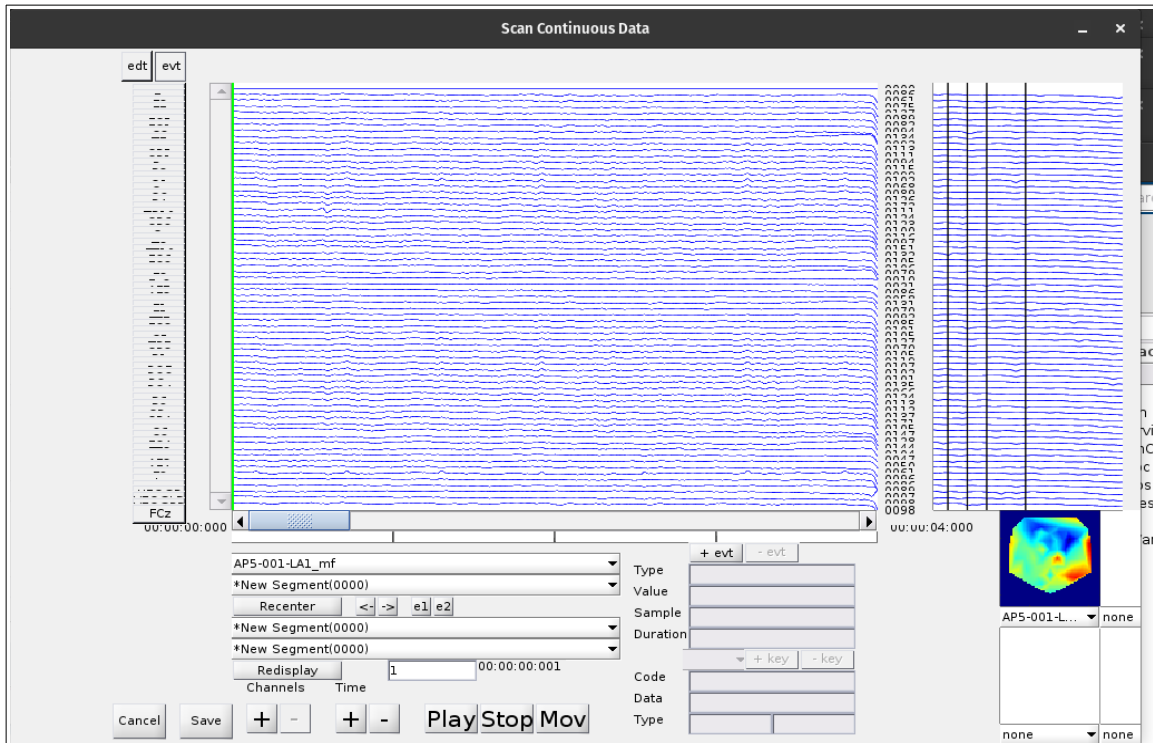


Once one averages the subject, one can then see the distribution of the blinks in the condition averages if one does not choose to drop the events. Here, one can see that there

is some overlap with the N400 effect, although topographical differences make it unlikely to be due to residual blink artifact. The timing is intriguing though, given that research suggests a tendency to blink after task processing is complete.



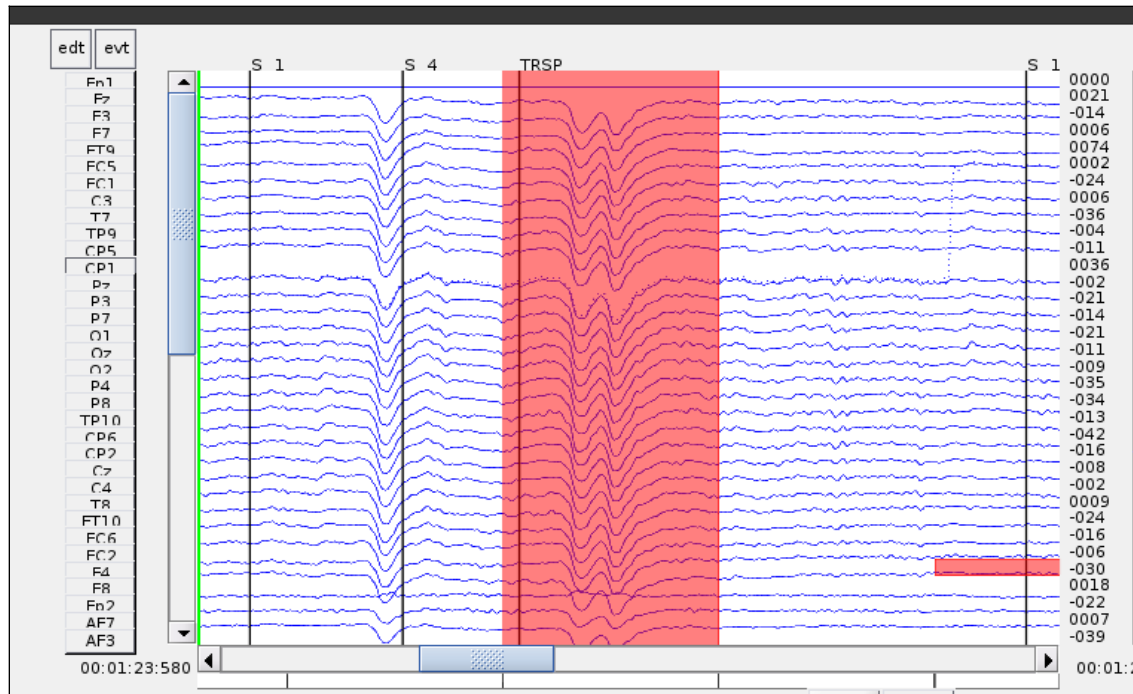
When the Scan function is utilized with a continuous dataset, such as the AP5-001-LA1_mf, then it will present it in the form of a graph plot.



This graph plot has the following features. The channels are presented in a scrollable window. The channel names are placed along the left boundary. As before, multiple datasets can be overplotted. Event labels are located along the top boundary and black lines indicate their latency and will correspond to the dataset specified in the first popupmenu below the graph figure. The display of the events can be turned off by clicking on the "evt" toggle in the top left of the figure. Doing so can be helpful if there are so many events that they are bogging things down. Next to the "evt" toggle is a "edt" toggle that can turn off the red bad data markings that can otherwise obscure the data. Discontinuities in the recording are marked with a red line. A green line marks the time marker. The time marker can be moved by right-clicking anywhere in the graph plot or manually using the field to the right of the Redisplay button. By default, a 2D plot provides the scalp topography at the moment of the time marker. There are four possible 2D plots, each of which can be set to one of the data colors, as well as eye-tracks or videos, if available in the data. Clicking on a 2D plot generates a 3D figure of it. Along the right boundary the voltage value for each channel at the moment of the time marker is provided (if the waves are not too closely packed). Along the bottom boundary below the scroll bar, markings delineate one-second segments. Along the right side of the screen, a chart provides the spectral graph of the one-second segment currently occupied by the time marker. It has vertical lines at 4, 8, 12, and 20 Hz.

As before, the Cancel button returns one to the main pane without keeping any bad data edits while the Keep button retains bad data edits. The Channels - and + buttons zoom out and zoom in the number of channels while the Time - and + buttons do the same for

the time axis. The Play button causes the waveforms to start automatically scrolling, animating the 2D topo plot, while the Stop button stops this process. The Mov button also starts animating the display but will additionally save a movie file of the 2D plot(s) once finished.



Bad data can be manually marked if the **edt** button is toggled on, which can then be corrected if the file is then submitted to the Preprocessing function with either the “both” or the “manual” options chosen. To toggle global bad channel status, click on the channel name, which dots the line. To toggle a bad channel in a single one-second segment, left-click on the channel row (in line with the channel name) within the one-second epoch delineated by the markings on the epoch bar below the horizontal scroll bar (for continuous data, the EP Toolkit divides it into one-second epochs for purposes of editing and spectral analysis). To mark the entire one-second epoch bad, left-click on the epoch bar itself.

One can specify the display period in terms of two events by selecting them the two popup menus above the controls. The upper popup menu indicates the event marking the start of the display period and the lower popup menu indicates the event marking the end of the display period. Events that fall within the current display period are marked with an asterisk. Once the events have been selected, click on the Redisplay button below the two popup menus and the display will adjust accordingly. The events in the two popup menus are listed in alphabetical order. In case there are multiple instances of an event code, then the ordinal rank of the event will be listed in parentheses.

Alternatively, there is another popup menu above them that lists all the events in chronological order. One can select an event of interest and then recenter the display on it by pressing the Recenter button. The time marker will be placed at the sample of the selected event. The information fields associated with that event are listed to the right of these controls and can be edited. One can also shift the time marker to the prior or succeeding event by clicking on the <- and the -> buttons. One can also set the upper or the lower popupmenu to be the same as that of the top Recenter popupmenu by pressing the e1 and the e2 buttons respectively.

Principal Components Analysis (PCA)

The next step is to obtain a PCA of the data. I typically recommend using a two-step sequential PCA (Spencer et al., 1999, 2001) in which the first step is a temporal Promax rotation and the second step is a spatial Infomax (ICA) rotation (Dien, 2010). Tutorials are available that discuss the principles of applying PCA to ERPs (Dien & Frishkoff, 2005) and the decisions that need to be made (Dien, 2012).

First click on Main to return to the Main Pane. Then click on PCA. I generally recommend using Promax with a covariance relationship matrix (Kayser & Tenke, 2003) and Kaiser weighting (Dien et al., 2005) for this first step. The kappa for the Promax is generally set at 3 but does not make much difference (Dien, 2010). For Promax, one may choose to use either the SAS or SPSS algorithms (SAS by default), although their results seem largely identical (note that default kappa for SAS is 3 and for SPSS is 4).

For the decomposition procedure, the default is singular value decomposition (SVD) but one can also choose non-linear iterative partial least squares or NIPALS (Wold, 1975), which is a newer alternative that is sometimes said to be more efficient and to produce comparable or even more accurate results. I haven't finished my evaluation of it, but it does appear to be promising.

PCA

Mode EP

temporal

Rotation

Promax

kappa 3 Algorithm SAS

Decomposition

SVD

Relationships covariance Loadings Kaiser

☐ Parametric Analysis

Factors (0=screen) 0

Title of PCA

1 AP5-005-LA1

2 AP5-001-LA1_mf_seg

3 *tutorial_avg_rb

4 AP5-001-LA1_mf_seg

5 AP5-001-LA1_mf_seg

6 AP5-001-LA1_mf_seg

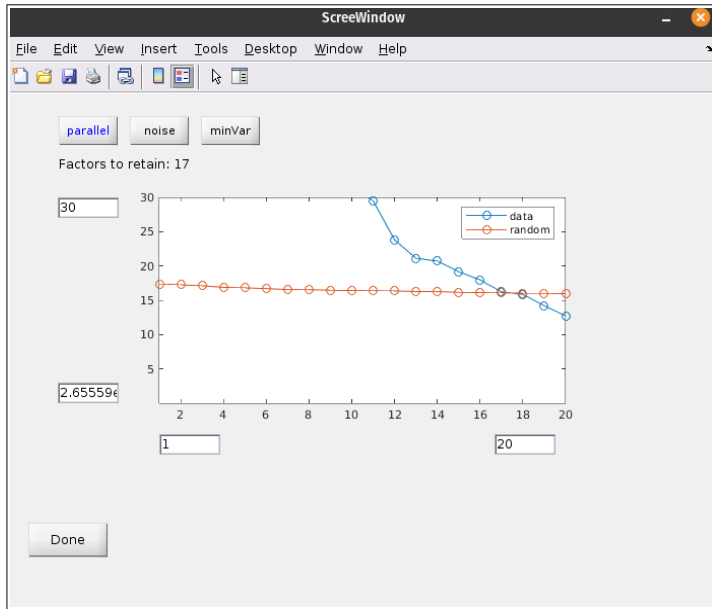
7 *AP5_avg_noDropFive

Main

You will need to determine how many factors to retain for the initial step. Unfortunately, the question of how to best proceed remains unsettled at this point. Indeed, one paper concluded it is best to keep all the factors (Kayser & Tenke, 2003). At this point I find the data insufficient to convince me to change my analysis procedures, given the potential drawbacks of such an approach such as multiple comparison problems (Dien, 2006), but agree that it is a reasonable position and the data provided are intriguing. I therefore consider the question open and in need of further investigation. Leave the # Factors as zero to indicate that a Scree chart is needed. Click on the name of the dataset to begin.

A Scree plot (Cattell, 1966; Cattell & Jaspers, 1967) is now displayed. The normal criterion is to choose the elbow but unfortunately, at least with ERP data, there are typically multiple elbows. The initial plot suggests 2 factors, for example, but if one expands the scaling then other elbows appear. Also whether one should choose the elbow or the point before it has been muddled.

Due to these ambiguities, I currently suggest the use of a parallel test (Horn, 1965), which compares the Scree of the dataset to that obtained from a fully random dataset. The buttons at the top allow for parallel, noise, and minimum variance criteria. The display automatically calculates the number of factors to retain. You can also examine the chart to manually determine the value. Adjust the y-scaling (on the left) to 20 to reveal the point at which the blue line intersects with the red line. In this case, 17 factors is the number indicated (the last point above the line).



The noise subpane displays the same scree chart but with a yellow line representing the Scree plot of the noise data produced by the +/- reference (Schimmel, 1967). While I once recommended using it as the comparison Scree (Dien, 1998a), I no longer recommend its use as an effective PCA needs to represent coherent noise sources as well as the ERP components and so the random noise dataset is a more appropriate comparison. I am currently working on a more systematic treatment of these issues.

For completeness sake, the minimum %age accounted criterion is also available. Once finished, use the Done button in the lower left hand corner.

Set the # of factors to 17 and provide a title for the PCA, as in "tAP5" and then click on the name of the dataset. It will first announce that it is computing jack-knife PCA Loadings. Basically, it is computing a set of PCAs in which each of the subjects in turn is let out of the sample, in order that one might later on determine how stable the PCA solution is to changes in the composition of the sample (see the Topos pane section of the tutorial for a further explanation). The PCA solution should be shortly added to the list of active datasets. It will have a star in front of the name to denote that it hasn't been saved yet.

PCA

Mode: EP

Rotation: temporal

Promax

kappa: 3

Algorithm: SAS

Decomposition: SVD

Relationships: covariance

Loadings: Kaiser

☐ Parametric Analysis

Factors (0=scree): 13

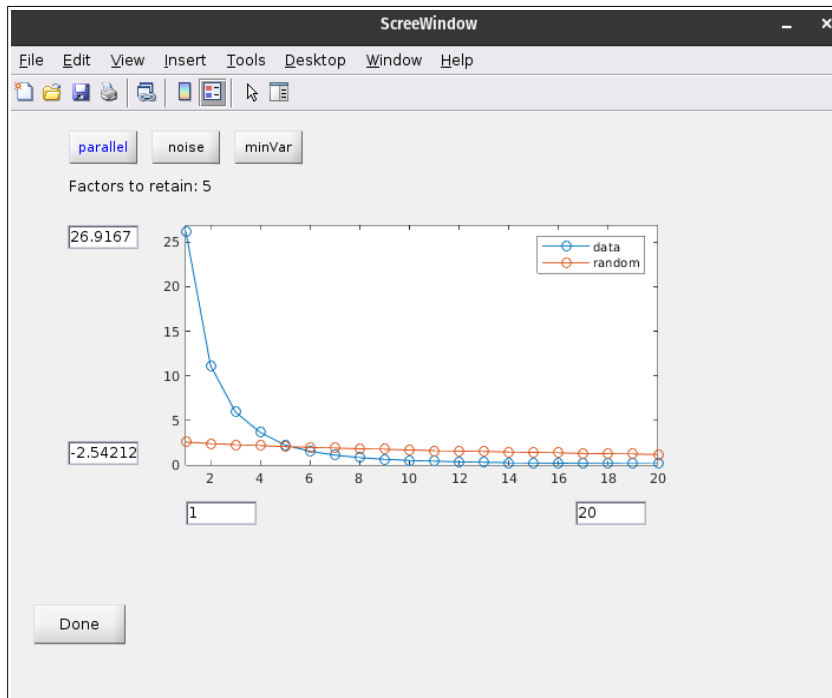
Title of PCA: tAP5

5	AP5-001-LA1_mr_seg
6	AP5-001-LA1_mf_seg
7	*AP5_avg_noDropEve
8	AP5-001-LA1_mf_seg
9	AP5-001-LA1_mf_seg
10	AP5-001-LA1_mf
11	*tAP5

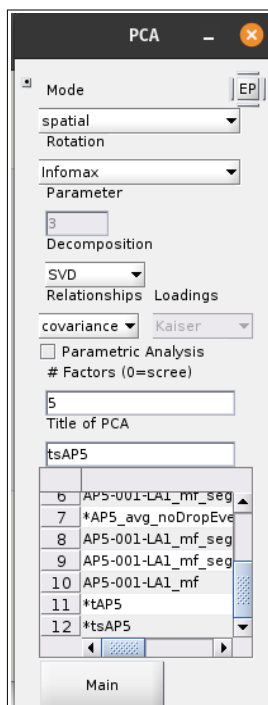
Main

Next, set it up for the second step by changing the settings to that of a spatial Infomax rotation. For Infomax, the rotation parameter and loading weighting does not apply. Make sure the number of factors is set at zero. You might as well go ahead and change the name to tsAP5. Now click on the temporal PCA dataset (tAP5).

The EP Toolkit generates a separate spatial PCA for each individual temporal factor, based on the argument that, for a given temporal factor, the other factors have nothing to contribute and can only distort the results (Dien, Spencer, et al., 2003). This means that in the current example seventeen separate spatial PCAs are run. Currently, for simplicity's sake the EP Toolkit retains the same number of factors for each of the spatial PCAs. The scree chart is the average of all seventeen scree charts. This is obviously an oversimplification and the intention is to address this issue in a future update. In this case, the chart suggests the five factors should be retained in this second step.



Enter five for the # of factors and then click on the name of the temporal PCA (tAP5).



At this point (with an earlier version) there was a memory fragmentation message:

??? Error using ==> zeros

Out of memory. Type HELP MEMORY for your options.

Error in ==> ep_PCAoutput at 434

```
reconFacs = zeros(numChans, numPoints,numOutputCells,numSubs,NUM_FAC);
```

Error in ==> ep>pickPCAdat at 3715

```
[PCAoutput, peakLatency, peakSamp, peakChan] = ep_PCAoutput(FactorResults,  
cnames, cellcoll,  
exclChan, parametricData.data, parametricData.colheaders);
```

??? Error while evaluating uitable CellSelectionCallback

I therefore quit out of MATLAB without bothering to quit out of the EP Toolkit itself by going to MATLAB's File menu and choosing Quit. I then changed the current directory back to the one I had been using and typed ep at the command line. The EPwork directory retained the datasets that were being analyzed and so all that was needed was to go back to the PCA Pane and try the spatial step again as if nothing had happened. This time it worked fine. To purge a dataset from the working set, return to the Read Pane and click on the name of the dataset.

One can also cross-verify a PCA solution by applying the factor scoring coefficients directly to a different dataset, as long as it has the same number of relevant variables (timepoints for temporal, channels for spatial, bins for frequency). Cross-verification can be helpful if one wishes to rule out the possibility that differences in the factor structure of two separate PCAs were due to variations in the rotational process rather than differences in the latent structure of the datasets (e.g., Simons et al., 2001).

It is important to understand that this procedure is not intended to allow one to apply a factor solution to entirely different datasets per se. A factor solution is specific to a dataset. For example, factor scoring coefficients reflect not just what is needed to measure a latent variable (e.g., ERP component) but also what is needed to disentangle it from other overlapping latent variables. So if both datasets had exactly the same P300 but only the first also had an N400, then applying the PCA results to the second dataset would be invalid. It would only be valid to apply the factor scoring coefficients from the PCA of one dataset to another dataset if the second dataset had exactly the same latent variables (e.g., same ERP components with exactly the same time course for a temporal PCA, differing only in amplitudes) and the same inter-factor correlations. This can rarely be said to be the case.

The sole intent is to allow one to compare two different factor solutions on the same dataset to quantitatively determine how similar they are (correlate the two sets of factor scores). I have chosen to call this procedure "cross-verification" to emphasize this point. To the extent that the latent structure of the second dataset varies from the first, you will also see the factor loading patterns (e.g., the waveforms in a temporal PCA) differ. Even

if you apply it to the original data, you will see some minor differences due to rounding errors.

If you were to do this, you would select cross-verification from the Mode popupmenu. Then click on one of the PCA solutions in the upper table (so that it's name is listed as being "chosen" underneath the table). Then click on a dataset in the lower table (only compatible datasets will be displayed there) and the same PCA solution will be applied to the new dataset. If the PCA was a two-step PCA then both steps will be applied to the new dataset.

Click on Main and then use the Edit function to examine the temporo-spatial PCA data, you will find that the Factors and the PCA panes are now active. The PCA Pane for the temporo-spatial PCA shows the basic results for the PCA. The Summary subpane displays the basic parameters of the PCA. Note how the second step accounted for less total variance (75% vs. 93%) since additional noise variance was dropped from each of the spatial PCAs conducted on each of the temporal factors. The variance accounted for by each spatial PCA of each temporal PCA is provided in the table. Factor Variance is how much of the temporal factor was retained and Total Var is how much of the total original variance was retained by that second step.

The screenshot shows the 'EditData' dialog box with the 'PCA' tab selected. The dialog is divided into two main sections: 'First PCA' and 'Second PCA'. The 'First PCA' section shows parameters for a temporal PCA, and the 'Second PCA' section shows parameters for a spatial PCA. A table on the right displays the results of the PCA, showing Factor Var and Total Var for 10 factors. The 'Summary' subpane is also visible at the bottom.

	Factor Var	Total Var
1	0.7820	0.3003
2	0.8012	0.1245
3	0.8309	0.0750
4	0.8196	0.0475
5	0.8210	0.0413
6	0.8313	0.0407
7	0.8367	0.0157
8	0.8823	0.0154
9	0.8789	0.0143
10	0.8677	0.0135

The factor matrices are also available for inspection. For example, the FacPat Subpane displays the factor pattern matrix (one type of factor loading) from the initial temporal PCA step. The FacPatST displays it from the second spatial step. FacStr is the factor structure matrix, FacScr is the factor scores, FacCof is the factor scoring coefficients, and FacCor is the factor correlations. I don't have time to provide tutorials on factor analysis by e-mail but you're welcome to read my tutorial chapter (Dien & Frishkoff, 2005) or to consult books on factor analysis (Gorsuch, 1983; Tabachnick & Fidell, 1989). Note that there is normally no reason to export the factor scores. As I have argued elsewhere (Dien, 2012), it is much more useful to view the factors in the form of factor waveforms (the portion of the ERPs accounted for by each factor). These factors are in microvolt scaling and can therefore be directly compared to the original data. If you wish to generate a figure with all the microvolt-scaled temporal factor loadings, a ScFacPat subpane is also available. Just use the Export button and then graph them in Excel or whatever. The PCA waveforms in the working set (in this case tAP5 and tsAP5) have the full factor score information so one can derive the necessary numbers to run ANOVAs from them (see the remainder of the tutorial).

The Factors pane provides information on the individual factors. Peaklatency is the time point with the greatest absolute voltage (out of all the cells after computing the grand average). PeakNegChan is the channel with the most negative voltage (out of all the cells after computing the grand average) and PeakPosChan is the channel with the most positive voltage. PeakPolarity is whether the negative or the positive peak channel had the greater absolute voltage. Variance is the %age of the overall variance accounted for by the factor. Unique Variance is the %age variance that is uniquely accounted for by the factor (they are the same for orthogonal rotations where factors are not correlated).

Notice that the factors from a two-step PCA are organized with the factors derived from the same first step factor clumped together (so the first five of the example are the five spatial factors derived from the first temporal factor).

EditData						
OverviewSubjectsCellsTrialsChannelsSamplesHzEvents FactorsQCPCASessionsSpecsHistoryMemory						
Undo Done Cancel New Delete Add 0.00 Clear All Append Export						
	order	select	weights	names	type	peakLatency
1	1	☐		0 TF01SF1	SGL	840
2	2	☐		0 TF01SF2	SGL	840
3	3	☐		0 TF01SF3	SGL	840
4	4	☐		0 TF01SF4	SGL	840
5	5	☐		0 TF01SF5	SGL	840
6	6	☐		0 TF02SF1	SGL	504
7	7	☐		0 TF02SF2	SGL	504
8	8	☐		0 TF02SF3	SGL	504
9	9	☐		0 TF02SF4	SGL	504
10	10	☐		0 TF02SF5	SGL	504
11	11	☐		0 TF03SF1	SGL	196
12	12	☐		0 TF03SF2	SGL	196
13	13	☐		0 TF03SF3	SGL	196
14	14	☐		0 TF03SF4	SGL	196
15	15	☐		0 TF03SF5	SGL	196
16	16	☐		0 TF04SF1	SGL	648
17	17	☐		0 TF04SF2	SGL	648
18	18	☐		0 TF04SF3	SGL	648
19	19	☐		0 TF04SF4	SGL	648
20	20	☐		0 TF04SF5	SGL	648
21	21	☐		0 TF05SF1	SGL	416
22	22	☐		0 TF05SF2	SGL	416
23	23	☐		0 TF05SF3	SGL	416

One can also view the factors directly by going to the View Pane from the Main Pane of the Main Window. One could, for example, examine what portion of the grand average is accounted for by TFSF1 (the first spatial factor formed from the first temporal factor) by overlaying them using the Waves function.

View EEG

tutorial_avg_rb

LAWR

grand

-none-

No Factors

tutorial_avg_rb

LAWU

grand

-none-

No Factors

tsAPS

LAWR

grand average

-none-

TF01SF1

tsAPS

LAWU

grand average

-none-

TF01SF1

-none-

Voltage Ms Hz Mark

4.9866 200 NaN

8.727 1000 NaN

Waves

Topos

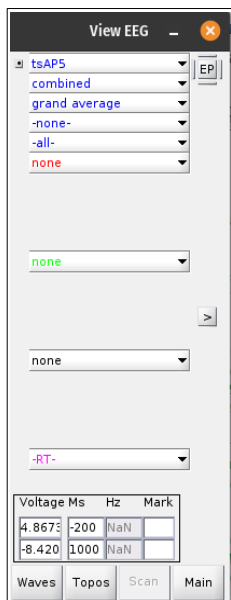
Scan

Main

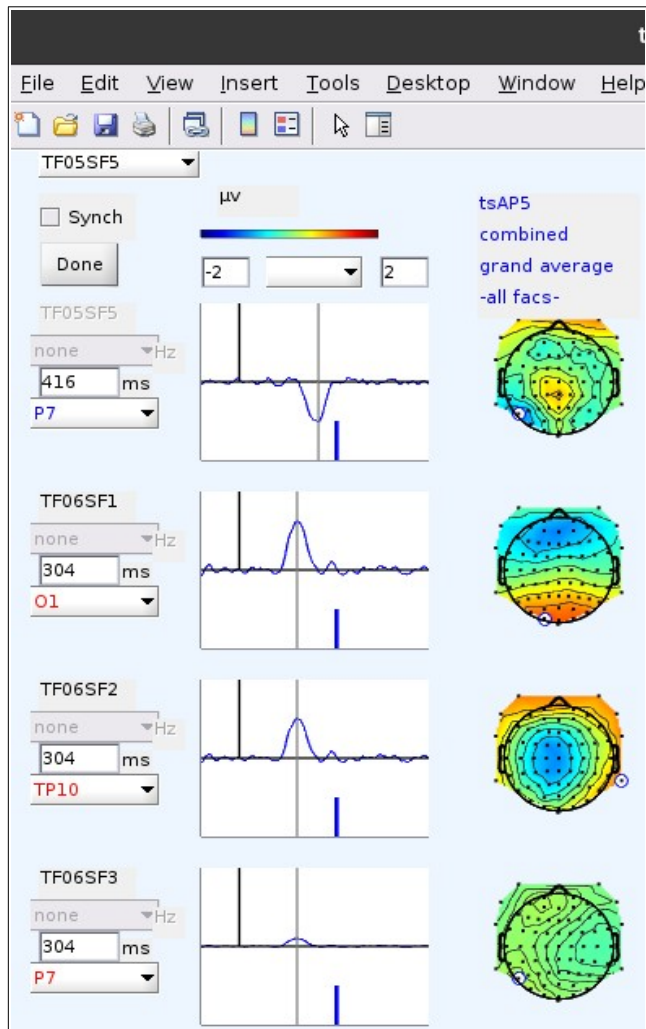
An expanded view of channel Fz (double click on it) reveals:

Potential (p07) with a latency of around 250 ms may also appear (Rudell, 1991). Each of these should be defined in advance in terms of their latency, peak channel, and polarity (e.g., N400 is a negativity peaking at 400 ms, with a peak channel near Pz). Note, though, that since the tutorial data is comprised of only a small subset of a complete study, we should expect that the quality of the PCA results should be impaired.

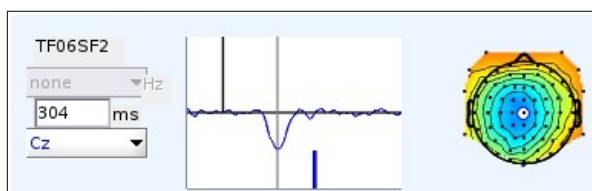
These definitions should be reported in any published report in the Methods section. Since the goal is not to test the researcher's guessing ability, it is permissible (in my opinion) to later add an ERP component to this list, but it would need to be just as concretely specified so that reviewers can evaluate whether the literature supports its *a priori* status. The -RT- option is also being selected as part of the P300 definition is that it should precede the response (as it reflects the completion of decision-making). Finally, the cells should be set as "combined", which is the average of all the cells. This should be done so that the identification of a priori factors can proceed blind to effects, which could otherwise cause the researcher to inadvertently capitalize on chance fluctuations.



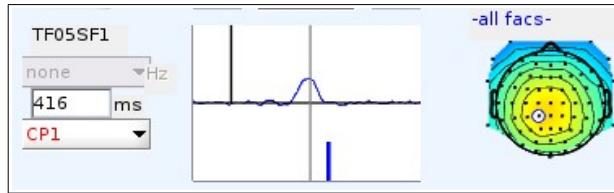
Press the Topos button. Plotting so much data will take some time so be patient. Only a few factors (depending on the monitor size) will be displayed on a page. The control at the top of the screen allows one to move between the pages of factors. The initial scaling left the color scheme too washed out so I changed the scale to -2 to +2 microvolts. In the following screenshot, TF06SF02 (the second spatial factor derived from the sixth temporal factor) had a peak latency of 304 ms. It is being initially presented with its absolute peak channel at TP10, its positive pole.



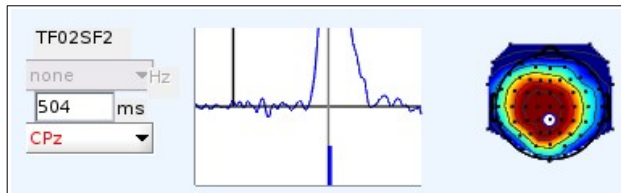
We can shift the plot to the negative pole by clicking on the channel popup menu and selecting the blue channel name, which is Cz. This is the factor most likely to correspond to the N400. One reservation is that it has a 304 ms latency rather than 400 ms. One possibility is that in this dataset the P300 had a very similar latency as the N400 and hence the PCA was unable to separate them as they were similar in both timing and topography (the RT line is consistent with this hypothesis). In that case, the TF06SF02 may reflect the early part of the N400, with the rest being subsumed into the P300 factor.



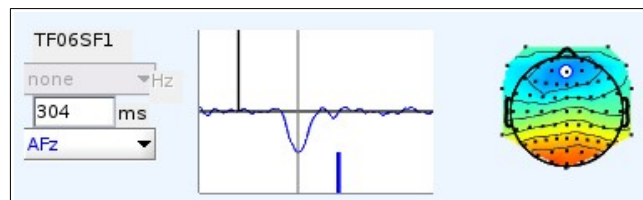
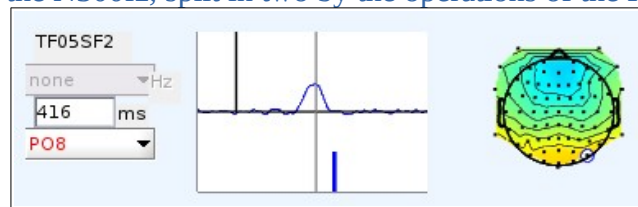
By this account, the combined P300/N400 would likely be the TF5SF1:



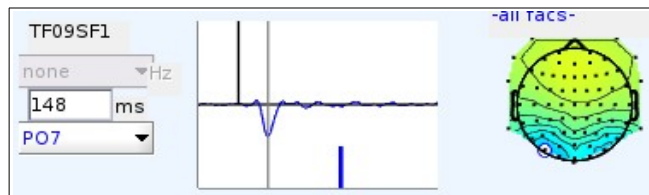
The pure, late P300 factor would then be TF02SF2:



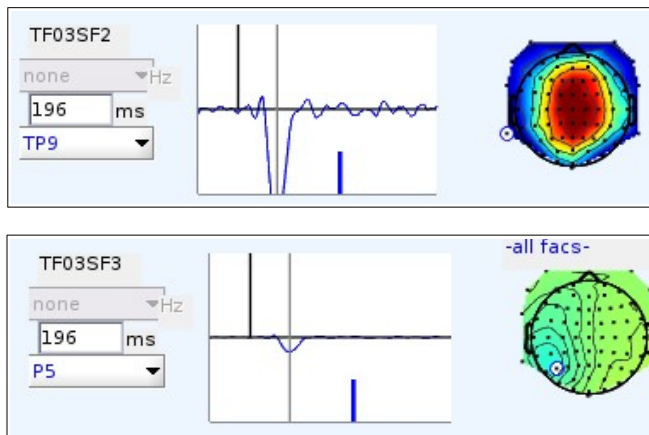
The N300fz candidates are the TF6SF1 and the TF5SF2. It is possible that they are both the N300fz, split in two by the operations of the PCA.



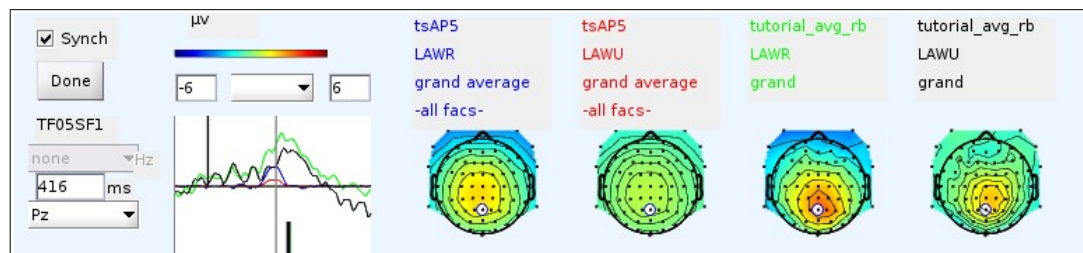
The N170 candidate is TF9SF1:



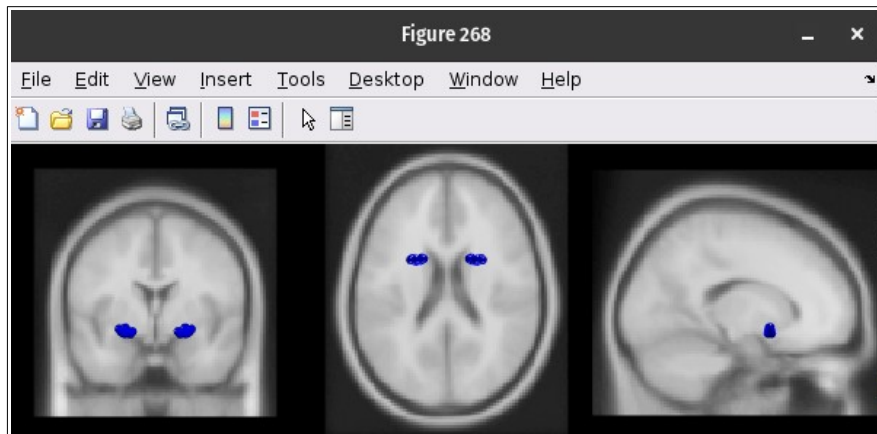
The Recognition Potential candidates are TF3SF2 and TF3SF3.



Now that we have established the a priori factors, we may now examine the data in a more exploratory fashion and indulge our curiosity. Looking at the P300/N400 (TF5SF1) factor overplotted on the grand average waveforms, we can see that it does indeed appear to be accounting for the N400 effect.



We can look at this factor close with the jack-knife option; what happens is that every time one does a PCA, a jack-knife analysis is automatically conducted in which a set of PCAs are computed in which one subject is left out (so twenty subjects results in twenty PCAs, each with nineteen of the subjects). For two-step PCAs, the jack-knife analysis is only conducted on the second step. The factor loadings from this are stored for later use. When the "jack-knife" option is chosen in the Topos function, a simple dipole analysis is conducted on each of the jack-knife solutions and the resulting cluster of solutions is generated, with the original location in red and the jack-knife solutions in blue. This provides a graphical representation of how stable the original solution was to individual subject variability (Foti et al., 2011). Since this procedure is conducted on the factor loadings, it will only be done for a spatial PCA or for a temporo-spatial PCA. In addition, the jack-knife PCA procedure allows for a significance test to be conducted on the relative strength of the hemispheric dipole amplitudes, using a t-test that has been adapted for jack-knife analyses (Miller, Patterson, Ulrich, 1998), which is then printed out at the command line. Unfortunately, the corresponding test statistic is not yet available using robust statistics.



The statistical result was:

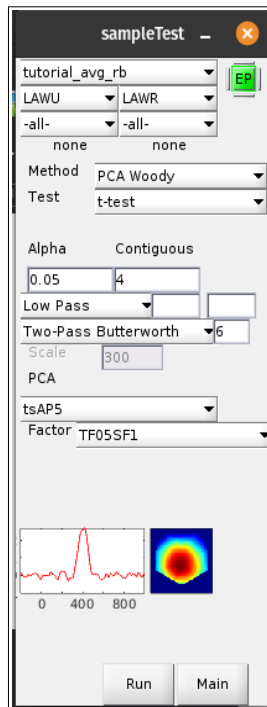
The t-value for the amplitude of the two hemispheric dipoles is 2.4783 with 19 degrees of freedom.

Right hemisphere larger than left.

It has a two-sided p-value of 0.022758.

Sample Test

A next step can be to example sample-by-sample test statistics. A limitation of temporal PCA is that the factors are by definition a specific time course and so cannot be used to examine subtle latency shifts. Spatial PCA is capable of doing so but it tends to conflate multiple ERP factors, hence the need for the two-step PCA procedure. One way of proceeding is to use the PCA to identify ERP components of interest and then to use sample-by-sample tests to determine if there are any latency shifts. Sample-by-sample test statistics suffer from high multiple comparison rates so they need to either be highly constrained by a priori hypotheses or they need to be treated as highly exploratory. Go back to the Main Menu and click on SampleTest. This button will be grayed out if the Statistics Toolbox from Mathworks is not installed.



The first line is to indicate the dataset to be analyzed. It has to be part of the working set. It can be either single-trial or multi-subject averaged data. Choose the “tutorial” dataset. The next two lines are used to indicate which two cells are to be compared. At present only pairwise comparisons are available. If the dataset is frequency domain then the tests will be conducted across the frequency bins instead of time (bin-by-bin). If the dataset is time-frequency domain then the tests will be conducted across time (sample-by-sample) and this will be done individually for each frequency bin. The third line specifies if a subset of the subjects is to be compared. One might, for example, choose to compare the LAWU condition in males versus females.

The Method menu allows one to choose one of four related methods:

- 1) Sample - is a sample-by-sample non-parametric in which each of the samples are individually tested. For single-trial data the Wilcoxon rank sum test is used and for averaged data the Wilcoxon signed rank test is used.
- 2) CWT - is a continuous wavelet transform analysis in which a Mexican Hat wavelet is generated and is slide across the epoch sample-by-sample. A cross-product fit statistic is generated at each sample and then tested. This is a simplified version of a published test procedure (Bostanov & Kotchoubey, 2006). The scale (or essentially the width) of the wavelet function can be specified in the Scale field and an image of the resulting wavelet is presented below.
- 3) Template Woody - uses an average waveform as a non-iterative Woody filter template (Woody, 1967) to slide across the epoch, generating a dot-product fit statistic (mean of

the elementwise multiplication of the two matrices) at each lag and then tested. The latency and the amplitude of the maximum fit statistic will be saved in the trial specs. This latency information can then be used to with the latency-lock option of the Average function. In this case you'd want to use the difference wave that you generated earlier (LAWU-LAWRof the grand average of the tutorial data). This works best if only a single ERP component differs between conditions and is especially appropriate if it varies in latency, such as the P300.

4) PCA Woody - uses a PCA factor as a non-iterative Woody filter template (Woody, 1967) to slide across the epoch, generating a dot-product fit statistic (mean of the elementwise multiplication of the two matrices) at each lag and then tested. This PCA factor can be either a temporal PCA factor (temporal template only) or a temporospatial PCA factor (both temporal and spatial template). The PCA dataset (from the working set) and the template factor are specified below and the time course and the scalp topography (if temporospatial) is illustrated. Since it shows the pure factor loadings, polarity is arbitrary (so it might show as positive rather than negative). If a spatial PCA is utilized on single-trial data, the latency and the amplitude of the maximum fit statistic will be saved in the trial specs. This latency information can then be used to with the latency-lock option of the Average function.

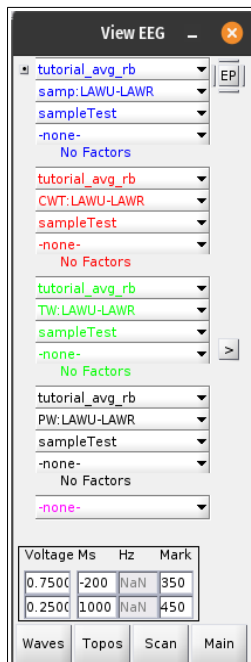
The test setting controls whether a sample-by-sample t-test is applied (either matched sample or two-sample depending on whether the data are multiple subject averages or single-trial data from a single subject) or a jack-knife test (Miller et al., 1998). The jack-knife test is applied to a single channel and will test whether there is a significant difference in when the waveform at that channel exceeds the threshold value (either above or below depending on whether the threshold value is positive or negative). The result of the test is output to the command window and is not saved with the dataset in any manner. This test seems to work quite well when the ERP is relatively robust, as in the lateralized readiness potential for which it was developed, but seems to me to be of less utility for other cases. If either of a subject's waveforms do not reach the threshold at all, the subject is counted as missing data and dropped from the test, as reflected in the smaller degrees of freedom. Also available is a t-map option which does much the same but instead of marking whether a time point is significant or not, it just saves the p-value, which can then be viewed in View Topos as an interpolated significance map.

The Alpha field controls the alpha threshold used to determine whether a sample is to be considered statistically significant.

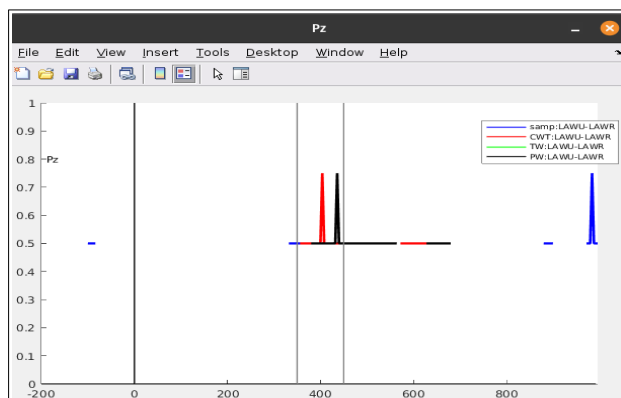
The Contiguous field controls a very crude multiple comparison control. If there is a series of contiguous samples at least as long as the value set in this field, then those samples will be highlighted when the results are viewed.

The next two lines are filtering controls that work as in the Transform function, except that they are only applied during the analysis and will not result in any lasting changes to the dataset itself.

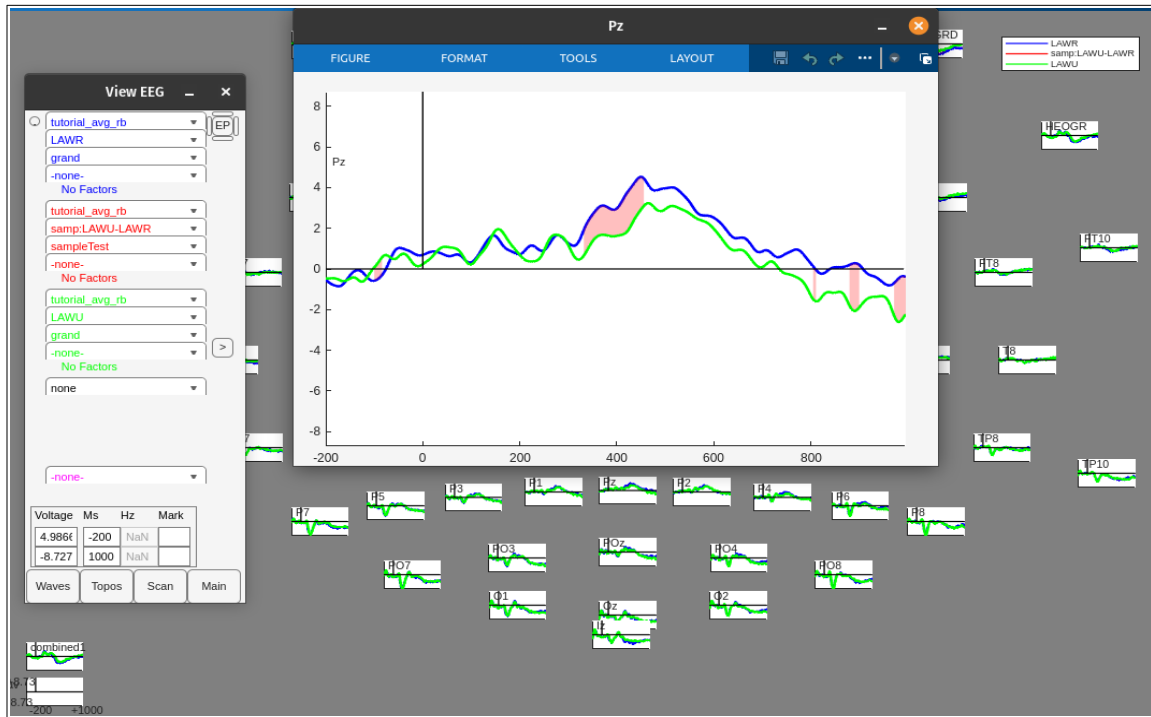
Try out all four tests on the LAWU and LAWR conditions in the tutorial dataset, using the t-test to look for support for the TF5SF1 putative N400 effect. For the Woody Filter, use the LAWU-LAWR difference wave. For the PCA Woody filter, use the TF01SF1 factor. The results will be saved to the tutorial dataset in a “subject” called “sampleTest” and cell names corresponding to the test and comparison made (e.g., samp:LAWU-LAWR). If the same type of test and comparison is repeated, the results will overwrite the previous result. They can be viewed using the View function (remember to choose "sampleTest" as the subject). The results take the form of waveforms where .25 marks a significant effect, .5 marks a significant effect which is part of a series of significant samples whose length exceeds the contiguity setting. A value of .75 marks the most significant sample.



Expand the Pz channel by clicking on it and then right-click and choose legend. In this case, they show effects at the N400 latency, as expected. It can also be seen how the four methods have different sensitivities.

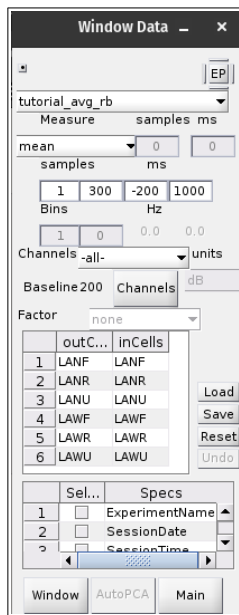


If the two colors on either side of sample test results are normal waveforms, then the sample test effects will be presented as a filled in area between the two waveforms. Naturally this will only be meaningful if the two ERP waveforms represent the two conditions used to generate the sample test results. While not suitable for rigorous statistical testing, this method is useful for highlighting where the meaningful differences might be, beyond a simple windowed measure. On the other hand, while there is a small prestimulus patch, it is probably due to uncontrolled MCP (multiple comparison problem) and thus not real.



Windowing

To test whether there are any statistically significant effects, go to the Window Pane of the Main Window. Let us first consider how subject averages would be analyzed and then address how to analyze PCA results.



Windowing Subject Average Data

First you need to indicate which of the datasets in the working set to analyze. Then which measure to use within the window. Min and max centroids (Dien et al., 2004) are a way of measuring the latency of the overall area under the curve rather than just the peak, much as a mean measure does for amplitude. Use minimum centroid for negative components and maximum centroid for positive components. Mean is just the mean voltage in the window. A methodology paper (Clayson, Baldwin, & Larson, 2013) has reported that the centroid measure is "biased" but failed to provide information about in what manner. Further examination of their findings and methods by myself after the authors generously allowed me to examine their data showed me that the measure has a tendency to err in the direction of the middle of the epoch as a function of noise levels. So whether this is an issue or not depends on the noise levels. If one is comparing two conditions with comparable noise levels, which is a normal statistical assumption, and the concern is just whether they are different, then this is not an issue. If one is seeking to obtain an absolute measure of latency for other purposes then this is a useful caution. This is a good example of why it is important to not just report findings but also to interpret them.

Min and max peak are the minimum and maximum voltage values within the window. One can specify that some number of points surrounding the peak value are averaged together instead of measuring just the peak sample. Min and max latency are the latencies of the minimum and maximum voltage points within the window. Following (Luck, 2005), for the peak measures only a peak with smaller voltages on either side will be counted as a peak. For peak amplitude measures only, there is also an option to have

it average together the adjoining samples after locating the peak. This option is activated by first specifying a peak amplitude measure and then typing into the adjoining fields the number of neighboring samples to include (e.g., "1 sample" would indicate that a total of three samples would be averaged together, the peak and the samples on either side of it).

In addition, there are three Measure options for assessing noise levels in a dataset, all utilizing the sample and channel settings. The first is meanSME, which provides the mean standardized measurement error (Luck et al., 2021), using the EEGlab function written by the authors. Since this measure relies on trial-level data, this option works somewhat differently from the others in that it will expect that the average file was generated by the EP Toolkit 2.96 or higher, so that the names of the session files will be encoded in the History records. It will ask for you to select a directory that contains these session files (with unchanged file names). It will apply any post averaging Transforms like filtering and rereferencing or Edit sampling rate changes to these session data to ensure that the resulting SME measures are comparable to the average file of interest. Note that the SME only assesses noise levels, not the signal-to-noise ratio, so it is not entirely a measure of signal quality as claimed (Dien & O'Hare, submitted).

The second is meanNoise, which provides a summary measure of the root mean square +/- noise reference data (Schimmel, 1967) generated by the EP Toolkit's Average function. Conceptually, the resulting numbers are comparable to the SME numbers.

The third is meanERA, which provides a psychometric measure of dependability for use with subject-level parameters like questionnaire data (Clayson & Miller, 2017). Since installation of the ERA function is somewhat involved and requires non-Matlab software, this option only generates the text files in a format appropriate for the ERA software but the EP Toolkit does not itself generate the measure. Like the meanSME option, it relies on trial-level data and will go through the same process with the original session files. The text files generate by this option are also appropriate for other procedures that rely on trial-level data, such as multi-level modeling (MLM).

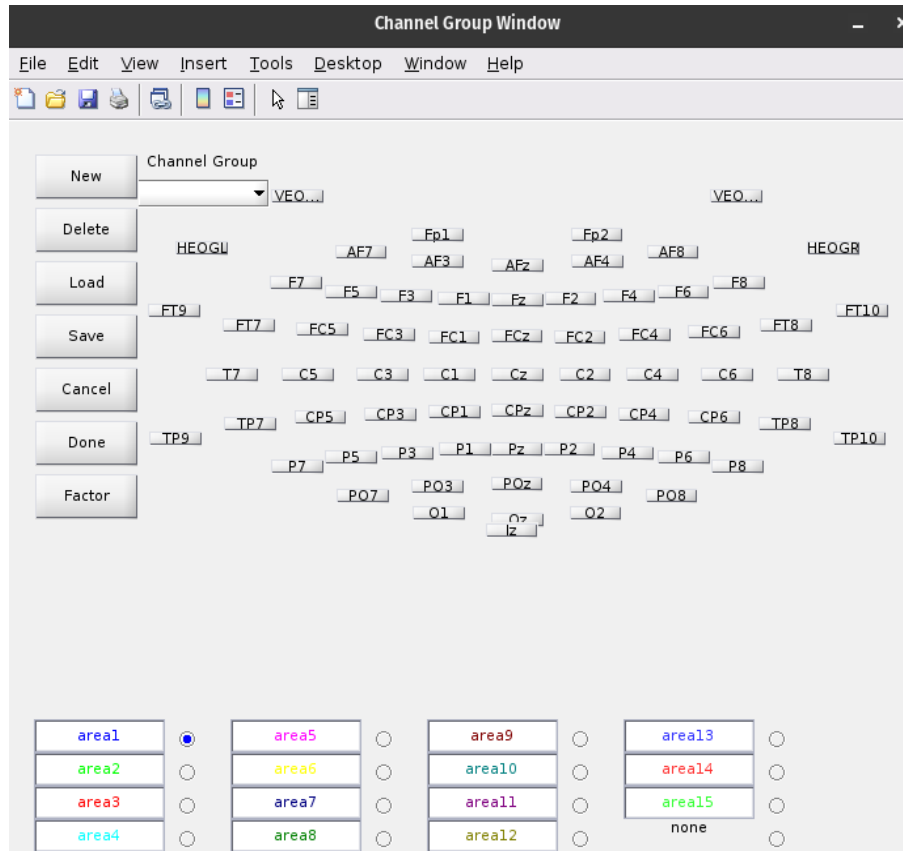
By default, the behavior is now to average together the channels in an Area and to measure the result, although there is an option under preferences to instead first measure each individual channel and then average together the resulting measures (the original behavior prior to EP Toolkit version 2.3).

Finally, if the trial specs include an RT field for reaction time and an ACC field for accuracy, one may choose "behavioral" to indicate that one wishes to generate text files suitable for ANOVAs for both the reaction time and the accuracy data (one for each). It will also generate task accuracy, D-Prime, and c signal detection figures if available from the averaging step.

Let's choose the "Mean" option.

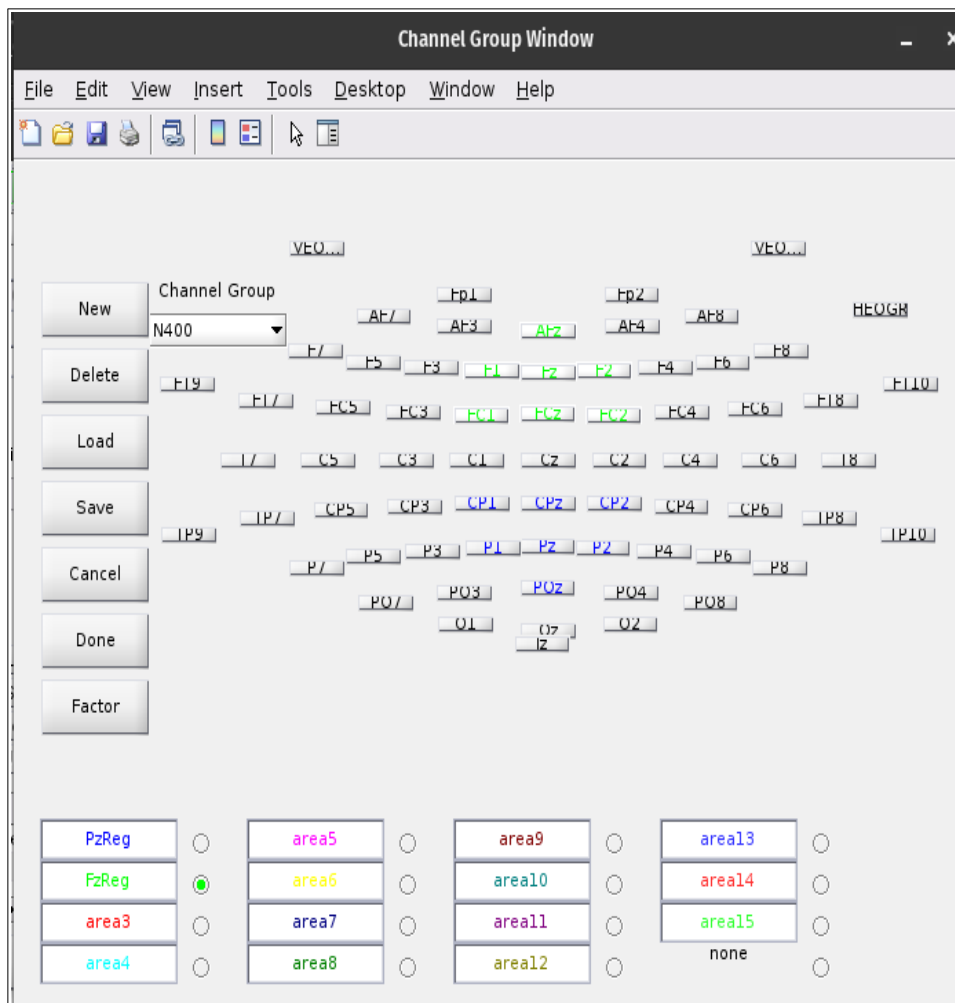
The time window also needs to be specified. Let's try a 350-450 ms window. The numbers will self-correct to 352 and 452 respectively, reflecting the fact that with a 250 Hz sampling rate, each sample is 4 ms long. These corrected numbers reflect the real timing (start of first sample to end of last sample). Authors often do not bother to report the actual numbers, which is not terrible but is certainly sloppy. Note also that the sample fields have automatically updated to reflect the chosen ms numbers.

Next, the channels are specified by clicking on the Channels button. It will then bring up the electrode montage.



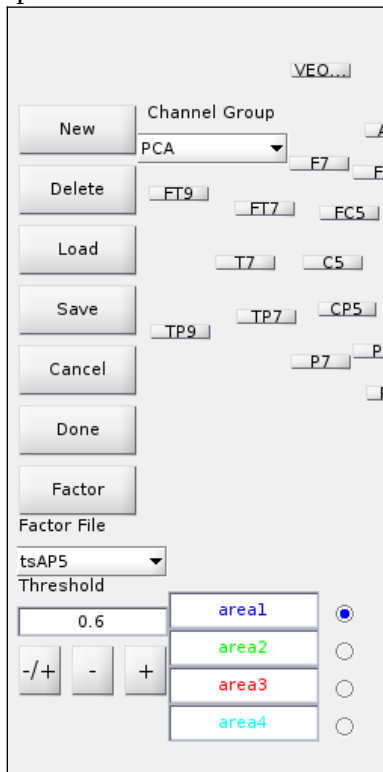
Click on the channels for the first electrode area surrounding Pz. After the first channel is clicked, it will ask for a name for the new electrode grouping (the partition scheme for the electrodes, which in turn is divided into electrode areas that will be averaged together). Type in N400. Continue clicking on the electrodes, as shown in the figure below. You should also give the channel area a name by typing in the "area1" box, say PzReg (make sure it is not the same as an actual channel name, like Pz). Let's also add a posterior grouping around Fz by typing in FzReg for "area2" and then clicking on the green box to activate the second area. Having clicked on the green box, any channels you click on will be added to the second area (and turned green) rather than the first box (and

turned blue). The channels of the currently selected area will be highlighted. Then click save and it will save these channel groupings for later use.



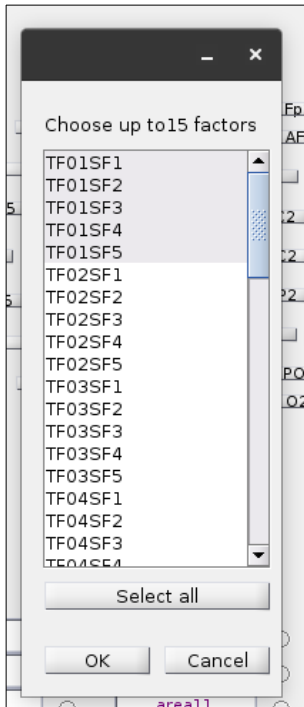
You could also define channel areas by using the results of a spatial PCA, if any are in the current working set. Click New to define a new channel grouping and give it the name PCA. Now click on the Factor button. It will give you a menu of the available

spatial PCA solutions and the factor loading threshold to use (0.6 by default).

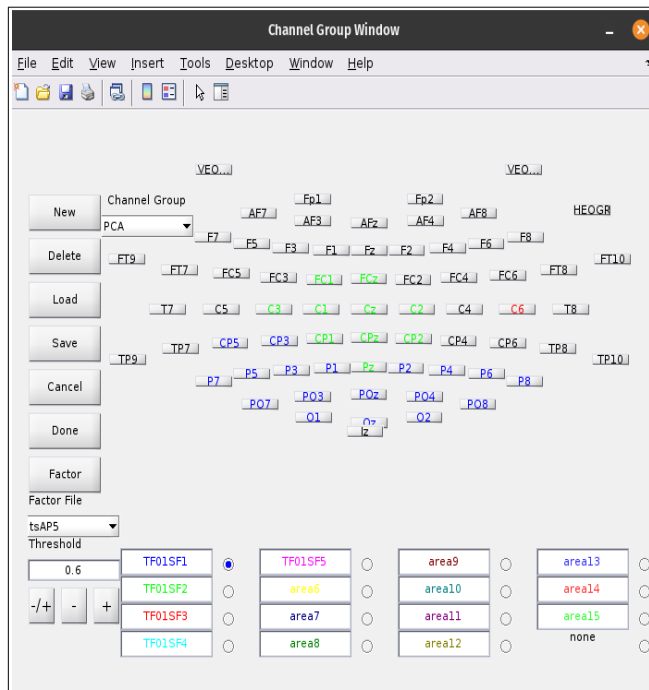


If you click on “-“ it will choose the factor loadings that are more negative than the threshold (e.g., -6). “+” will choose the factor loadings that are more positive than the threshold (e.g., +6). “+/-“ will choose the factor loadings whose absolute value is greater than the threshold. A negative threshold will be treated as a positive value. Note that the sign of the factor loading has no intrinsic meaning as it is essentially randomly determined. It is only the product of the factor loading and the factor score that has a direct relationship to the sign of the voltage data. Consider that for a spatial PCA, a channel with a negative loading can vary from having a positive to negative voltage reading over the course of the epoch. By itself, a negative loading only signifies that it correlates inversely with channels having a positive loading (and vice versa). The “-“ and “+” options are only provided so that a user can have more control over which set of channels are selected.

Click on +/- . It will then give you a listing of all the factors, of which you can choose 15. For this example, just choose the first five (typically, holding down shift allows you to select a range while holding control or command lets you select non-contiguous options).



It will now choose label each channel according to which factor had the highest loading, as long as it passed the threshold that you specified (0.6 in this case). If a factor has both positive and negative loadings that pass the threshold and you pressed +/-, it will determine which polarity had the highest absolute amplitude and mark only loadings with that sign (otherwise you could end up with a windowed measure with both positive and negative voltages which would then cancel out when averaged together).



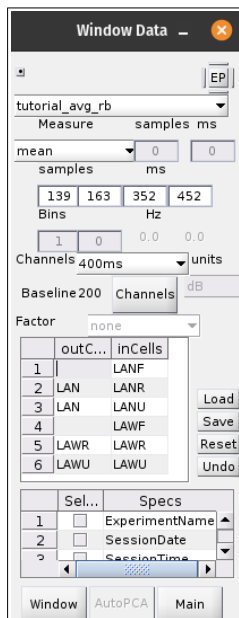
Once done, you can Save these channel groupings for later use and then press Done to go back to the Window Pane of the Main Window. For the tutorial, we'll just use the 400 ms channel grouping, so we'll only save that one.

Back to the Window pane, the 400ms channel group is now presented as an option. The Channels pop-up menu also offers an “-all-” option which will simply window each and every EEG channel separately. This -all- channel grouping cannot be edited.

The Units setting is for frequency-domain data only and so will be grayed out at this point.

If you are windowing PCA data, then you will also need to specify the factor.

You will also need to specify the cells. It will list all the cells in the dataset (excluding any combined cells that you added). If you want any of them to be combined, rename the outcells accordingly. For this example, let's combine together the LANR and LANU cells. If you want to drop a cell, just delete the corresponding name in the outCells column (leaving it blank) and that cell will be left out; let's do that for the LANF and LAWF filler cells. You can also freely change the order of the cells by retyping them. If you misname a cell, the function will let you know during the windowing process. There are Load and Save buttons next to the Cells table to allow you to save and load the settings in the Cells table, as well as a reset button to restore the table to its original state. Save the Window table and name it “tutorial_avg_rb_WindowTable.mat”. Finally, you can specify if you want any of the subject specs (specifications) to be included in the output file. Let's click on Gender.



Once ready, click on the Window button and then specify the name of the text file to be output (AP5-N400.txt). The windowing routine will check to see if the names of the output cells are different from those already in the dataset. If so, then it will add combined cells with those names to the dataset so that you can examine what the waveforms corresponding to the windowed measures looks like. Likewise, if the channel area names are different, then it will add new regional channels to the dataset so that you can again examine them.

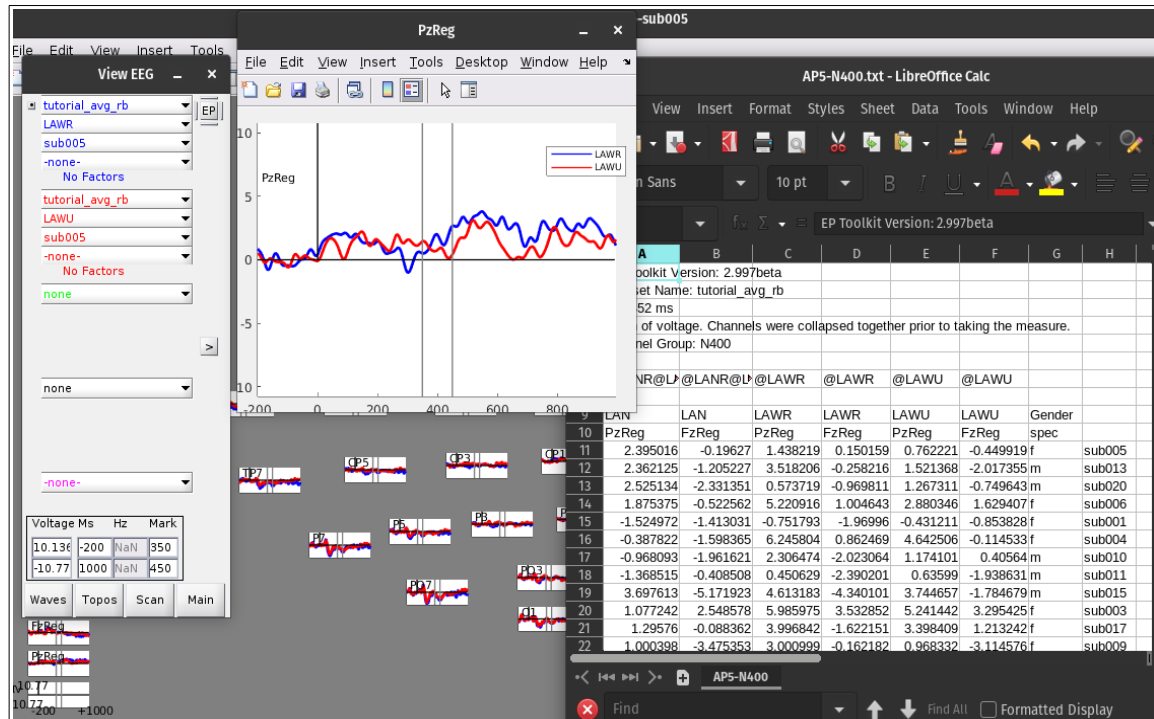
Turning to the output of the windowing procedure, it generates a tab-delimited text file that can be examined in Excel or similar programs. After some header information it provides the windowed measures in a form suitable for direct use in a statistics program, along with the subjects specs requested (Gender) and the subject names. There are three levels of organization, reflected in rows 8-10. Firstly, if there multiple sessions in the data, then the eight row will contain an automatically generated within-group session factor, structured as the highest level factor. The ninth row will contain the condition factors. The tenth row will contain the electrode factors, if any.

Note that “adds” (added summary data like grand averages and regional channels and combined cells) are excluded from windowing (and all other analyses like PCA and sample test) so they can be added to a dataset for inspection without concern that they might mess up the analyses.

EP Toolkit Version: 2.997beta							
Dataset Name: tutorial_avg_rb							
352-452 ms							
mean of voltage. Channels were collapsed together prior to taking the measure.							
Channel Group: N400							
@LANR@L@LANR@L@LAWR @LAWR @LAWU @LAWU							
LAN	LAN	LAWR	LAWR	LAWU	LAWU	Gender	
PzReg	FzReg	PzReg	FzReg	PzReg	FzReg	spec	
2.395016	-0.19627	1.438219	0.150159	0.762221	-0.449919f		sub005
2.362125	-1.205227	3.518206	-0.258216	1.521368	-2.017355m		sub013
2.525134	-2.331351	0.573719	-0.969811	1.267311	-0.749643m		sub020
1.875375	-0.522562	5.220916	1.004643	2.880346	1.629407f		sub006
-1.524972	-1.413031	-0.751793	-1.96996	-0.431211	-0.853828f		sub001
-0.387822	-1.598365	6.245804	0.862469	4.642506	-0.114533f		sub004
-0.968093	-1.961621	2.306474	-2.023064	1.174101	0.40564m		sub010
-1.368515	-0.408508	0.450629	-2.390201	0.63599	-1.938631m		sub011
3.697613	-5.171923	4.613183	-4.340101	3.744657	-1.784679m		sub015
1.077242	2.548578	5.985975	3.532852	5.241442	3.295425f		sub003
1.29576	-0.088362	3.996842	-1.622151	3.398409	1.213242f		sub017

As verification of the numbers, one can directly compare them with the waveforms. Go to View. Set markers at 350 ms and 450 ms. See how the waveforms for the subjects (using View>Waves) correspond to the measures for LAWR and LAWU. A complication is that the channel regions defined in the Windowing function will be difficult to relate to the individual channels, unless the channel region consisted of only a single channel. In order to facilitate comparisons, the EP Toolkit will add regional channel waveforms to the dataset. Since a regional channel (consisting of multiple

channels averaged together) can't have a specific location on the head, it will instead be located in the lower left hand corner.



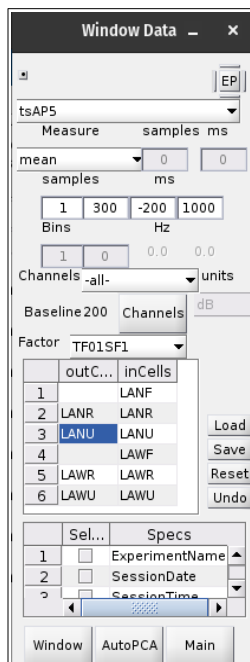
Windowing PCA Data

Analysis of PCA data is conducted in much the same way. The only difference is that one needs to specify the Factor in addition to everything else. Alternatively, one can click on the AutoPCA button and the Toolkit will run through all the factors, choosing the peak channel and the peak time point for each factor. When using this option, the pane's settings for channel and for window will be ignored. It will only generate a windowed file for each factor whose size (variance accounted for) meets a minimum threshold (set in Preferences, .5% by default) to screen out factors that account for only small clumps of noise. This kind of selectivity is, in turn, helpful when controlling for multiple comparisons as one will not have to be as stringent as if one had included the noise factors. The command line will announce the number of factors that met the criterion.

The output of a PCA is expressed in voltages, corresponding to the voltage accounted for at the chosen time points and channels (the peak time point and channel if the AutoPCA option was chosen). The Toolkit has automatically converted the factor scores into microvolt scaling. One can use the View option to examine the actual factor waveforms, if desired, to see how they correspond.

It is important to understand that when windowing a temporal factor (or a two-step factor), which sample you choose will not affect the inferential statistics in any way. The only way it matters is it presents the measure in a readily interpreted form, comparable to what one obtains from windowed measures of the averaged waveforms (in this case the peak point). One way to understand this is that the temporal PCA has converted 300 individually variable time points (in this example) into 300 factors, each one equally variable across all of their time points. Another way to think about this is that the PCA process is itself a type of windowing. Behind the scenes, each temporal factor consists of a single column of factor scores and a set of factor loadings that describes how this factor score relates to each of the time points. When you apply an ANOVA to the factor, you are testing the same set of factor scores no matter which time point you choose. The only effect of choosing a time point is that the factor score is multiplied by that factor loading (e.g., .5). Since this multiplication affects equally both the numerator and the denominator of the test statistic, they cancel out, and so you end up with the same p-value. In this, the factor loading is just scaling the factor scores, just like changing the measures from microvolts to volts (by multiplying them by .0001) will not change the p-values. A similar logic applies to spatial factors and both apply to two-step factors, such as in this tutorial.

Change the file to tsAP5. Again set the two filler cells LANF and LAWF to blank and also select Gender to be included in the ANOVA files. For AutoPCA, the sample and the channel settings will be ignored so they do not need to be set. You'll be asked to provide a root name for these files (let's call them AP5fac). AutoPCA will generate one file per factor. 29 factors meet the .5% threshold.



As can be seen in the resulting file for TF05SF01 (temporal factor 5/spatial factor 1), which is the 21st factor, the peak channel is named as being electrode VEOGLD and the

peak time point was at 416-420 ms (start and end of the 4 ms long sample). The gender is also included as you had clicked on its checkbox. If one wished, one could rerun this factor, manually choosing Pz rather than VEOGLD; it wouldn't change the p-value, but might be preferable if one were presenting the measures in a table. Likewise, one might manually change the time window to 352-452 ms to directly correspond to the conventional windowing measure. An AutoPCA run won't add channels to the dataset as it will be using existing channels.

EP Toolkit Version: 2.997beta					
Dataset Name: tsAP5					
416-420 ms					
One sample of voltage.					
Channel Group: autoPCA					
Factor: TF05SF1 (21)					
@LANR	@LANU	@LAWR	@LAWU		
LANR	LANU	LAWR	LAWU	Gender	
VEOGLD	VEOGLD	VEOGLD	VEOGLD	spec	
-0.54115	-2.288216	-0.795191	1.273509	f	sub005
0.760334	-0.122065	-2.707211	1.615131	m	sub013
-1.106946	-1.067921	0.51153	0.224065	m	sub020
-1.725716	1.012764	-6.005463	-2.960199	f	sub006
1.279481	1.499928	-2.129402	-1.779703	f	sub001
1.466758	3.7017	-10.260944	-5.468041	f	sub004
3.044591	0.887751	-2.097835	1.410687	m	sub010
3.242512	4.268448	-0.117043	0.928908	m	sub011
0.522402	1.483849	-1.921725	0.101667	m	sub015
1.211452	-1.97969	-10.332632	-9.096153	f	sub003
-0.416919	1.718218	-2.498155	-4.210549	f	sub017

ANOVA

The ANOVA function provides the choice of either conventional or robust ANOVAs. While users are encouraged to consider using the more advanced robust form, the conventional statistics are also supported for three reasons: 1) for those not feeling adventurous, 2) to enable direct comparisons with the robust ANOVA output, 3) and for features not yet available for robust ANOVAs, namely covariates and G*POWER compatible effect sizes. The conventional ANOVAs rely on Mathwork's Statistics and Machine Learning Toolbox and match exactly the output of both SAS and SPSS for my test datasets.

About Robust Statistics

The robust statistics function generates inferential statistical tests comparable to ANOVAs that are designed to be more robust against violations of statistical assumptions. The statistical routines were translated from the SAS/IML routines posted by Lisa Lix (<http://www.usaskhealthdatalab.ca/sas-programs/>) and exactly matched their output with test datasets. The routines were fully described in the following publications (Keselman et al., 2003, 2008). Also, the accompanying file "robust.pdf" is also included. A front-end has been added that automatically generates the necessary contrast matrices

for omnibus ANOVAs and that organizes the results in an easy to read manner. Note that the routines take some time to run.

The effect size literature is rather confused at present (Lakens, 2013). For robust statistics, the effect sizes are provided care of routines translated from Rand R. Wilcox's R WRS library. For two-level tests, the d_R effect size statistic is provided for between-group (Algina et al., 2005a) and within-group (Algina et al., 2005b) effects. With this type of statistic, the suggestion is that .2 is a small effect, .5 is a medium effect, and .8 is a large effect (Cohen, 1988). For multi-level between-group tests, the Explanatory Measure ξ (ξ) is provided (Wilcox & Tian, 2011). For multi-level within-group tests, Projection Distance D is provided (Wilcox, 2022, p. 468). No effect size is yet available for between-within group interactions. The EP Toolkit also provides the 95% bootstrapped confidence interval for each effect size measure (although confidence intervals are not currently available for the Project Distance measure and not for the Explanatory Measure CI if there are more than eight levels involved). For conventional ANOVAs, effect sizes comparable to those of SPSS are provided.

The random aspect of bootstrapping results in some variability of the p-values. Based on experience, I'm now recommending 499,999 simulation runs, even more than my prior suggestion of 4,999 runs (Dien, 2017). The EP Toolkit runs this set of simulations eleven times to compute the standard deviation of the resulting p-values and reports the median one. If twice the standard deviation of the p-values plus the median p-value exceeds the alpha threshold then a notation is given after the p-value that it was “(not confirmed)” and it should be treated as a borderline significant result. The reasoning is that this instability is more akin to limits on the number of significant decimals than it is akin to issues with the Type I error rate. As seen in my 2017 paper, with a typical ERP dataset, instability was only seen starting with the fourth decimal point, so for the most part only a p-value of about .049 would be affected. With smaller sample sizes some or all of the bootstrap runs may suffer from near-singularity, in which case they are dropped from the bootstrap computation. The EP Toolkit will report the %age of such runs on the next line, if they occur; if there is a sizable %age, then the results may be inaccurate.

Psychophysiology has recommended the use of these robust statistics (specifically, statistical tests using non-pooled error terms) in its Guidelines to Authors. In brief, this robust statistic has the following three features: 1) trimmed means and winsorized variances/covariances to minimize effects of outliers; 2) bootstrapping routine to estimate the sample mean distribution rather than making the assumption that the data is normally distributed; 3) Welch-James approximate degrees of freedom statistic (resulting sometimes in decimal degrees of freedom) that avoids the assumption of homogeneous error variances/covariances. The latter also makes it unnecessary to use epsilon correction like G-G or H-F since sphericity is not assumed. In practice, it seems to generate results that are largely comparable to normal ANOVAs but that are more robust against violations of assumptions. For a direct comparison of ERP robust statistical results against conventional univariate and multivariate ANOVAs, see (Dien et al., 2006). A very approachable treatment of this robust statistic is available (Wilcox, 2001). A

description of the issues involved in using conventional ANOVAs with ERP data is also available (Dien & Santuzzi, 2005). Further discussion of the issues in applying robust statistics to ERP data is also available (Dien, 2017).

General Procedures

To run statistical tests, first go back to the Main Pane of the Main Window and then click on the ANOVA button. The ANOVA function is set up to assume that there will be a series of windowed files that have a similar structure (in terms of the cells). The first step is therefore to choose one of these files to serve as the template. Click on the Load button and then select the AP5fac-TF05SF1.txt that you just generated with the AutoPCA option.

The screenshot shows the 'Robust ANOVA' dialog box. It has several sections: 'Data Columns' with two dropdowns (LANR-VEOGRD, LAWU-VEOGRD) and a status 'have: 0 need: 4'; 'Between Group' with six rows of 'none' dropdowns; 'Factor' and 'Levels' with two empty tables and buttons 'Load', 'Save', 'Auto', 'Clear', 'Undo'; a 'Robust' dropdown; 'MCP' set to '0' and 'All Posthoc' unchecked; and a bottom row of buttons 'View', 'Contrast', 'Load', 'Run', 'Main'.

In this file, as you saw earlier, there were four data columns plus the Gender column. The first two menus are the leftmost and rightmost data columns to be analyzed. By default the Toolkit assumes that you will be analyzing all of the data columns but you can adjust the settings if you wish. Note that it indicates that 4 are needed. It means that the factor structure needs to accommodate four total within group cells. The Between Group section allows you to designate a column as holding the labels for a between-group factor. If you are using conventional ANOVAs (controlled by the pop-up menu near the bottom of the pane), there will also be a control next to each row with the values of either F for between-group factor or C for covariate.

The next section is for the within-group factors. Enter in a LEX (lexical) factor with a N and a W level (nonword vs. word) and a REL (related) factor with a R and an U level (related vs. unrelated). The lowest factor varies fastest as one goes across the columns. If there is a session factor, it should be the first factor. If there are electrode factors, they

should be the last factors. Note that the ANOVA factor name needs to have three letters and the ANOVA factor level names need to be one letter each. It is best if the letters chosen for the levels are different across factors to avoid confusion (e.g., don't use N for both the LEX and the REL factors).

Also select Gender as a between group factor by selecting Gender-Spec and provide it with a three letter name, in this case "GEN". You do not need to specify the level codes as the EP Toolkit will harvest them from the ANOVA files. Between group level names (provided by the specs and editable by the Edit function) need to be a single letter. If longer labels were provided, only the first letter will be used. It is therefore important that such labels start with different characters. Finally, enter the number of factors that met the variance criterion minus the number of a priori factors, $29-8=21$, into the MCP field. The Dunn-Šidák post-hoc is a minor improvement on the Bonferroni post-hoc, changing the corrected threshold typically around the fourth decimal for the sorts of numbers you will be using here.

Finally, there is a checkbox, All Posthoc, that indicates that all post-hoc pairwise contrasts should be provided even if the effect was not significant, for cases where doing so might be helpful for better understanding the data or where there were a priori hypotheses.

The pane should end up looking like the following figure:

Factor	Levels
LEX	NW
REL	RU

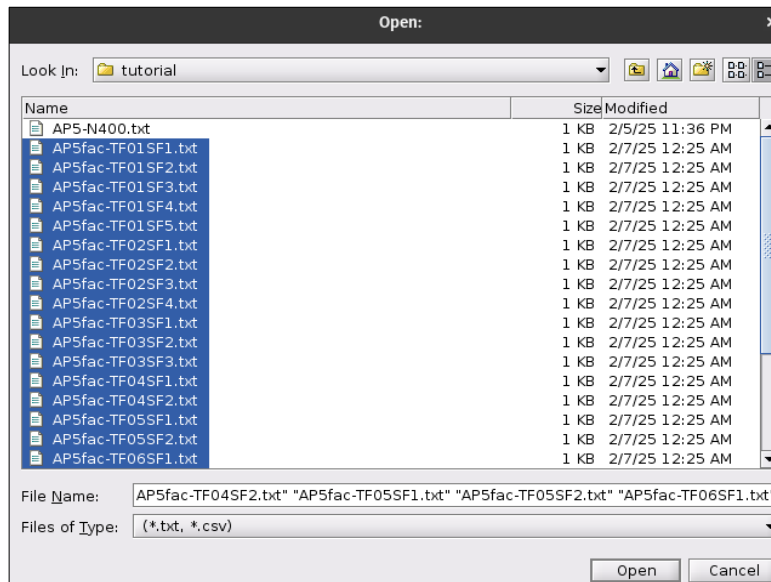
There are also a set of buttons to the right of this table to allow you to save the contents of the table for later, load them in, clear the table, and undo the last change to the table.

There is also an Auto button that will automatically populate the table if the cell names are all the same length and each letter position is either the same for all the cell names or specifies the level of a factor (e.g., OVT to specify O for oddball, V for visual, and T for target).

You can verify that your factor structure mapped correctly onto the data by clicking on the View button.

	N R	N U	W R	W U	Gender-spec
	LANR-VEOGLD	LANU-VEOGLD	LAWR-VEOGLD	LAWU-VEOGLD	
1	-0.541150	-2.288216	-0.795191	1.273509	f
2	0.760334	-0.122065	-2.707211	1.615131	m
3	-1.106946	-1.067921	0.511530	0.224065	m
4	-1.725716	1.012764	-6.005463	-2.960199	f
5	1.279481	1.499928	-2.129402	-1.779703	f
6	1.466758	3.701700	-10.260944	-5.468041	f
7	3.044591	0.887751	-2.097835	1.410687	m
8	3.242512	4.268448	-0.117043	0.928908	m
9	0.522402	1.483849	-1.921725	0.101667	m
10	1.211452	-1.979690	-10.332632	-9.096153	f
11	-0.416919	1.718218	-2.498155	-4.210549	f

Finally, click on the Run button. Enter in "AP5facResults" as the ANOVA output name. Then click on the set of windowed files that you wish to run. In this case, we'll run all of the factor files.



The Toolkit will start running a full ANOVA on the data. The output goes into an .html file with the name that you specified. The analysis will take some time, even on a fast computer, although the EP Toolkit now comes with mex files for Mac, Windows, and Linux that speed it up by a factor of about six. You can check on its progress by double-clicking on the html file. It should come up in a browser window that will show its

current state. Click on the browser's refresh button periodically to follow its progress, if desired.

Some typical output follows. First it provides the basic parameters. Then each combination of between factor effects is presented. For each between factor effect combination, the within factor effects are provided. The between factors are named in italics. Each effect lists the statistical test numbers (appropriate for directly copying into a manuscript). If the test reaches one-tailed significance (normally .10) then it is listed in green. If it reaches two-tailed significance but not post-hoc corrected significance then it is listed in orange. If it reaches two-tailed significance even with the post-hoc correction then it is listed in red.

A further feature of the Toolkit's output is that when an interaction is significant (uncorrected), the Toolkit will provide all the possible follow-up tests (i.e., for a three-way interaction, it will provide all the possible two-way interactions, where one level is held constant). If any of those are significant then the follow-up tests for that test are provided as well. A Dunn-Šidák correction is applied at each of these steps.

If a significant test was an interaction or was a main effect with more than two levels, then it would provide further follow-ups, also providing the appropriate post-hoc alpha for these tests as well. Note that follow-up tests are in a smaller font to help set them off from the main tests. Also, follow-ups are indented deeper than the preceding level of results. If the "All Posthoc" option is checked, then all the follow-ups will be generated regardless of significance level. Once the follow-ups to a test are completed then the results proceed to the next test.

For the N400/P300 factor TF05SF1, the critical LEXical x RELatedness interaction is significant with $p=.0031$, although the MCP correction would have made it not significant if it weren't identified as being of a priori interest, and hence is colored yellow. The red LEX main effect ($p=.0021$) is considered significant regardless of whether it was *a priori*. Since it was significant (yellow or red), the EP Toolkit provides follow-up contrasts. The first is testing the REL main effect only for the N (non-word) level of LEX. Here we see that relatedness made no effect for non-words, as it should (the test only being of interest to verify that no confounding effects are present, as these non-words were matched for sub-lexical characteristics with the word stimuli). We see that the REL effect is, as expected, significant for the W (word) level of LEX. When a follow-up contrast is significant, it too will have contrasts to clarify its nature, unless it is already a simple t-test. Finally, after a significant within-group interaction, a full set of pairwise contrasts will be generated to facilitate making bar charts where the significant pairwise contrasts are marked.

The test lines also provide an effect size appropriate for the statistic and a confidence interval, if they are available. For conventional statistics, f values are provided that can be entered directly into G*POWER for post-hoc power calculations. If doing so, be sure to use the Options button to indicate that the f statistic is of the type provided by SPSS

(with the variable correlations included in the calculation). Also note that when using conventional ANOVAs, interactions between covariates and between-group factors (if present) are included whereas in SPSS these are omitted, which will lead to different results from SPSS when both covariates and between-group factors are present.

Next, the MSe (mean-squared error) is provided as this information is often required by journals as it facilitates meta-analyses. Finally, the mean of all the bivariate correlations (winsorized for robust) between cells (for within-group factors) is provided as some power analysis software (e.g., Superpower) requires this information.

```
#####
AP5fac-TF05SF1
EP Toolkit version: 2.9974
Time Started: 12-Jun-2025 18:46:51
Robust ANOVA based on SAS/IML code by Keselman, Wilcox, and Lix (2003),
and R effect size code by Wilcox (2022), and translated to Matlab by Dien (2010).
WELCH-JAMES APPROXIMATE DF SOLUTION
TRIMMED MEANS & WINSORIZED VARIANCES
PERCENTAGE OF TRIMMING: 0.05
BOOTSTRAP CRITICAL VALUE FOR SINGLE TEST STATISTIC
NUMBER OF BOOTSTRAP SAMPLES: 499999
MEDIAN OF BOOTSTRAP RUNS: 11
STARTING SEED (multiplied by the run number): 1000
Number of subjects in each group: 10 9
Number trimmed from extremes of each cell: 0
Number trimmed from extremes of each group cell: 0 0
Uncorrected alpha criteria: 0.05
Familywise (e.g., a set of PCA factors) corrected alpha criteria: 0.0024396
Post-hocs use the Dunn-Šidák MCP correction.
Per commonly accepted practice, post-hoc MCP correction not applied to simple main
effects and interactions.
Familywise (e.g., a set of PCA factors) Dunn-Šidák MCP correction factor: 21
#####
ANOVA Header Information
#####
EP Toolkit Version: 2.997beta
Dataset Name: tsAP5
416-420 ms
One sample of voltage.
Channel Group: autoPCA
Factor: TF05SF1 (21)
@LANR @LANU @LAWR @LAWU

LANR LANU LAWU LAWU
VEOGLD VEOGLD VEOGLD VEOGLD
```

LEX

$T_{WJt/c}(1.0,18.0)=16.96$, $p=0.0021$, $d_R=0.86$ ($CI_{95}=0.56$ to 1.40), $MSe=4.81$, Mean $R=0.41$.

Averaged Trimmed Cell Means (Winsorized Standard Deviations):

N W

+0.91 -2.02 (+2.14) (+3.29)

REL

$T_{WJt/c}(1.0,18.0)=9.35$, $p=0.015$, $d_R=-0.64$ ($CI_{95}=-1.04$ to -0.33), $MSe=0.69$, Mean $R=0.88$.

Averaged Trimmed Cell Means (Winsorized Standard Deviations):

R U

-0.96 -0.14 (+2.41) (+2.33)

LEX * REL

$T_{WJt/c}(1.0,18.0)=12.03$, $p=0.0031$, $D=1.76$, $MSe=2.02$, Mean $R=0.51$.

Averaged Trimmed Cell Means (Winsorized Standard Deviations):

NR NU WR WU

+1.07 +0.76 -2.99 -1.04 (+2.05) (+2.51) (+3.62) (+3.24)

#####

Post-hoc alpha with Dunn-Šidák: 0.00061045
Holding level N of factor LEX constant.

REL

$T_{WJt/c}(1.0,18.0)=0.64$, $p=0.43$, $d_R=0.17$ ($CI_{95}=-0.25$ to 0.65), $MSe=1.39$, Mean $R=0.75$.

Averaged Trimmed Cell Means (Winsorized Standard Deviations):

R U

+1.07 +0.76 (+2.05) (+2.51)

#####

Holding level W of factor LEX constant.

REL

$T_{WJt/c}(1.0,18.0)=18.05$, $p=0.00094$, $d_R=-0.89$ ($CI_{95}=-1.46$ to -0.52), $MSe=2.01$, Mean $R=0.83$.

Averaged Trimmed Cell Means (Winsorized Standard Deviations):

R U

-2.99 -1.04 (+3.62) (+3.24)

#####

Holding level R of factor REL constant.

LEX

$T_{WJt/c}(1.0,18.0)=27.37$, $p=0.0010$, $d_R=1.09$ (CI₉₅=0.80 to 1.74), MSe=5.72, Mean R=0.40.

Averaged Trimmed Cell Means (Winsorized Standard Deviations):

N W

+1.07 -2.99 (+2.05) (+3.62)

#####

Holding level U of factor REL constant.

LEX

$T_{WJt/c}(1.0,18.0)=5.20$, $p=0.036$, $d_R=0.48$ (CI₉₅=0.09 to 0.98), MSe=5.91, Mean R=0.31.

Averaged Trimmed Cell Means (Winsorized Standard Deviations):

N W

+0.76 -1.04 (+2.51) (+3.24)

#####

Full set of pairwise contrasts for bar charts

Post-hoc alpha with Dunn-Šidák: 0.00040701

Contrasting LANR-VEOGLD versus LANU-VEOGLD. $T_{WJt/c}(1.0,18.0)=0.64$, $p=0.43$, $d_R=0.17$ (CI₉₅=0.35 to 0.35), MSe=1.39, Mean R=0.75.

Contrasting LANR-VEOGLD versus LAWR-VEOGLD. $T_{WJt/c}(1.0,18.0)=27.37$, $p=0.0010$, $d_R=1.09$ (CI₉₅=1.23 to 1.23), MSe=5.72, Mean R=0.40.

Contrasting LANR-VEOGLD versus LAWU-VEOGLD. $T_{WJt/c}(1.0,18.0)=10.00$, $p=0.012$, $d_R=0.66$ (CI₉₅=0.65 to 0.65), MSe=4.21, Mean R=0.47.

Contrasting LANU-VEOGLD versus LAWR-VEOGLD. $T_{WJt/c}(1.0,18.0)=19.72$, $p=0.0018$, $d_R=0.93$ (CI₉₅=1.03 to 1.03), MSe=6.79, Mean R=0.32.

Contrasting LANU-VEOGLD versus LAWU-VEOGLD. $T_{WJt/c}(1.0,18.0)=5.20$, $p=0.036$, $d_R=0.48$ (CI₉₅=0.44 to 0.44), MSe=5.91, Mean R=0.31.

Contrasting LAWR-VEOGLD versus LAWU-VEOGLD. $T_{WJt/c}(1.0,18.0)=18.05$, $p=0.00096$, $d_R=-0.89$ (CI₉₅=-1.04 to -1.04), MSe=2.01, Mean R=0.83.

GEN

$T_{WJt/c}(1.0,15.0)=5.35$, $p=0.036$, $d_R=-0.98$ (CI₉₅=-2.18 to -0.21), MSe=3.47,

Averaged Trimmed Cell Means (Winsorized Standard Deviations):

f +0.13 (+1.59)

m +2.11 (+2.08)

*GEN * LEX*

T_{WJt/c}(1.0,14.1)=1.56, p=0.23, MSe=4.41, Mean R=0.07.

Averaged Trimmed Cell Means (Winsorized Standard Deviations):

	N	W
f	-0.08 -3.82 (+1.78) (+3.05)	
m	+2.02 -0.01 (+2.03) (+2.29)	

*GEN * REL*

T_{WJt/c}(1.0,15.8)=0.00, p=0.99, MSe=0.70, Mean R=0.27.

Averaged Trimmed Cell Means (Winsorized Standard Deviations):

	R	U
f	-2.36 -1.54 (+1.71) (+1.87)	
m	+0.59 +1.42 (+2.15) (+1.78)	

*GEN * LEX * REL*

T_{WJt/c}(1.0,16.4)=0.13, p=0.72, MSe=2.05, Mean R=0.16.

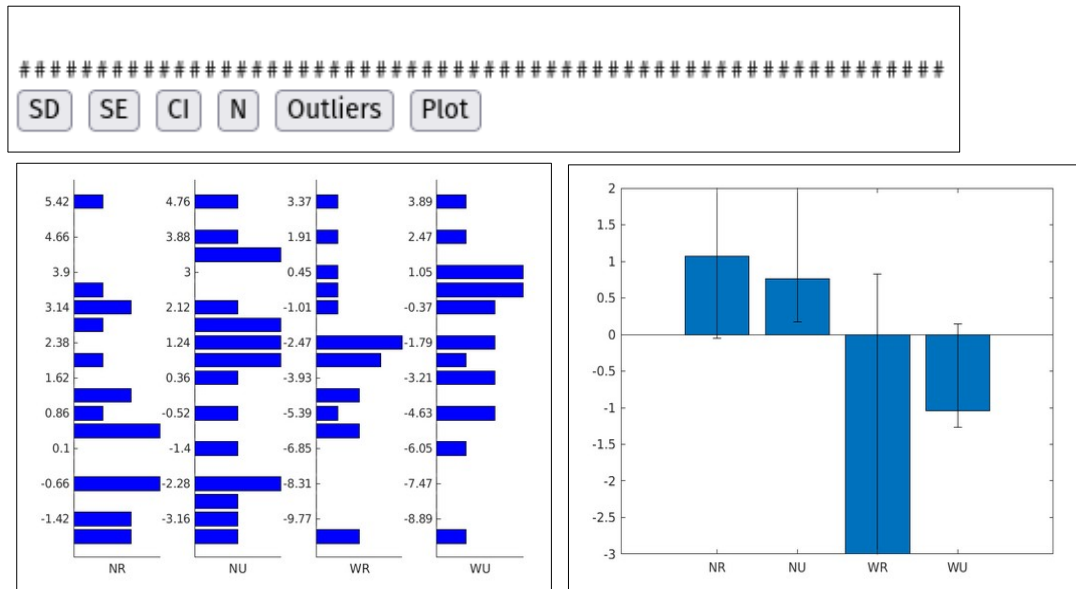
Averaged Trimmed Cell Means (Winsorized Standard Deviations):

	NR	NU	WR	WU
f	+0.13 -0.29 -4.85 -2.79 (+1.59) (+2.47) (+3.25) (+3.22)			
m	+2.11 +1.93 -0.93 +0.90 (+2.08) (+2.09) (+2.92) (+1.98)			

#####

Finally, each effect lists the means (trimmed if robust) and, by default, the standard deviations of each cell. At the very bottom of the output are six buttons that controls the content of the cell tables. The SD button provides the standard deviations (winsorized if robust statistics). The SE button provides the standard errors (winsorized if robust statistics and with the Cousineau-Morey correction if within-group). The CI button provides the 95% confidence intervals (winsorized and bootstrapped if robust and with the Cousineau-Morey correction if within-group). The N button provides the number of observations in each cell (trimmed if robust statistics and with missing data dropped). The Outlier button provides histograms depicting the distribution of the scores in each cell and marks in red which ones were dropped by the trimmed mean (here, the sample

size of 18 was too small to perform the 5% trim; for a sample size of 10-19, one might wish to use a 10% trim). Finally, the Plot button provides simple histogram plots with confidence interval error bars.



Another feature of the Toolkit is that when an effect involves an electrode factor, it will provide the McCarthy and Wood (1985) vector test to determine if it does indeed indicate a change in the scalp topography rather than being artifact of the ANOVA model. For an explanation of this issue, see (Dien & Santuzzi, 2005). Such vector tests are set off from the rest of the results tables with italics.

Vector Scaling Test (is the scalp topography effect genuine?):

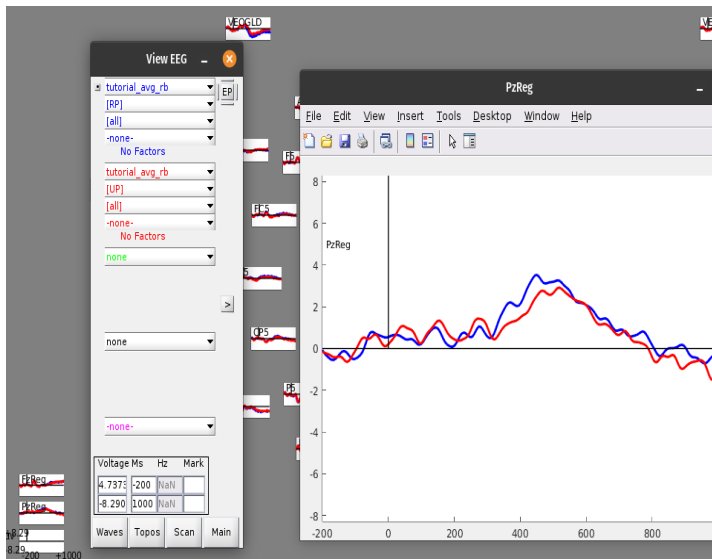
$T_{WJ}/c(1.0,7.8)=295.24, p=0.00012$

When there is a session factor, an intra-class correlation (ICC) reliability statistic will also be provided. It will only be provided for the session main effect as ICC for multi-factorial designs are not currently available. There are many types of ICC (McGraw & Wong, 1996). The one provided here is one that has been recommended for test-retest designs (Qin et al., 2019).

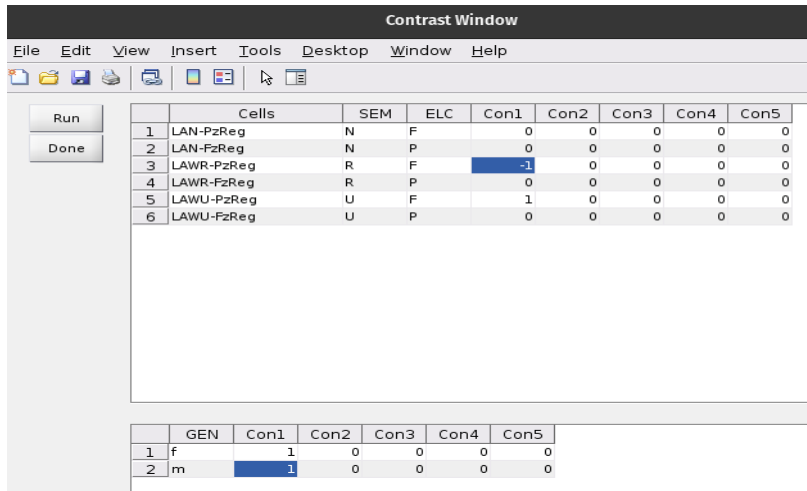
Intra-class coefficient reliability Single Fixed Raters ICC(3,1):
 $Rho=0.46, F(20.0, 220.0)=11.32, p<0.00000001$

An issue when using trimmed averages is that the grand average waveforms do not exactly correspond to the trimmed means that are being tested. The ANOVA function will therefore generate a trimmed grand average corresponding to the trimmed means and add it to the dataset (if present in the current working set). In the tutorial, the sample is too small for trimming, but it still generates the grand averages. Note that for lower level cells (e.g., LAW), the trimming is conducted after collapsing the cells whereas the

trimmed waveforms are based on the separate cells (e.g., LAWR, LAWU) and so simply combining them might not yield exactly the same result. Secondly, as a convenience, the ANOVA function will also generate trimmed grand averages corresponding to the between group cells. There will also be an “all” average representing the full set of subjects or cells as the case may be. These additions are in brackets (e.g., [all]). If regional channels were used (as in windowed averages, not a typical PCA), then these will be present as well. Here, we see the grand averages corresponding to the ANOVAs of R vs. U at the parietal regional channel (PR vs. PU). FR and FU represent the averages from the frontal regional channel. This can make a difference if robust ANOVAs are being used and different subjects are being trimmed for these contrasts. The statistical reasoning is that the measures are all estimates of the true mean, but a given measurement may suffer from excessive error (as in artifacts), so it is reasonable to be trimming from different measurements. The EP Toolkit allows one to see exactly what the corresponding grand average looks like.



If needed, one can also perform follow-up contrasts. To do so, click on the Contrast button. From there one can specify up to five contrasts to be done. The numbers in the contrast should add up to zero. If no factors are desired of a certain type (no within or no between) then either put in all ones (the one exception to the add to zero rule) or leave them as zeroes and the Toolkit will change them to all ones.



One then clicks on Run. As with the regular ANOVA, one is asked where to output the results and what ANOVA file to use. The results are rather barebones but functional:

AP5-N400.txt
 Contrast: 1
 Within: 0 0 -1 0 1 0
 Between: 1 1
 $T_{WJt/c}(1.0, 18.0) = 8.05, p = 0.029, d_R = 0.59$ (CI₉₅ = 0.34 to 1.02), MSe = 1.14, Mean R = 0.76.

Once finished with contrasts, click on Done to return to the Main Window.

Behavioral Data

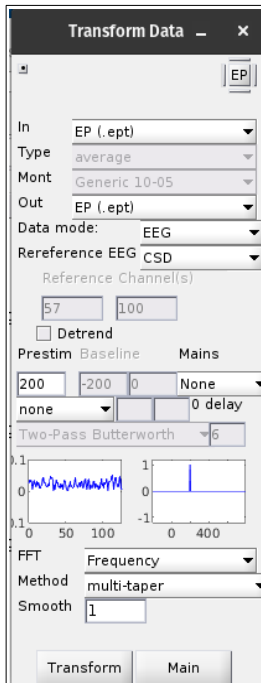
To use the ANOVA pane to analyze non-ERP data, use a spreadsheet program like Excel to have the following format:

- 1) The first line can be blank.
- 2) The second line should contain the keyword "behavioral" without the quotation marks.
- 3) There should be five more lines whose contents will be ignored.
- 4) The next line contains the column header names of the sessions if this is multi-session data, separated by tabs. If not, then this line should be left empty.
- 5) The next line is the column header line and should contain the names of the columns, separated by tabs. The Toolkit will ignore any additional columns that have no column name.
- 6) The next line will be ignored.
- 7) The remainder of the file should have the data with the values separated by tabs.

The Toolkit can also handle text files where there are no header lines, in which case it will assume it is behavioral data and will generate single letter headings for each column. The EP Toolkit can handle a wide range of variations in text file formats including unicode and both space and comma delimiters. By default, the value "-999" is treated as missing data and this value can be changed in the Preferences menu for ANOVAs. Values of NaN are also treated as missing data.

Spectral Analyses

The EP Toolkit can also be utilized to conduct spectral analyses. To do so requires a different stream of analysis. One can run the two streams of time-domain (ERP) and frequency-domain (FFT) analyses in parallel to obtain a more complete understanding of the data.



First set the appropriate FFT domain in the Transform function. “Frequency” means using Fourier analysis (with a Hanning filter) to measure oscillatory activity, such as alpha waves, using a function from FieldTrip. This option converts the data to frequency measures (complex amplitude), collapsing over the timepoints. One can also choose whether the method used is multi-taper or the more traditional Hanning method to restrict the FFT to discrete one-second epochs. The FieldTrip team recommends the Hanning method when measuring frequencies below 30 Hz.

For continuous data, the data is divided into one-second epochs and each one serves as a separate “trial” that can be separately artifact-rejected or averaged together. The conventional Welch approach does so with each one second epoch overlapping half of the adjoining one second epochs so that the edges of the epochs are fully represented (the Hanning and multi-taper filters reduce the sensitivity of the FFT measures to zero towards the edges of the epoch to avoid edge artifacts and so the overlap approach allows those parts of the EEG recording to be fully utilized). However, taking this approach complicates artifact rejection so the EP Toolkit uses fully non-overlapping epochs. Whether this approach is optimal or not in this context will require further evaluation.

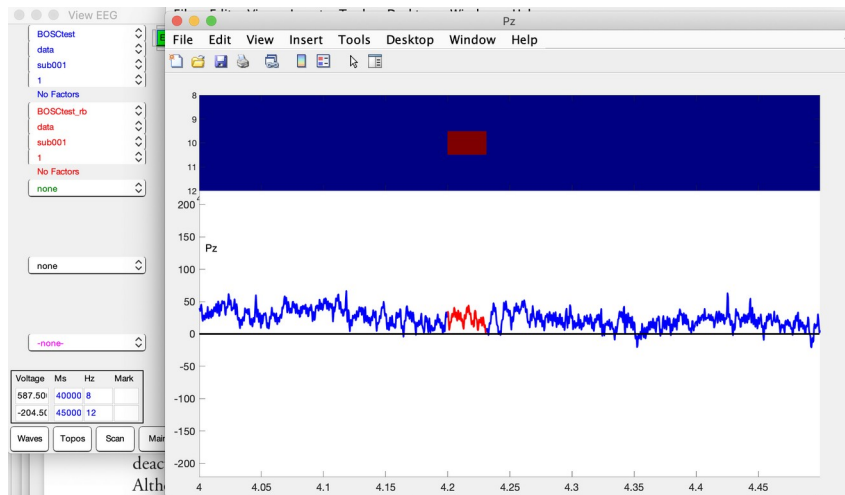
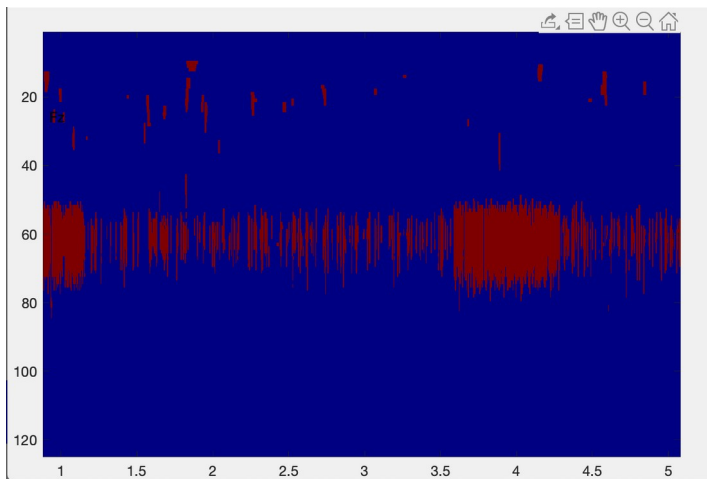
The Smoothing field is used to indicate the degree of smoothing in the spectral domain, with greater smoothing causing frequency peaks to become broader and more spread out.

The "Time-Frequency" option converts the data to time-frequency measures any of four different methods: Hanning, multi-taper, wavelet multiplication, and wavelet convolution. A half second window is used to generate the spectral measures and it is slide in 50 ms increments to produce a time course. The latency of each measure is attributed to the midpoint of the time window.

The "BOSC" option uses a new approach for identifying periods of oscillatory activity (Hughes et al., 2012; Whitten et al., 2011). Basically, one first defines a baseline period to determine the level of background frequency-domain activity. The prototypical example would be alpha and using "eyes open" as the background condition. BOSC then runs a wavelet in the chosen frequency window across each channel of the data of interest. Periods where the power exceeds the threshold determined from the background activity and which has a duration that exceeds a minimum number of cycles is marked. An extended BOSC or "eBOSC" option is also available that is meant to be an improvement upon the BOSC option (Kosciessa et al., 2020), although I found it to be quite slow.

For both the BOSC and eBOSC options, one defines a dataset in the working set and then the condition from that dataset that will provide the background estimate. All the frequency bands will be computed with one Herz bins if the sampling rate allows. One must also specify the width of the Morelet wavelet in cycles, the minimum duration in cycles to be considered an oscillatory episode, and the degree of deviation from the background level to be considered sufficient (in terms of percentile).

This procedure generates a new dataset where the channel types are "BSC" and the values are either zeros or ones (or values in between if one then averages this data). If one uses the View function to view the BOSC data, it will provide a time-frequency plot of the BOSC markings. Alternatively, if one views the raw EEG data in conjunction with the corresponding BOSC data, then the waveforms will be shown with the identified oscillatory periods highlighted in the BOSC color. Note that one cannot display the TFT transform and the BOSC output in the same figure as the TFT data will have reduced time resolution whereas the BOSC data will still have the time resolution of the original voltage data. Also, note that the BOSC procedure requires very wide shoulders for the data segments so it will generally require at least one minutes worth of data (dependent on the frequency and the Width and Duration settings), so not appropriate for the tutorial data. Here are some snapshots from a different dataset.



For the tutorial, choose the preprocessed (`_seg_e`) session files (not the average file) and generate both “frequency” and “time-frequency” versions, using the CSD reference. This procedure will generate frequency versions of each session file (“`_cs`” for CSD frequency and “`_cw`” for CSD time-frequency). For each set, then run the average function to generate an average file for analysis. Leave out #16, which you’ve already determined is a bad session. Note that you already low-pass filtered the tutorial data at 30Hz. Normally one would not do so for data intended for frequency analyses if one wished to inspect spectral analyses higher than or even in the vicinity of the 30Hz threshold.

As with the ERP data, read in the resulting averages and add a grand average using the Edit function. One can then view the results using the View function. One can view the data using four different common scaling: complex (cm), absolute amplitude (am), power (pw), and dB. Spectral results start in complex form with the real and imaginary parts coding for the cosine and sine components. Absolute amplitude converts these two parts into a single amplitude value. Power is the square of the amplitude.

Spectral measures follow a $1/f$ distribution, such that low frequencies have a high amplitude and higher frequencies have a much lower amplitude, which makes it harder to examine and gives the lower frequencies a disproportionate weight when averaging together frequency bands. The dB option helps address this problem by making the output log10-transformed (e.g., a value of 100 is 10^2 and the log10 of that is the exponent “2”, and then multiplied by a factor of 10 is “20”), evening out the frequency distribution. By the original definition of dB, it the log transform is applied to the power values. The dB option will result in negative numbers for values less than one (e.g., the log10 of .1 is “-1” and the dB is “-10”).

Internally, the EP Toolkit will represent the data in complex form to preserve the full information and will convert to the desired scaling upon output (whether as figures or for statistical analysis). Thus all the internal calculations are performed in complex scaling. There are two exceptions to this.

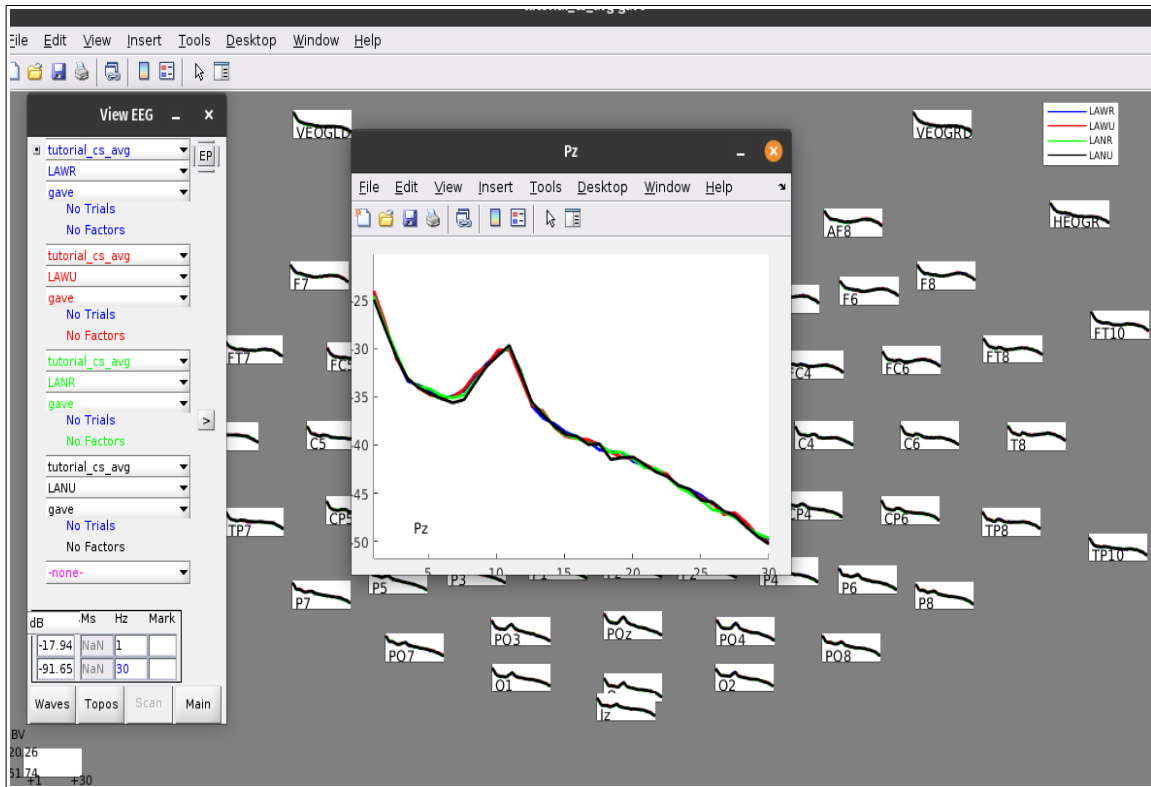
The first is that when data are added together (as in averaging or using the flexible segmenting or edit functions) there is a problem in that opposite signs can cancel out. While meaningful with voltages, when in complex scaling the signs just indicate phase differences. Thus two subjects might have equally large alpha waves but be 180 degrees out of synch. For most analyses this phase difference is irrelevant so when you add the data together you don't want them to cancel out. The EP Toolkit will therefore convert to absolute amplitude in such cases, although at the cost of losing the phase information. The exception to this is when adding channels together from the same segment. In this case the phase differences are meaningful as they reflect the bipolarity of the oscillating voltage fields (the two ends of the dipole will be 180 degrees out of phase) and preserving the sign information allows one to rereference the data. Thus, in this case the EP Toolkit will not convert the data to amplitude form.

The second exception is that when windowing data preparatory to running ANOVAs, the scaling is applied to the individual numbers (cells and subjects) that are output to the text file. Thus, if one chose dB scaling, each number will be in dB and when the ANOVA is calculated, it will be performing these calculations on numbers that are already in dB form. The EP Toolkit does it this way to respect standard practice in the field.

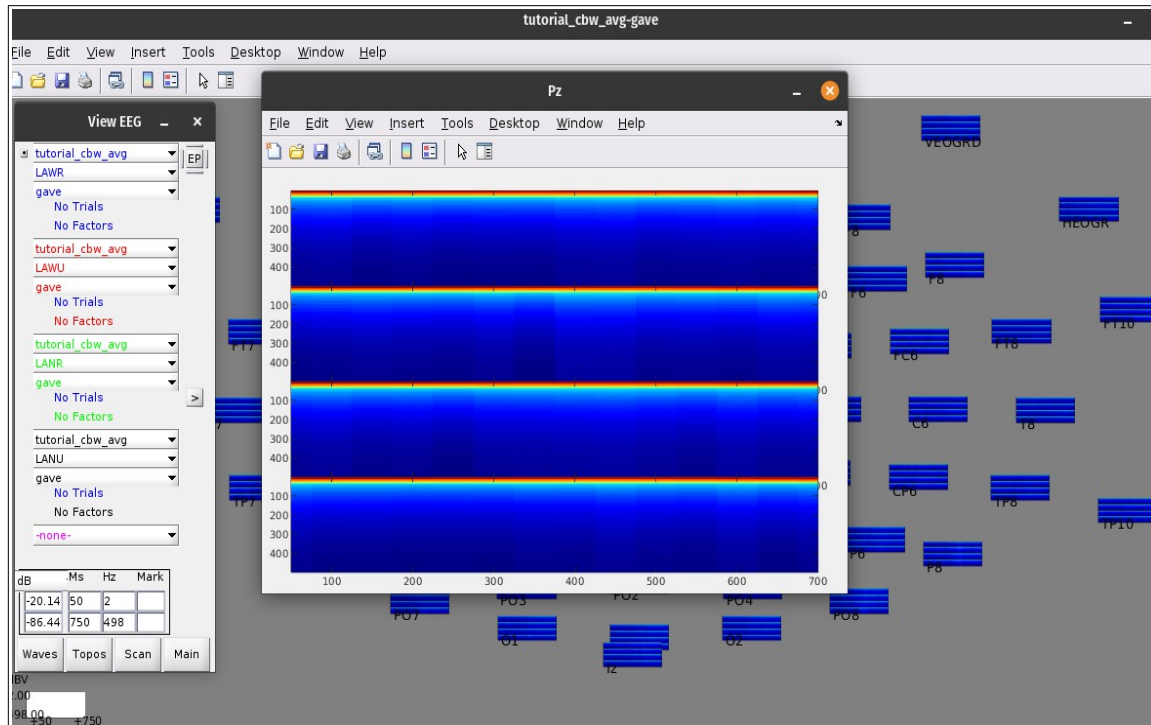
Regardless of which of the four options is chosen, the EP Toolkit will always apply a spectral density correction to the output. The spectral density measure is intended to correct for a distortion produced by digitized as opposed to continuous data. This is computed by dividing by the size of the spectral bins, which in turn is determined by 1 over the epoch length in seconds (e.g., 1 Hz for 1 second epochs). By convention this is done in terms of power so amplitude measures are divided by the square root of the spectral bin size.

One can then view the results using the View function. For Frequency domain data, the results look much the same except that the lines represent the distribution across

frequencies rather than across time. Since there will not be any frequencies above 30 Hz (due to the low-pass filtering), the frequency window should be set to 1-30 Hz.



For the Time-Frequency data, the display takes the form of color-coded charts in which the y-axis is the frequencies and the x-axis is the time. Each box is divided into the four different conditions, with Blue on top and Black on the bottom. If one sets the Hz control to only one frequency then the View function will present that one frequency bin as a line in the same manner as for time domain data. If one sets the time control to only one sample then the View function will present that one time point in the same manner as for frequency domain data. If you do both, the View function will not be able to present the data.

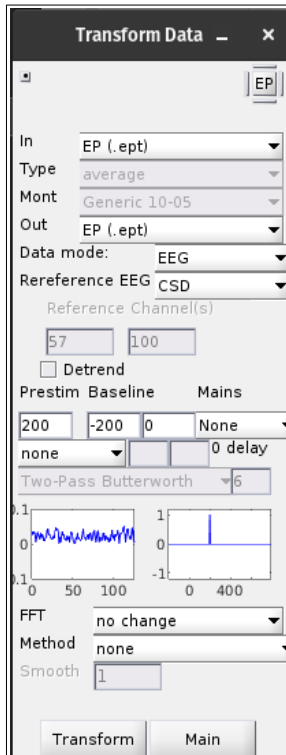


From here on out, the frequency data can be analyzed in a manner analogous to the voltage data.

Finally, inter-channel spectral coherence may be calculated. The EP Toolkit implementation should be considered a work in progress. The approach illustrated should not be considered a recommended procedure. There is much about the math that does not seem clear to me and I have had difficulty finding authoritative reference works. In this procedure, what is essentially a correlation is computed between channels in each frequency band, producing a number between -1 (inversely synchronized) and 1 (synchronized). Since it needs to be computed across trials and results in a summary measure, the EP Toolkit treats it as a type of averaging. To invoke it, use the Average Data panel with the Procedure set on Frequency-Coherence. As with the spectral transforms, one can choose either the Hanning or the multi-taper options. The measure indicates the degree of synchrony between channels in a given spectral band, although it cannot distinguish between degree of synchrony and amplitude (vis a vis the background activity).

Before the inter-channel relationships are computed, a decision needs to be made about the reference scheme. Unlike with voltage data, the data can't be rereferenced after the averaging procedure. A widely cited paper (Thatcher, 2012) recommended against average reference or CSD, although I didn't find its arguments to be especially persuasive; in fact, the review cited my own paper (Dien, 1998b) in a manner that did not reflect what it actually reported. However, this otherwise very informative review

(Thatcher, 2012) also made a very well-taken caution that just like with ERPs, oscillating fields also have both positive and negative poles, so if two distant electrodes have high negative coherence or high phase-lock value, it may just reflect the measurement of two sides of the same field. I'll therefore use the CSD transform for this part of the tutorial as well as it helps avoid the opposite pole problem (in that it filters out activity distant from the electrode so it is likely to attenuate or eliminate the more distant pole). First run Transform on the _seg_e session files, with rereference set on CSD and the Domain set on Time (so no spectral transform yet) and a 200 ms baseline, yielding files ending in "_cb".



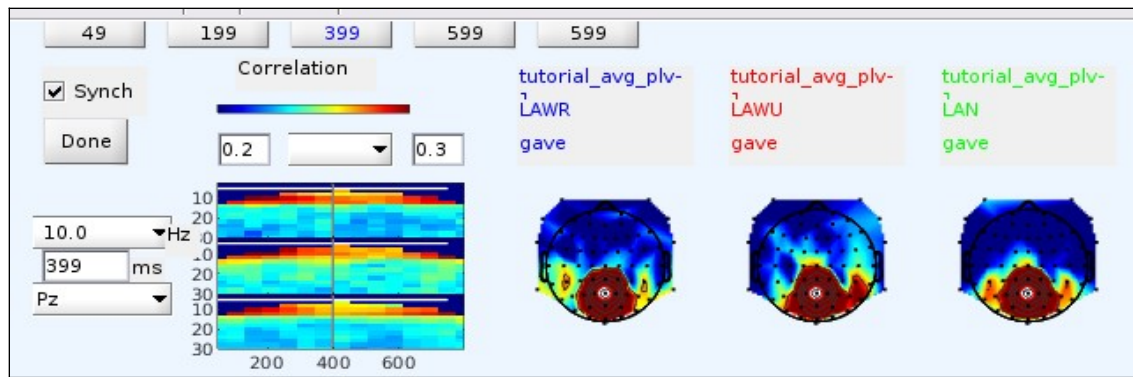
The next step then is to use the Average function to convert the data into coherence form, which will have a name ending in "_avg_coh".

The imaginary row (with the dotted waveform lines) also suggests temporal effects. While the imaginary component of coherence is often ignored, it is of interest because by definition it represents activity that is coherent with the index channel but at a fixed lag and hence is not due to the opposite side of the dipole (Nolte et al., 2004). The index channel itself will always have an imaginary value of zero, by definition.

When windowing coherence data, if there is one channel in an area then the correlations with all other channels will be used. The absolute value will be taken since otherwise they would tend to sum to zero (at least if average referenced). If a single common reference is used, it will be excluded since coherence with a flat channel produces a not-a-number result. If there is more than one channel in an area, then only the correlations between the channels in that area will be used. The absolute value will not be taken since the sign is meaningful and will not necessarily sum to zero. Separate files will be generated for the real and imaginary values.

Alternatively, one can use the coherence equivalent to the time-frequency transform, generating a time course of coherence changes, with a file name ending in “_avg_plv”. In this case the phase-lock statistic (Lachaux et al., 1999) is used as, unlike the coherence measure, it can help separate between amplitude and synchronization effects. As in the original report, the wavelet approach is used, although the Hilbert transform is often used, the results are the same if equivalent parameters are chosen (Bruns, 2004). The phase-lock statistic ranges from zero (no relation) to one (perfect phase synchrony). Also, unlike the coherence measure, phase-lock does not necessarily mean degree of synchrony but rather degree of stable phase-relationship. Thus, a phase-lock value would be a perfect one regardless of whether the oscillations at two electrodes are exactly in synchrony or exactly opposite to each other (180 degrees out of phase) as long as the phase relationship remains stable.

In the following figure, an area of phase-locked electrodes can be seen in the alpha band. The time-course clearly reveals the effects of approaching the edges of the epoch in blue (Not-a-Number due to falling off the edge of the segment), with the effects greater on low frequencies since they have a longer period. Here, a possible interesting effect can be seen involving CP5 and CP6 when the index channel is Pz in the alpha band at about 400 ms. Or perhaps it is artifactual in some manner? Many questions await!



Some Publications Using The EP Toolkit

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